

Generalized probability problems

with applications in quantitative finance

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Abstract

These notes develop a collection of classical problems in probability, each treated as a self-contained chapter: records and the constant e , the secretary problem, the Pólya urn, pattern hitting times, the ballot problem and the arcsine law, the longest run of heads, and the Kelly criterion, among others. The organizing principle is that a well-chosen problem is the best entry point into a body of theory. Every chapter starts from a concrete question and builds exactly the machinery its solution requires, so that exchangeability, martingales and optional stopping, dynamic programming, generating functions, renewal theory, and the Poisson approximation each appear where they are first genuinely needed, and recur across chapters. A running theme is the distance between intuition and calculation: many of the answers, such as the U-shape of the arcsine law, the rarity of records, or the appearance of π in coin tossing, are precise, elementary, and wrong to guess. Results are proved in full wherever feasible; the few that are not are stated precisely and referenced. Every numerical claim in the text has been verified by simulation. Where the mathematics bears on trading, in volatility estimation, backtest evaluation, position sizing, and the reading of P&L paths, the applications are developed alongside the theory rather than appended to it. An appendix collects the background results the chapters share.

Contents

1	Geometric Paradoxes and Classical Combinatorial Problems	5
1.1	Bertrand's Paradox	5
1.2	The Coupon Collector's Problem	7
1.3	The Derangement Problem	8
1.4	Variant: The Expected Number of Correct Envelopes	9
1.4.1	The Indicator Method	9
1.4.2	The Second Moment by the Same Device	10
1.4.3	All Factorial Moments Equal One	10
2	Records and the Constant e	12
2.1	The Motivating Problem	12
2.2	The Probability of a Record	12
2.3	The Expected Number of Records	12
2.4	Independence of the Record Indicators	13
2.5	The Sister Problem: How Many Uniforms to Exceed One?	14
2.6	Records in Trading: Which Clock Are You On?	17
3	The Secretary Problem and Optimal Stopping with Rankings	19
3.1	Problem Statement	19
3.2	The Optimal Strategy	19
3.3	Derivation	19

3.4	Variant: Minimizing Expected Rank	20
3.4.1	The Rank Lemma	20
3.4.2	The Value Function	20
3.4.3	A Worked Example: $n = 3$	21
3.4.4	The General Solution	22
3.5	Variant: Selecting Any of the Top k	22
3.6	Connection to the Sample-Buying Problem	23
4	Urn Models and the Pólya Urn	26
4.1	The Classical Pólya Urn Scheme	26
4.2	The Martingale Structure	26
4.3	Exchangeability of the Draw Sequence	27
4.4	The Limiting Distribution	27
4.5	Interpretation: The Reinforcement as a Dial	29
4.6	The Sign of c : Reinforcement, Independence, Exhaustion	30
4.6.1	The Brownian Bridge	30
4.6.2	The Card Deck as a Discrete Bridge	31
5	The Egg Drop Problem	33
5.1	Problem Statement	33
5.2	The Recursion	33
5.3	The Closed-Form Solution	33
5.4	Application: Two Eggs and 100 Floors	34
6	The Red–Black Card Stopping Problem	36
6.1	Problem Statement	36
6.2	Why This Is Not a Martingale Problem	36
6.3	Dynamic Programming Formulation	37
6.4	Structure of the Optimal Policy	38
6.5	Fixed-Threshold Approximations	38
7	Pattern Hitting Times: Setup and Autocorrelation	39
7.1	The Problem	39
7.1.1	The Autocorrelation Polynomial	39
7.2	The Guibas–Odlyzko Theorem	40
7.2.1	Proof I: Probability Generating Functions	40
7.2.2	Proof II: The Infinite Casino	42
7.3	Conway’s Trick: Pattern Races	44
7.3.1	The Penney Ante Game	44
7.3.2	Penney’s Strategy	45
7.3.3	Multi-Pattern Races	46
7.4	ABRACADABRA: Pattern Hitting Times via Martingales	46
7.4.1	The Casino of Fair Gamblers	47
7.4.2	Stopping at the Moment of Victory	47
8	Order Statistics: Maximum, Minimum, and Their Symmetries	49
8.1	The Pointwise Identities	49
8.2	The Tail-Sum Formula	49
8.3	Expected Maximum and Minimum: Continuous Uniform	50
8.4	Expected Maximum and Minimum: Discrete Uniform	51
8.5	The Discrete Sum Identity	51
8.6	The Rule of Thumb: Approximating Discrete by Continuous	52

8.7	Application: Solving $E X - Y $ for Dice in One Line	53
8.8	Example of verification: The Same Problem by Direct Summation	54
8.9	Application: The Expected Median of Three Dice	55
8.9.1	Method 1: Direct Computation	55
8.9.2	Method 2: The Sum Identity	56
8.9.3	Method 3: The Reflection Symmetry	56
8.10	Application: Auction Revenue and the Equispaced-Sample Formula	57
8.11	Where Order Statistics Appear in Markets	57
9	Size-Biasing: When Random Sampling Lies	59
9.1	The Waiting Time Paradox	59
9.2	The Length-Biased Distribution	59
9.3	Renewal Theory: The Density of Points	60
9.4	The Generalised Dice Problem	61
9.5	A Gallery of Size-Biasing Paradoxes	62
9.6	Applications to Trading	63
10	Random Walks and Gambler's Ruin	66
10.1	The Symmetric Random Walk on $\{0, 1, \dots, N\}$	66
10.2	Hitting Probability: The Linear Solution	66
10.3	Expected Hitting Time: The Quadratic Solution	67
10.4	Conditional Hitting Time	68
10.5	The Asymmetric Random Walk	68
10.6	Worked Examples	70
11	The Reflection Principle and the Ballot Problem	72
11.1	Lattice Paths and the Setup	72
11.2	The Reflection Principle	72
11.3	The Ballot Problem	73
11.4	First-Passage Probabilities for the Random Walk	74
11.5	Connection to Brownian Motion	75
12	First Crossing of the Majority: Catalan Numbers and $\pi/4$	77
12.1	Problem Statement	77
12.2	Reduction to a First-Passage Problem	77
12.3	The Shape of the Stopping State	77
12.4	The Analytic Toolbox: Two Generating Functions	78
12.5	The First-Passage Distribution: Catalan Numbers Appear	79
12.6	Computing the Expected Final Proportion	80
12.7	Convergence and the Tail	81
12.8	Generalizations and Connections	82
13	The Arcsine Law for the Time Spent Positive	84
13.1	Problem Statement	84
13.2	Strategy of the Proof	84
13.3	Tools Recalled from the First-Crossing Section	85
13.4	Two Combinatorial Lemmas	85
13.5	The Prefix Generating Function	86
13.6	The Last-Zero Decomposition	87
13.7	The Main Theorem	87
13.8	The Continuous Limit: The Arcsine Distribution	89
13.9	The Counterintuitive Shape	90

13.10	Why π Appears	91
13.11	Applications to Trading	91
13.12	Connection to Brownian Motion	92
14	The Longest Run of Heads: A Tour of Approximations	93
14.1	Problem Statement	93
14.2	Why This Problem Is Interesting	93
14.3	Method 1: The Poisson Approximation	93
14.3.1	Counting Potential Runs	93
14.3.2	The Poisson Step	94
14.3.3	From One Person to Thirty	94
14.4	Method 2: The Pooling Principle	95
14.5	Method 3: Schilling's Asymptotic Formula	96
14.5.1	The Biased Coin	96
14.6	Method 4: Exact Computation via a Markov Chain	97
14.6.1	The Run-Length State Machine	97
14.6.2	The Distribution of L_{162}	98
14.6.3	The Distribution of the Maximum	99
14.7	Comparison of the Four Methods	99
14.8	Connections to Trading	100
15	The Kelly Criterion and Volatility Drag	102
15.1	The Motivating Problem	102
15.2	The Seduction of Expected Value	102
15.3	The Right Objective: The Growth Rate	102
15.4	The Kelly Fraction	104
15.5	Volatility Drag	104
15.6	Continuous Kelly and the Growth Parabola	105
15.7	Fractional Kelly and the Price of Safety	106
15.8	Kelly and Information Theory	107
15.9	What Kelly Does Not Say	107
A	Mathematical Background	109
A.1	Asymptotic Estimates	109
A.1.1	Stirling's Approximation	109
A.1.2	Harmonic Numbers and the Euler–Mascheroni Constant	109
A.2	Recurring Distributions	109
A.2.1	The Geometric Distribution: Two Conventions	109
A.2.2	The Poisson Paradigm	110
A.2.3	The Beta Distribution	110
A.2.4	The Gumbel Distribution and Extreme Values	111
A.3	Structural Tools	111
A.3.1	Exchangeability and de Finetti's Theorem	111
A.3.2	Martingales and the Optional Stopping Theorem	112
A.3.3	Generating Functions	113

This Part collects a set of representative quant problems organized not by topic but by the *technique* required to solve them. The aim is to build pattern recognition: when given an unfamiliar problem, the first task is to recognize which tool to reach for.

1 Geometric Paradoxes and Classical Combinatorial Problems

1.1 Bertrand's Paradox

The Question

Consider a circle of radius 1 and an equilateral triangle inscribed in it (side length $\sqrt{3}$). Draw a “random” chord. What is the probability that the chord is longer than the side of the triangle?

The paradox is that the answer depends entirely on what “random” means. Three natural interpretations give three different answers: $1/2$, $1/3$, and $1/4$. This is not a contradiction in mathematics; it is a warning that “uniform at random” on a non-trivial geometric object requires explicit specification.

Before the three constructions, we record the elementary geometry that all of them share. The number $\sqrt{3}$ is not arbitrary: it is the side length of the equilateral triangle inscribed in the unit circle, and every method below reduces to comparing a chord against that triangle.

Lemma 1.1 (Chord Geometry in the Unit Circle). *In a circle of radius 1:*

- (a) *A chord whose endpoints are separated by central angle $\varphi \in [0, 2\pi)$ has length $2\sin(\varphi/2)$.*
- (b) *A chord whose midpoint lies at distance r from the centre has length $2\sqrt{1-r^2}$.*
- (c) *The inscribed equilateral triangle has side $\sqrt{3}$, and a chord is longer than $\sqrt{3}$ if and only if*

$$\varphi \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right) \iff r < \frac{1}{2}.$$

Proof. (a) Let O be the centre and A, B the endpoints. The triangle OAB is isosceles with $|OA| = |OB| = 1$ and apex angle φ . Dropping the perpendicular from O to AB bisects both the apex angle and the chord, producing a right triangle with hypotenuse 1 and opposite angle $\varphi/2$; the half-chord is therefore $\sin(\varphi/2)$ and the chord is $2\sin(\varphi/2)$.

(b) Let M be the midpoint of the chord, at distance $r = |OM|$. The segment OM is perpendicular to the chord (the perpendicular from the centre to a chord bisects it, and conversely the bisecting segment is the perpendicular). The radius to an endpoint, the segment OM , and the half-chord form a right triangle with hypotenuse 1 and legs r and the half-chord; by Pythagoras the half-chord is $\sqrt{1-r^2}$.

(c) The vertices of the inscribed equilateral triangle are separated by central angle $2\pi/3$, so by (a) its side is $2\sin(\pi/3) = \sqrt{3}$. For the thresholds:

$$2\sin(\varphi/2) > \sqrt{3} \iff \sin(\varphi/2) > \frac{\sqrt{3}}{2} \iff \frac{\varphi}{2} \in \left(\frac{\pi}{3}, \frac{2\pi}{3}\right) \iff \varphi \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right),$$

$$2\sqrt{1-r^2} > \sqrt{3} \iff 1-r^2 > \frac{3}{4} \iff r < \frac{1}{2}. \quad \square$$

Proposition 1.2 (Method 1: Random Endpoints). *Choose two points independently and uniformly on the circle and connect them. The probability that the chord is longer than $\sqrt{3}$ is $1/3$.*

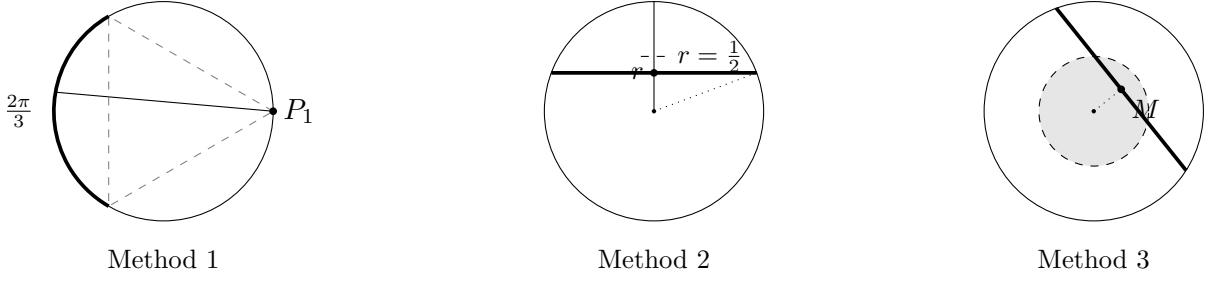


Figure 1: The three constructions. *Left:* with P_1 fixed, the chord beats $\sqrt{3}$ iff the second endpoint falls in the thick arc of width $2\pi/3$ opposite P_1 (the arc cut off by the inscribed triangle). *Centre:* the chord perpendicular to a radius at distance r has its midpoint at r and beats $\sqrt{3}$ iff $r < 1/2$. *Right:* a random midpoint M determines its chord; the chord beats $\sqrt{3}$ iff M lands in the shaded concentric disk of radius $1/2$.

Proof. By rotational symmetry, fix the first endpoint P_1 at angle 0; the second endpoint then has angle $\varphi \sim \text{Uniform}[0, 2\pi)$. By Lemma 1.1(a,c), the chord exceeds $\sqrt{3}$ iff $\varphi \in (2\pi/3, 4\pi/3)$ — the arc of angular width $2\pi/3$ centred at the antipode of P_1 , namely the arc that the inscribed equilateral triangle with a vertex at P_1 cuts off on the far side (Figure 1, left). Since φ is uniform,

$$P(\text{chord} > \sqrt{3}) = \frac{2\pi/3}{2\pi} = \frac{1}{3}. \quad \square$$

Proposition 1.3 (Method 2: Random Radius). *Choose a radius uniformly at random, then a point uniformly along that radius, and draw the chord through the point perpendicular to the radius. The probability that the chord is longer than $\sqrt{3}$ is $1/2$.*

Proof. The chord is perpendicular to the segment joining the centre to the chosen point; since the perpendicular from the centre bisects a chord, the chosen point is the midpoint of the chord, at distance $r \sim \text{Uniform}[0, 1]$ from the centre. By Lemma 1.1(b,c), the chord exceeds $\sqrt{3}$ iff $r < 1/2$ (Figure 1, centre). Since r is uniform on $[0, 1]$,

$$P(\text{chord} > \sqrt{3}) = \frac{1/2}{1} = \frac{1}{2}. \quad \square$$

Proposition 1.4 (Method 3: Random Midpoint). *Choose a point M uniformly at random in the disk and let it be the midpoint of a chord. The probability that the chord is longer than $\sqrt{3}$ is $1/4$.*

Proof. For M different from the centre, there is exactly one chord with midpoint M : the chord through M perpendicular to OM (the case $M = O$ has probability zero). By Lemma 1.1(b,c), this chord exceeds $\sqrt{3}$ iff $|OM| < 1/2$, i.e. iff M falls inside the concentric disk of radius $1/2$ (Figure 1, right). Since M is uniform with respect to area,

$$P(\text{chord} > \sqrt{3}) = \frac{\pi(1/2)^2}{\pi \cdot 1^2} = \frac{1}{4}. \quad \square$$

Remark 1.5 (Resolution: Different Probability Measures). The three methods correspond to three different probability measures on the space of chords. There is no canonical “uniform measure on chords” because the chord space is not a homogeneous symmetric object. Edwin Jaynes argued for Method 2 as the most natural based on a principle of maximum invariance (the only measure invariant under translation, rotation, and scaling).

Why This Matters for Trading

Bertrand's paradox is a cautionary tale about the ambiguity of “random” in continuous settings. In finance, this manifests whenever you specify a model with hidden assumptions:

- **Sampling intervals:** choosing log returns vs. arithmetic returns, daily vs. minute frequency, business days vs. calendar days—each “random time” specification gives a different statistical picture.
- **Distribution choice:** “the implied volatility is random”—random under what measure? Risk-neutral, real-world, or local-volatility?
- **Order arrival models:** “orders arrive at random times”—Poisson, Hawkes, or simply self-exciting? Each leads to different microstructure conclusions.

The lesson: always make explicit the probability measure or sampling scheme. “Random” is not self-defining.

1.2 The Coupon Collector's Problem

The Problem

Cereal boxes contain one of n different coupons (uniformly at random, independently). How many boxes do you need to buy, on average, to collect all n coupons?

Coupon Collector

Theorem 1.6. *Let T_n be the number of boxes needed to collect all n coupons. Then*

$$E[T_n] = n H_n = n \sum_{k=1}^n \frac{1}{k} \sim n \log n$$

Proof. Decompose by phases. Phase k begins when you have $k - 1$ distinct coupons and ends when you get the k -th. Let X_k be the number of boxes in phase k .

During phase k , each box reveals a missing coupon with probability $(n - k + 1)/n$. Hence $X_k \sim \text{Geom}(\frac{n-k+1}{n})$ with $E[X_k] = \frac{n}{n-k+1}$.

By linearity:

$$E[T_n] = \sum_{k=1}^n \frac{n}{n-k+1} = n \sum_{j=1}^n \frac{1}{j} = n H_n.$$

Since $H_n \sim \log n + \gamma$, $E[T_n] \sim n \log n$. □

Proposition 1.7 (Variance of Coupon Collector).

$$\text{Var}(T_n) = \sum_{k=1}^n \frac{n(k-1)}{(n-k+1)^2} \sim \frac{\pi^2}{6} n^2.$$

Practical Implications

The variance is $O(n^2)$, comparable to the mean's $O(n \log n)$ squared—the distribution has a heavy right tail. Collecting the last few coupons takes disproportionately long.

Applications:

- **Algorithm design:** random sampling coverage.
- **Market making:** at least one trade in each of n instruments takes $\sim n \log n$ time.
- **Network testing:** hitting all servers.
- **Backtest data sufficiency:** how many independent regimes must you observe before generalization confidence?

1.3 The Derangement Problem

The Problem

n letters are placed at random into n pre-addressed envelopes. What is the probability that *no* letter ends up in its correct envelope? A permutation with no fixed points is a *derangement*.

Derangement Probability

Theorem 1.8.

$$D_n = \sum_{k=0}^n \frac{(-1)^k}{k!} \rightarrow \frac{1}{e} \quad \text{as } n \rightarrow \infty.$$

Proof. By inclusion-exclusion. Let A_i be the event that letter i ends in envelope i . We want $P(\cap A_i^c) = 1 - P(\cup A_i)$.

For any set S of size k , the probability that all letters in S go correctly is $(n-k)!/n!$. By inclusion-exclusion:

$$P\left(\bigcup A_i\right) = \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{(n-k)!}{n!} = \sum_{k=1}^n \frac{(-1)^{k+1}}{k!}.$$

Therefore $D_n = \sum_{k=0}^n (-1)^k/k! \rightarrow e^{-1}$. □

Remark 1.9 (Convergence Speed). The convergence to $1/e$ is extremely rapid: $D_4 \approx 0.3750$, $D_7 \approx 0.3679$, and D_n rounds to 0.3679 for all $n \geq 7$. Faster than the $1/e$ in the secretary problem.

Why Two $1/e$'s?

Both the derangement problem and the secretary problem yield $1/e \approx 0.3679$, but for different reasons:

- In derangements, $1/e$ arises from the exponential series $\sum (-1)^k/k!$.
- In the secretary problem, $1/e$ arises from maximizing $-x \log x$ over $x \in (0, 1)$.

The coincidence is fundamentally about the exponential function appearing in two ways: as a Taylor series and as a solution to an optimization.

Geometric and Combinatorial Classics

1. **Bertrand's paradox:** “random” on continuous objects requires explicit measure specification. Three answers $(1/2, 1/3, 1/4)$ for three natural measures.
2. **Coupon collector:** $E[T_n] = nH_n \sim n \log n$. Proof by phases: each phase is geometric with success probability $(n - k + 1)/n$.
3. **Derangements:** $D_n \rightarrow 1/e$, via inclusion-exclusion.
4. **Common theme:** combinatorial problems reduce to harmonic numbers, exponentials, or optimal stopping. The same structures appear repeatedly.

1.4 Variant: The Expected Number of Correct Envelopes

The derangement problem asks for the probability that *no* letter reaches its intended recipient. A companion question is often easier and more instructive: *how many letters reach the right recipient, on average?*

Example 1.10 (The Secretary's Envelopes). A secretary places n letters into n pre-addressed envelopes uniformly at random. Let X be the number of letters that end up in the correct envelope. Find $E[X]$.

The natural first instinct — compute the distribution of X and sum $\sum_k k P(X = k)$ — is a trap: that distribution involves derangement numbers and is far harder than the question. The efficient route dismantles X into pieces so simple that no distribution is needed at all.

1.4.1 The Indicator Method

Expected Number of Fixed Points

Proposition 1.11. For every $n \geq 1$,

$$E[X] = 1.$$

On average exactly one letter reaches its destination — whether $n = 5$ or $n = 5,000,000$.

Proof. For $i = 1, \dots, n$ define the indicator

$$\mathbb{I}_i = \begin{cases} 1 & \text{if letter } i \text{ lands in envelope } i, \\ 0 & \text{otherwise,} \end{cases} \quad \text{so that} \quad X = \sum_{i=1}^n \mathbb{I}_i.$$

The defining property of an indicator is that its expectation is the probability of the event: $E[\mathbb{I}_i] = 1 \cdot P(A_i) + 0 \cdot P(A_i^c) = P(A_i)$. By symmetry letter i is equally likely to land in any of the n envelopes, so $P(A_i) = 1/n$. Linearity of expectation gives

$$E[X] = \sum_{i=1}^n E[\mathbb{I}_i] = \sum_{i=1}^n \frac{1}{n} = 1. \quad \square$$

Why This Works: Linearity Needs No Independence

The events A_i are emphatically *not* independent: if $n - 1$ letters are correctly placed, the last one must be too, so $P(A_n \mid A_1, \dots, A_{n-1}) = 1 \neq 1/n$. A method requiring independence would be unusable here.

Linearity of expectation does not care. $E[Y + Z] = E[Y] + E[Z]$ holds for *any* random variables on a common probability space, dependent or not. This is the single most useful fact in elementary probability, and the indicator method is its systematic exploitation: *to count something, write the count as a sum of indicators, and add up the individual probabilities*. The dependence structure — however tangled — never enters.

The recipe generalises far beyond this problem. Whenever a quantity can be phrased as “the number of times something happens,” decompose it into one indicator per opportunity. We have already used the device to count records ($E[R_n] = \sum 1/k = H_n$) and expected runs; we will use it again for expected coupon draws and expected pattern occurrences.

1.4.2 The Second Moment by the Same Device

Indicators handle variance just as cleanly, provided one computes the *factorial* moment $E[X(X - 1)]$ rather than $E[X^2]$ directly.

Proposition 1.12 (Variance of the Number of Fixed Points). *For every $n \geq 2$,*

$$\text{Var}(X) = 1.$$

Proof. Counting ordered pairs of distinct correct letters,

$$X(X - 1) = \sum_{i \neq j} \mathbb{I}_i \mathbb{I}_j,$$

where the sum runs over the $n(n - 1)$ ordered pairs. Now $\mathbb{I}_i \mathbb{I}_j$ is itself the indicator of the event “letters i and j are both correctly placed,” whose probability is

$$P(A_i \cap A_j) = \frac{1}{n} \cdot \frac{1}{n - 1} = \frac{1}{n(n - 1)},$$

since once letter i is fixed, letter j has $n - 1$ equally likely envelopes. Hence

$$E[X(X - 1)] = n(n - 1) \cdot \frac{1}{n(n - 1)} = 1,$$

so $E[X^2] = E[X(X - 1)] + E[X] = 1 + 1 = 2$ and

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 2 - 1 = 1. \quad \square$$

1.4.3 All Factorial Moments Equal One

The two computations above are the first cases of a pattern that fully identifies the limiting law.

Factorial Moments

Proposition 1.13. *For every $r \leq n$,*

$$E[X(X - 1) \cdots (X - r + 1)] = 1.$$

Proof. The falling factorial counts ordered r -tuples of distinct correctly-placed letters:

$$X(X-1)\cdots(X-r+1) = \sum_{\substack{(i_1, \dots, i_r) \\ \text{distinct}}} \mathbb{1}_{i_1} \cdots \mathbb{1}_{i_r}.$$

There are $n(n-1)\cdots(n-r+1)$ such tuples, and each product is the indicator that r specified letters are all correctly placed, of probability $\frac{1}{n(n-1)\cdots(n-r+1)}$. Multiplying gives 1. $\square$

Remark 1.14 (Why the Limit Is Poisson(1)). A Poisson random variable with parameter λ has factorial moments $E[Y(Y-1)\cdots(Y-r+1)] = \lambda^r$. Proposition 1.13 says all factorial moments of X equal $1 = 1^r$ — exactly those of Poisson(1). Since a distribution on $\mathbb{N}$ with suitably controlled moments is determined by them, this forces

$$X \xrightarrow{d} \text{Poisson}(1), \quad P(X = k) \rightarrow \frac{e^{-1}}{k!}.$$

The convergence is extraordinarily fast. Exactly,

$$P(X = k) = \binom{n}{k} \frac{D_{n-k}}{n!} = \frac{1}{k!} \cdot \frac{D_{n-k}}{(n-k)!},$$

(choose which k letters are correct, derange the rest), and already at $n = 10$ the values 0.367879, 0.367879, 0.183941, 0.061310 for $k = 0, 1, 2, 3$ agree with Poisson(1) to five decimal places.

Closing the Circle with Derangements

Setting $k = 0$ in the exact formula recovers the previous subsection: $P(X = 0) = D_n/n! \rightarrow e^{-1}$. The derangement problem is precisely the $k = 0$ atom of the fixed-point distribution, and the constant $1/e$ appearing there is the Poisson(1) probability of *zero* events.

The two subsections therefore illuminate one another. Inclusion–exclusion delivers the exact probability of the single event $\{X = 0\}$ at the cost of an alternating sum; indicators deliver the entire moment structure — mean, variance, all factorial moments, and hence the limiting law — at the cost of almost no computation. The lesson is about tool selection: *when the question is about a count, reach for indicators before reaching for the distribution.*

2 Records and the Constant e

2.1 The Motivating Problem

An asset's price series sets a *new all-time high* whenever today's value exceeds every previous value. Records of this kind feel newsworthy — financial media report “record highs” as though they were extraordinary events. How often should records actually occur?

We formalise the cleanest version first. Let $X_1, X_2, \dots$ be i.i.d. continuous random variables (the continuity assumption removes ties, which occur with probability zero). Observation n is a *record* if

$$X_n > \max(X_1, \dots, X_{n-1}),$$

with X_1 counting as a record by convention. Two natural questions:

- (a) What is the probability that observation n is a record?
- (b) How many records do we expect among the first n observations?

The answers require no knowledge whatsoever of the underlying distribution — only the assumption of exchangeability — and they lead, by a short detour, to one of the most charming appearances of the constant e in all of probability.

2.2 The Probability of a Record

Record Probability

Theorem 2.1. For i.i.d. continuous $X_1, \dots, X_n$,

$$P(X_n \text{ is a record}) = \frac{1}{n}.$$

Proof. By exchangeability, the maximum of $(X_1, \dots, X_n)$ is equally likely to sit at each of the n positions. The event “ X_n is a record” is precisely the event “the maximum sits at position n ,” which therefore has probability $1/n$. $\square$

This one-line argument is worth savouring: no densities, no integrals, no distributional assumptions beyond symmetry. The result is *distribution-free* — it holds for Gaussian returns, Cauchy returns, or any exotic continuous law, because it is a statement about relative ranks only.

2.3 The Expected Number of Records

Records Are Logarithmically Rare

Theorem 2.2. Let R_n be the number of records among $X_1, \dots, X_n$. Then

$$E[R_n] = \sum_{k=1}^n \frac{1}{k} = H_n \approx \ln n + \gamma,$$

where H_n is the n -th harmonic number and $\gamma \approx 0.5772$ is the Euler–Mascheroni constant.

Proof. Write $R_n = \sum_{k=1}^n I_k$, where I_k is the indicator that observation k is a record. By Theorem 2.1, $E[I_k] = 1/k$, and linearity of expectation (which requires no independence) gives the sum. $\square$

Example 2.3 (Concrete Numbers). Among 100 observations: $H_{100} \approx 5.19$ records. Among 10,000: $H_{10,000} \approx 9.79$. Among a million: ≈ 14.39 . The pattern is worth stating precisely: *multiplying the sample size by a fixed factor c adds a fixed number of records, namely $\ln c$* — each hundredfold increase above buys only $\ln 100 \approx 4.6$ more. (Squaring n , by contrast, adds $\ln n$, which itself grows without bound: 2.3 records when $n = 10$ becomes 9.2 when $n = 10^4$.) Records become progressively, relentlessly rarer: each one leaves a newcomer more values to beat.

2.4 Independence of the Record Indicators

The indicators I_k have known means, but a far stronger and genuinely surprising statement holds.

Independence of Records

Theorem 2.4. *The record indicators $I_1, I_2, I_3, \dots$ are mutually **independent**, with $P(I_k = 1) = 1/k$.*

Proof. Condition on the set of values $\{X_1, \dots, X_n\}$ (without their order). By exchangeability, all $n!$ assignments of these values to positions are equally likely; equivalently, the relative-rank sequence is a uniformly random permutation.

Let $\rho_k \in \{1, \dots, k\}$ denote the rank of X_k among the first k observations ($\rho_k = k$ meaning X_k is the largest so far). We claim the ranks $\rho_1, \rho_2, \dots, \rho_n$ are independent, with ρ_k uniform on $\{1, \dots, k\}$. Indeed, the map from permutations of $\{1, \dots, n\}$ to rank sequences $(\rho_1, \dots, \rho_n)$ is a bijection onto $\{1\} \times \{1, 2\} \times \dots \times \{1, \dots, n\}$ (the sequence of ranks reconstructs the permutation by sequential insertion), and the uniform distribution on the $n!$ permutations pushes forward to the uniform distribution on this product set — which is precisely the claim that the ρ_k are independent uniforms.

Since $I_k = \mathbb{1}\{\rho_k = k\}$ is a function of ρ_k alone, the indicators inherit independence, and $P(I_k = 1) = P(\rho_k = k) = 1/k$. $\square$

Why Independence Is Surprising — and Then Obvious

At first sight, independence seems impossible: if X_7 set a towering record, shouldn't that make a record at time 8 *less* likely? The resolution is that “record at time 8” is a statement about *ranks*, not values. Whether X_8 exceeds the current maximum depends only on where X_8 falls among the first 8 values — and by exchangeability, X_8 is equally likely to occupy any of the 8 relative positions, *regardless* of the internal ordering of the first 7. The height of the previous record is irrelevant to the rank of the newcomer. Records are rank events, and ranks decouple.

Remark 2.5 (Fluctuations: a Free Central Limit Theorem). Independence upgrades Theorem 2.2 from a statement about the mean to a full asymptotic distribution.

The mean $E[R_n] = H_n$ required only linearity of expectation, which holds whether or not the indicators are dependent. Variances are different: in general $\text{Var}(\sum_k I_k) = \sum_k \text{Var}(I_k) + \sum_{j \neq k} \text{Cov}(I_j, I_k)$, and it is precisely Theorem 2.4 that annihilates every covariance term. With $I_k \sim \text{Bernoulli}(1/k)$ and $\text{Var}(I_k) = \frac{1}{k}(1 - \frac{1}{k})$,

$$\text{Var}(R_n) = \sum_{k=1}^n \frac{1}{k} \left(1 - \frac{1}{k}\right) = H_n - H_n^{(2)} \approx \ln n + \gamma - \frac{\pi^2}{6},$$

where $H_n^{(2)} = \sum_{k \leq n} k^{-2}$ is the harmonic number of order two. The two sums behave oppositely: H_n diverges like $\ln n$, whereas $H_n^{(2)}$ converges to $\pi^2/6$ (Basel). The subtraction therefore costs only a constant, and the variance grows at the same logarithmic rate as the mean. At $n = 5000$ the formula gives 7.45 against a simulated 7.48.

Since the summands are bounded and $\text{Var}(R_n) \rightarrow \infty$, the Lindeberg condition holds trivially and the central limit theorem applies:

$$\frac{R_n - \ln n}{\sqrt{\ln n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Centring at $\ln n$ rather than at $H_n = \ln n + \gamma + o(1)$ is immaterial here: an $O(1)$ shift vanishes against a $\sqrt{\ln n}$ normalisation.

Both the level and the fluctuation of the record count are governed by the same slow clock, $\ln n$.

Remark 2.6 (Records and Cycles: A Hidden Bijection). The identity $E[R_n] = H_n$ has an exact counterpart: a uniform random permutation of $\{1, \dots, n\}$ has, on average, H_n cycles. This is no coincidence, and the agreement is far stronger than a shared mean — the two counts have the *same distribution*, because there is an explicit bijection carrying one into the other.

Foata's fundamental bijection. Given a permutation in cycle notation, write each cycle starting with its largest element, order the cycles by increasing largest element, and erase the parentheses. The result is a permutation in one-line notation, and the operation is reversible: the cycle maxima are exactly the left-to-right maxima — the *records* — of the resulting word, so the parentheses can always be restored.

Example. The permutation with cycles (13)(2)(45) becomes, after writing each cycle max-first, (31)(2)(54); ordering by maxima gives (2)(31)(54); erasing parentheses gives the word 23154, whose records are 2, 3, 5. Three cycles, three records.

Since the map is a bijection on the $n!$ permutations,

$$\#\{\text{cycles of } \sigma\} \stackrel{d}{=} \#\{\text{records of } \sigma\},$$

and both are therefore distributed as $\sum_{k=1}^n \text{Bernoulli}(1/k)$ with independent summands — the structure established in Theorem 2.4. Their common distribution is given by the unsigned Stirling numbers of the first kind.

A word of caution about a different H_n : the coupon collector's mean nH_n is *not* an instance of this phenomenon. There the harmonic sum arises from adding geometric waiting times $n/(n-k)$, a mechanism unrelated to ranks or permutations. Harmonic numbers are common enough to appear from several independent causes; only the records–cycles agreement is structural.

2.5 The Sister Problem: How Many Uniforms to Exceed One?

The same exchangeability engine drives a problem whose answer is unreasonably beautiful.

Example 2.7 (The Uniform Sum Problem). Draw i.i.d. $U_1, U_2, \dots \sim \text{Uniform}(0, 1)$ and keep a running sum. Let

$$N = \min\{n : U_1 + U_2 + \dots + U_n > 1\}$$

be the number of draws needed for the sum to exceed 1. What is $E[N]$?

The Expected Count Is e

Theorem 2.8.

$$E[N] = e \approx 2.71828.$$

Proof. We proceed in four steps, each elementary on its own.

Step 0: reduce everything to one probability. By the tail-sum formula for non-negative integer variables,

$$E[N] = \sum_{n=0}^{\infty} P(N > n).$$

Now, $N > n$ says that the threshold has *not yet* been crossed after n draws — that is, $U_1 + \cdots + U_n \leq 1$. (For $n = 0$ the empty sum is $0 \leq 1$, so $P(N > 0) = 1$.) Everything therefore hinges on the single quantity

$$p_n := P(U_1 + \cdots + U_n \leq 1),$$

and we claim $p_n = 1/n!$.

Step 1: the probability is a volume. The vector $(U_1, \dots, U_n)$ is uniformly distributed on the unit cube $[0, 1]^n$, and for a uniform distribution the probability of a region is its volume. Our region is the *corner simplex*

$$\Delta_n = \{u : u_1, \dots, u_n \geq 0, u_1 + \cdots + u_n \leq 1\}$$

(the constraints $u_i \leq 1$ are automatic: non-negative numbers summing to at most 1 cannot exceed 1 individually). For $n = 2$ this is the triangle of Figure 2, left, with area $1/2$; for $n = 3$ it is the tetrahedron with vertices at the origin and the three unit points, with volume $1/6$. The pattern to prove: $\text{vol}(\Delta_n) = 1/n!$.

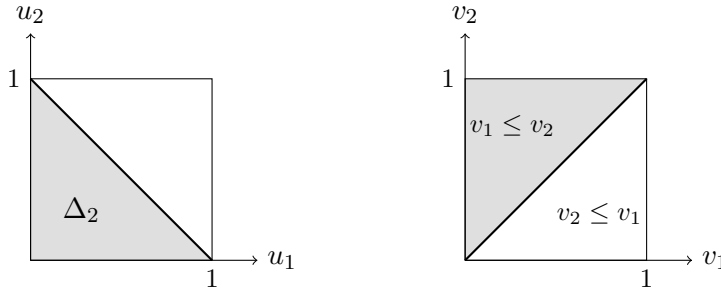


Figure 2: The case $n = 2$. *Left:* p_2 is the area of the corner triangle Δ_2 , visibly $1/2$. *Right:* the running-sum map $(u_1, u_2) \mapsto (v_1, v_2) = (u_1, u_1 + u_2)$ carries Δ_2 onto the upper triangle $\{0 \leq v_1 \leq v_2 \leq 1\}$ without changing area, and the diagonal $v_1 = v_2$ cuts the square into $2! = 2$ congruent triangles — so each has area $1/2!$.

Step 2: change variables to the running sums. Define the partial sums

$$v_1 = u_1, \quad v_2 = u_1 + u_2, \quad \dots, \quad v_n = u_1 + \cdots + u_n,$$

in matrix form (shown for $n = 3$):

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}}_A \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \quad \text{with inverse} \quad u_1 = v_1, \quad u_2 = v_2 - v_1, \quad u_3 = v_3 - v_2.$$

Each constraint on u translates into one on v :

$$u_k \geq 0 \iff v_k - v_{k-1} \geq 0 \iff \text{the } v_k \text{ are increasing}; \quad u_1 + \cdots + u_n \leq 1 \iff v_n \leq 1.$$

So the map carries Δ_n exactly onto the *order simplex*

$$S_n = \{0 \leq v_1 \leq v_2 \leq \cdots \leq v_n \leq 1\}.$$

And it carries volumes faithfully: a linear map scales all volumes by $|\det A|$, and A is triangular with ones on the diagonal, so $\det A = 1$. Hence $\text{vol}(\Delta_n) = \text{vol}(S_n)$, and it remains to find the volume of S_n .

Step 3: the order simplex is one of $n!$ identical tiles. Classify each point v of the cube $[0, 1]^n$ by the *ordering* of its coordinates: for each permutation σ of $\{1, \dots, n\}$, let

$$S_\sigma = \{v \in [0, 1]^n : v_{\sigma(1)} \leq v_{\sigma(2)} \leq \dots \leq v_{\sigma(n)}\}.$$

Three observations:

- (i) The $n!$ regions S_σ *cover the cube*: every point's coordinates admit some ordering.
- (ii) They overlap only where two coordinates are *equal* — a set of volume zero.
- (iii) They are *congruent*: relabelling the axes by a permutation is an isometry of the cube that maps any one region onto any other.

So the cube of volume 1 is tiled by $n!$ pieces of equal volume, and each piece — in particular our $S_n = S_{\text{id}}$ — has volume $1/n!$. For $n = 2$ this is the right panel of Figure 2: the diagonal splits the square into two congruent triangles. (This is the same symmetry that powered Theorem 2.1: there, all $n!$ orderings of ranks were equally *likely*; here, all $n!$ ordering regions are equally *large*.)

Step 4: assemble. Combining the steps, $p_n = \text{vol}(\Delta_n) = \text{vol}(S_n) = 1/n!$, and the tail sum evaluates to the exponential series:

$$E[N] = \sum_{n=0}^{\infty} \frac{1}{n!} = \underbrace{1}_{n=0} + \underbrace{1}_{n=1} + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \dots = e.$$

The first two terms both equal 1 because with zero or one draw the sum can never exceed 1 — reflecting the fact that $N \geq 2$ always. $\square$

Remark 2.9 (An Alternative Proof Without Geometry). Readers who prefer analysis to geometry can obtain $p_n = 1/n!$ by a one-line induction. Let $F_n(t) = P(U_1 + \dots + U_n \leq t)$ for $t \in [0, 1]$; we claim $F_n(t) = t^n/n!$. Indeed $F_1(t) = t$, and conditioning on the last draw,

$$F_{n+1}(t) = \int_0^t F_n(t-u) du = \int_0^t \frac{(t-u)^n}{n!} du = \frac{t^{n+1}}{(n+1)!}.$$

Setting $t = 1$ gives $p_n = 1/n!$. The geometric proof explains *why* the factorial appears (it counts the orderings of n coordinates); the inductive proof merely confirms that it does.

Remark 2.10 (The Full Distribution). Differencing the tails yields

$$P(N = k) = \frac{1}{(k-1)!} - \frac{1}{k!} = \frac{k-1}{k!}, \quad k \geq 2,$$

so $P(N = 2) = 1/2$, $P(N = 3) = 1/3$, $P(N = 4) = 1/8$, decaying factorially. The answer e is an average dominated by 2's and 3's with a whisper-thin tail.

Why These Two Problems Are Siblings

Records and the uniform-sum problem look unrelated, but both are powered by the identical mechanism: *counting orderings under exchangeability*. For records, the uniform permutation of ranks gives $P(\text{record at } n) = 1/n$ and the harmonic series $H_n \approx \ln n + \gamma$. For the uniform sum, the $n!$ orderings of the cube give $P(N > n) = 1/n!$ and the exponential series $\sum 1/n! = e$. One combinatorial idea, two of the most fundamental constants in mathematics — and in the next section, the same machinery will produce a third: the threshold $1/e$ of the secretary problem, whose optimal strategy is precisely “wait, then accept the first *record*.”

2.6 Records in Trading: Which Clock Are You On?

The i.i.d. theory above applies to *returns*-like quantities. Prices, however, are cumulative sums, and this distinction changes the frequency of records dramatically. The comparison is one of the most instructive calibration exercises in these notes.

Remark 2.11 (The Three Regimes of Record Frequency). Counting new all-time highs over n observations, three regimes appear, distinguished by how the observations are generated.

(i) **I.i.d. observations** (daily returns, or any stationary signal). Records are *rank* events: observation k is a record with probability $1/k$, so

$$E[\#\text{records}] = H_n \approx \ln n.$$

Logarithmically rare. Over forty years of daily data ($n \approx 10^4$), expect roughly ten record daily returns.

(ii) **Driftless random walk** (a price with no trend). Records of the *cumulative sum* are no longer rank events; they are the *ascending ladder points* of the walk, and the times between them are i.i.d. by the strong Markov property. Their common law has the heavy tail $P(T > k) \sim 1/\sqrt{\pi k}$ — a consequence of the Sparre Andersen theory developed later, in the arcsine-law section — so $E[T] = \infty$. A renewal process with infinite mean gap has *zero* asymptotic density, and the count grows sublinearly:

$$E[\#\text{records}] \approx 2\sqrt{n/\pi}.$$

For $n = 2000$ this predicts 50.5; simulation gives 52.8. Compare with $H_{2000} \approx 8.2$ for the i.i.d. case — the cumulative structure produces six times as many records.

(iii) **Random walk with positive drift** (a trending price). The ladder points remain a renewal process, but now the drift makes $E[T]$ *finite*. By the renewal theorem of the size-biasing section, records then occur with a constant asymptotic density $1/E[T]$, and the count grows *linearly*:

$$E[\#\text{records}] \approx \frac{n}{E[T]}.$$

The rate depends on the drift-to-volatility ratio μ/σ and is bounded by 1. For Gaussian steps of unit variance:

drift μ	0	0.05	0.10	0.25	0.50	2.00
records per step	$\rightarrow 0$	0.076	0.134	0.307	0.530	0.976

The limiting behaviour is intuitive at both ends: as $\mu/\sigma \rightarrow \infty$ nearly every step is positive and therefore a record (rate $\rightarrow 1$), while as $\mu \rightarrow 0$ the rate collapses to zero and regime (ii) takes over. The linearity is genuine, not asymptotic hand-waving: at $\mu = 0.25$ the simulated rate is 0.307, 0.305, 0.307, 0.305 for $n = 500, 1000, 2000, 4000$.

The diagnostic. The hierarchy

$$\underbrace{\ln n}_{\text{i.i.d.}} \ll \underbrace{\sqrt{n}}_{\text{driftless sum}} \ll \underbrace{n}_{\text{drifting sum}}$$

means the *empirical frequency of new highs identifies the regime*. Counts consistent with $\ln n$ indicate rank-driven, stationary behaviour; $\sqrt{n}$ -type counts indicate driftless cumulative dynamics; linear counts indicate genuine drift. “The index set another record high this month” is, for a drifting index, not news — it is the baseline, and its *rate* is the only informative part.

Remark 2.12 (Record Bias in Backtests and Headlines). Theorem 2.2 quantifies why superlatives are cheap. Track m independent statistics (strategies, desks, sectors) over n periods. Each accumulates H_n records on average, so by linearity of expectation the expected number of “best-ever” announcements is simply mH_n — with 20 metrics over 250 trading days, roughly $20 \times 6.1 \approx 120$ record-setting events per year *under pure noise*. A “best month in the fund’s history” therefore carries almost no information: some record is essentially always freshly set somewhere.

And mH_n is the *conservative* count, because it applies regime (i) above: it presumes each tracked series is stationary. Equity curves are cumulative sums, living in regime (ii) or (iii), where records arrive at rate $\sqrt{n}$ or faster — an equity curve with genuine drift sets new highs a constant fraction of the time. Measured against that baseline, “the strategy reached an all-time high” is not evidence of anything at all.

The honest question is never *whether* a record occurred, but whether records occur *more often than the null clock for the relevant regime* predicts. This is the record-count analogue of the multiple-testing inflation quantified in the longest-run section.

Remark 2.13 (Drawdown Records). The same logic applies to bad news. A strategy’s “worst drawdown in history” is a record of the drawdown process, and records must keep arriving — ever more slowly, but forever ($H_n \rightarrow \infty$). A new maximum drawdown after a long quiet period is not, by itself, evidence that the strategy is broken; for a stationary strategy it is expected at rate $1/n$. Risk frameworks that trigger reviews on “record drawdown” events should calibrate against this baseline hazard rather than treat every record as an anomaly.

Key Takeaways

Records and the Constant e

1. **Record probability.** For i.i.d. continuous observations, $P(\text{record at } n) = 1/n$ — one line, by exchangeability, distribution-free.
2. **Logarithmic scarcity.** $E[R_n] = H_n \approx \ln n + \gamma$. Squaring the sample adds a constant number of records.
3. **Independence.** The record indicators are mutually independent — records are rank events, and sequential ranks are independent uniforms. Consequence: $(R_n - \ln n)/\sqrt{\ln n} \Rightarrow \mathcal{N}(0, 1)$ for free.
4. **The uniform-sum problem.** $P(U_1 + \dots + U_n \leq 1) = 1/n!$ (simplex volume), hence the number of uniforms needed to exceed 1 satisfies $E[N] = \sum 1/n! = e$.
5. **One mechanism, three constants.** Exchangeability alone produces $\ln n$ (records), e (uniform sums), and — next section — $1/e$ (secretary threshold).
6. **The three clocks of record frequency.** Stationary series: $\ln n$ records. Driftless cumulative sums: $\sim 2\sqrt{n/\pi}$, since the gaps between records have infinite mean. Drifting sums: linear, at the constant rate $1/E[T]$ set by μ/σ . Record frequency diagnoses the regime.
7. **Superlatives are cheap.** Records accumulate at rate H_n per series, so by linearity m tracked series produce mH_n “best-ever” events under pure noise — about 120 per year for 20 metrics over 250 days, all meaningless. For cumulative series (equity curves) the count is far higher still. Judge a record against the null clock of its regime, never against zero.

3 The Secretary Problem and Optimal Stopping with Rankings

3.1 Problem Statement

You are presented with n candidates one at a time, in a random order. After interviewing each candidate, you must either accept (stop) or reject (move on irrevocably). At the moment of decision, you only know the relative rank of the current candidate among those seen so far. Your goal is to maximize the probability of selecting the overall best candidate.

3.2 The Optimal Strategy

The 37% Rule

Theorem 3.1. *The optimal strategy is to reject the first $r^* - 1 = \lfloor n/e \rfloor$ candidates outright (the “observation phase”), then accept the first candidate who is better than all previous ones. This succeeds with probability approaching $1/e \approx 0.3679$ as $n \rightarrow \infty$.*

3.3 Derivation

Proof. Let r be the number of candidates rejected outright ($r \geq 1$). Define $p(r)$ = probability that this strategy selects the best candidate.

Condition on the position k of the best candidate (uniform over $\{1, \dots, n\}$). The strategy succeeds iff:

- (1) $k > r$ (the best appears after the observation phase), AND
- (2) the best candidate among positions $1, \dots, k-1$ appears in positions $1, \dots, r$ (otherwise we would have accepted a non-best candidate first).

Given k , the best of $\{1, \dots, k-1\}$ is uniformly distributed over those $k-1$ positions. So condition (2) has probability $r/(k-1)$. Hence

$$p(r) = \sum_{k=r+1}^n \frac{1}{n} \cdot \frac{r}{k-1} = \frac{r}{n} \sum_{k=r+1}^n \frac{1}{k-1} = \frac{r}{n} \sum_{j=r}^{n-1} \frac{1}{j}.$$

For large n , set $r = xn$ with $x \in (0, 1)$:

$$p(xn) \approx x \int_x^1 \frac{dt}{t} = -x \log x.$$

Maximising over x : $\frac{d}{dx}(-x \log x) = -\log x - 1 = 0$ gives $x = 1/e$. At this value, $-x \log x = 1/e$. $\square$

Why $1/e$ Appears

The $1/e$ result arises from the tension between two competing forces:

- **Observe longer:** by waiting, you learn more about the distribution of candidate qualities.
- **Wait too long:** the best candidate has likely already passed.

The optimal compromise turns out to be “observe 37%, then take the next best.” The $1/e$ probability of success is remarkable: even though candidates appear in random order and we never see future ones, we can identify the best with probability 37%—vastly better than the $1/n$ rate of random selection.

3.4 Variant: Minimizing Expected Rank

The classical problem is unforgiving: choosing the second best scores no better than choosing the worst. A natural refinement replaces this all-or-nothing payoff by the *expected true rank* of the chosen candidate, where rank 1 is best and rank n is worst. The objective is now to *minimize* that expected rank. This change of goal transforms the optimal strategy completely.

Why the classical rule fails here. The $1/e$ -rule is disastrous for expected rank. With probability $\approx 1/e$ the best candidate falls inside the observation window, no later candidate ever beats the recorded maximum, and the rule is forced onto the *last* candidate — of expected rank $(n+1)/2$. The expected rank of the classical rule therefore grows *linearly* in n . An objective that rewards near-misses demands a strategy willing to settle.

3.4.1 The Rank Lemma

The one fact we need converts a candidate’s *observed* relative rank into its *expected true rank*.

Lemma 3.2 (Expected True Rank). *If the candidate at position i has relative rank r among the first i interviewed (rank 1 = best so far), then its expected rank among all n candidates is*

$$\rho(r, i) := E[\text{true rank} \mid \text{relative rank } r \text{ at position } i] = r \frac{n+1}{i+1}.$$

Proof. Each of the remaining $n - i$ candidates is, by exchangeability, equally likely to fall into any of the $i + 1$ gaps created by the i already seen. The current candidate’s true rank is r plus the number of future candidates who outrank it. A future candidate outranks it precisely when it lands in one of the r gaps at or above the current candidate’s position, which happens with probability $r/(i + 1)$. Hence

$$E[\text{true rank} \mid \text{relative rank } r \text{ at position } i] = r + (n - i) \frac{r}{i + 1} = r \cdot \frac{i + 1 + n - i}{i + 1} = r \frac{n + 1}{i + 1}. \quad \square$$

3.4.2 The Value Function

The problem is solved by backward induction on a value function, which we define precisely — it is the object most easily misread.

Definition 3.3 (Value Function). For $1 \leq i \leq n$, let

$V_i :=$ the smallest achievable expected final rank, given that optimal play begins at position i , *before* the candidate at position i has been seen. Equivalently, V_i is the expected true rank of the candidate ultimately selected, assuming every decision from position i onward is optimal.

Remark 3.4 (What V_i Is and Is Not). Three points prevent the standard misreadings:

- V_i is **not the rank of the candidate at position i** . It is an *expected value over the whole remaining game*, a single number attached to the *state* “about to interview candidate i ,” not to any candidate.
- The subscript indexes **where optimal play begins**, not which candidate is meant.
- In particular, V_1 is the **value of the entire game**: the expected rank the optimal strategy secures, starting from scratch. It is the quantity we ultimately want.

The Bellman Recursion

Theorem 3.5. *The value function satisfies the terminal condition $V_n = \frac{n+1}{2}$ (the last candidate must be taken, and its rank is uniform on $\{1, \dots, n\}$) and the backward recursion*

$$V_i = \frac{1}{i} \sum_{r=1}^i \min(\underbrace{\rho(r, i)}_{\text{accept}}, \underbrace{V_{i+1}}_{\text{continue}}),$$

with the optimal rule: at position i , seeing relative rank r ,

$$\text{accept} \iff \rho(r, i) = r \frac{n+1}{i+1} \leq V_{i+1}.$$

Proof. At position i the candidate’s relative rank r is uniform on $\{1, \dots, i\}$ (each of the first i candidates is equally likely to be the current one’s rank), whence the factor $1/i$ and the sum over r . For each r the optimal action compares the expected rank of stopping now, $\rho(r, i)$ (Lemma 3.2), against the expected rank of playing on, V_{i+1} ; the optimum takes the smaller, giving the min. Acceptance is optimal exactly when stopping is the smaller of the two. $\square$

3.4.3 A Worked Example: $n = 3$

With three candidates the recursion can be run entirely by hand, and it already exhibits the qualitative shape of the general solution.

Terminal stage $i = 3$. Forced to take the last candidate: $V_3 = \frac{3+1}{2} = 2$.

Stage $i = 2$. Here $\rho(r, 2) = r \cdot \frac{4}{3}$. The candidate’s relative rank r is 1 or 2, each with probability $\frac{1}{2}$:

$$r = 1 : \rho = \frac{4}{3} \leq V_3 = 2 \Rightarrow \text{accept}; \quad r = 2 : \rho = \frac{8}{3} > 2 \Rightarrow \text{continue}.$$

Therefore

$$V_2 = \frac{1}{2} \min(\frac{4}{3}, 2) + \frac{1}{2} \min(\frac{8}{3}, 2) = \frac{1}{2} \cdot \frac{4}{3} + \frac{1}{2} \cdot 2 = \frac{5}{3}.$$

Stage $i = 1$. The first candidate is trivially a relative best ($r = 1$), with $\rho(1, 1) = 1 \cdot \frac{4}{2} = 2$. Compare to $V_2 = \frac{5}{3}$:

$$\rho(1, 1) = 2 > \frac{5}{3} = V_2 \Rightarrow \text{continue}.$$

Hence

$$V_1 = \min(2, \frac{5}{3}) = \frac{5}{3}.$$

The optimal policy and its value. Never accept the first candidate; at position 2 accept only a relative best; at position 3 take whatever remains. The optimal expected rank is

$$V_1 = \frac{5}{3} \approx 1.667.$$

A direct check over all $3! = 6$ orderings confirms this: the policy yields final ranks averaging exactly $5/3$. For comparison, always taking the first candidate gives expected rank $\frac{1+2+3}{3} = 2$, and always waiting for the last also gives 2; the optimal rule beats both.

3.4.4 The General Solution

Running the recursion for growing n shows the optimal policy is *not* a single time threshold. Instead, the acceptance standard *relaxes in stages*: reject entirely at first; then accept relative rank 1; later also rank 2; then rank 3; and so on, as the shrinking horizon makes holding out for the best too risky. For $n = 100$, optimal play rejects through position ≈ 27 , accepts only a relative best through ≈ 47 , admits relative rank 2 from ≈ 48 , rank 3 from ≈ 60 , and loosens further toward the end.

Remark 3.6 (The Chow–Moriguti–Robbins–Samuels Constant). The optimal expected rank V_1 increases with n — $V_1 = \frac{5}{3}$ at $n = 3$, then 3.41 at $n = 50$, 3.72 at $n = 200$, 3.83 at $n = 1000$ — converging to

$$\lim_{n \rightarrow \infty} V_1 = 3.8695 \dots$$

(Chow, Moriguti, Robbins and Samuels, 1964). The contrast with the classical problem is the memorable part: chasing *the best* succeeds only 37% of the time, yet minimizing *expected rank* secures, on average, roughly the fourth-best candidate out of arbitrarily many — a constant, independent of n . Relaxing the objective from “the best or nothing” to “as good as possible on average” converts a coin-flip gamble into a near-certainty of an excellent outcome.

3.5 Variant: Selecting Any of the Top k

Suppose success means choosing *any* of the top k candidates, not necessarily the single best. One might guess the optimal rule is the classical one with a shorter observation phase; the truth has more structure, and getting it right requires being careful about what the cutoffs mean.

Proposition 3.7 (Top- k Secretary; Gusein-Zade 1966). *The optimal strategy is a ladder of k cutoffs*

$$c_1 < c_2 < \dots < c_k,$$

*defining $k + 1$ phases. Through position c_1 the rule observes and **rejects everything**, including a relative best. For positions in $(c_j, c_{j+1}]$ it accepts any candidate whose relative rank is at most j . Thus the acceptance standard loosens by one relative rank at each cutoff: only relative-best after c_1 , best-or-second after c_2 , and so on, until after c_k any of the top k relative ranks is accepted.*

For $k = 2$, the asymptotic cutoffs are

$$c_1 \approx 0.347n, \quad c_2 = \frac{2}{3}n,$$

and the success probability converges to ≈ 0.574 . As k grows the first cutoff shrinks and the success probability tends to 1.

Remark 3.8 (Reading the Cutoffs Correctly). The cutoff c_1 is the **end of pure observation**, not a position where anything is accepted: through c_1 the rule rejects every candidate, *including a relative best*, exactly as the classical $1/e$ -rule rejects its first n/e candidates. Acceptance of a relative best begins only *after* c_1 . For $k = 2$ this means: reject through $0.347n$; between $0.347n$ and $\frac{2}{3}n$ accept only a relative best; after $\frac{2}{3}n$ accept a relative best *or* second.

Why Exactly k Cutoffs, and Why a Ladder

Two facts pin down the structure.

Only relative ranks $1, \dots, k$ are ever acceptable. A candidate of relative rank r (already worse than $r - 1$ of those seen) can belong to the true top k only if $r \leq k$: if it is already worse than k earlier candidates, it cannot be among the k best overall. So relative ranks $r > k$ are *never* accepted — which is why the ladder has k rungs and not more.

Each acceptable rank has its own optimal activation time. The evidence that a given relative rank provides about true top- k membership *strengthens as the horizon shrinks*. A relative best is the strongest such evidence and becomes worth accepting first, at c_1 . A relative second is weaker evidence — early on, too many candidates remain for it to be a safe bet — so it becomes acceptable only later, at c_2 , once the risk of ending empty-handed outweighs the loss of precision. Each further rank j is admitted at its own c_j , later still. This is why the cutoffs are *ordered* ($c_1 < \dots < c_k$) and why the standard relaxes *one rank at a time*: neither $2k$ cutoffs (there are only k acceptable ranks to activate) nor a single cutoff (each rank has a distinct optimal moment) would be optimal. A single-threshold rule that accepts the relative top 2 from position $0.347n$ onward achieves only ≈ 0.45 , well short of the 0.574 of the two-phase ladder — the phases are not a refinement but a necessity.

The special case $k = 1$ recovers the classical problem: one cutoff $c_1 = n/e$, accept the first relative best thereafter, success $1/e$.

Remark 3.9 (Why the Values Are Quoted, Not Derived). The *structure* above — a monotone ladder of k cutoffs, each admitting one more relative rank — follows from backward induction on the value function, exactly as in the expected-rank variant: one shows the acceptance condition is monotone in relative rank (accepting rank r forces accepting all ranks $< r$) and that the activation times are time-ordered. The *numerical* cutoffs ($c_1 \approx 0.347n$, $c_2 = \frac{2}{3}n$) are another matter. They arise from equating the value of accepting against the value of continuing at each boundary and passing to the continuous limit $n \rightarrow \infty$; the resulting boundary equations are transcendental, without the clean closed form of the classical $1/e$. We therefore quote Gusein-Zade’s solution rather than reproduce its derivation, which belongs to the research literature (Gusein-Zade, 1966) and would sit above the level of these notes. The point worth carrying away is not the digits 0.347 but the shape: *optimal top- k selection is a staircase of steadily relaxing standards, one step per admissible rank*.

3.6 Connection to the Sample-Buying Problem

Remark 3.10 (When You See Values, the Problem Collapses to a Threshold). The secretary problem is hard partly because of an informational handicap: only *relative ranks* are observed, so no rule of the form “stop when the value exceeds t ” is even expressible. Remove the handicap and optimal stopping becomes almost trivial.

The problem. Draw i.i.d. $U_1, U_2, \dots \sim \text{Unif}(0, 1)$, paying a cost c for each draw, and stop whenever you like, keeping the best value seen. If you stop after N draws, your payoff is

$$\max(U_1, \dots, U_N) - cN,$$

the best value obtained minus the total cost paid. The objective is to maximise its expectation. (Assume $c < 1/2$, else even the first draw is not worth its cost.)

The one-step (myopic) rule. Suppose your current best is x , and consider buying *one* more draw U . It costs c , and it improves your best by $(U - x)^+ = \max(U - x, 0)$: you gain $U - x$ if

the new draw beats x , and nothing otherwise. The expected gain is

$$E[(U - x)^+] = \int_x^1 (u - x) du = \frac{(1 - x)^2}{2},$$

the integral running only over $[x, 1]$ because the integrand vanishes elsewhere. Comparing marginal benefit against marginal cost — note that x itself is *not* at stake, since drawing again never destroys the best already secured — the rule is

$$\text{draw again} \iff \frac{(1 - x)^2}{2} > c \iff x < 1 - \sqrt{2c}.$$

Hence the optimal policy is the fixed threshold

$t^* = 1 - \sqrt{2c} : \quad \text{stop as soon as the best value reaches } t^*.$

Why myopic is exactly optimal. A one-step rule is generally only a heuristic: it could be worth accepting a locally bad step for a longer-run gain. Not here. The state x (the running best) is *non-decreasing*, so once $x \geq t^*$ it remains so forever: the stopping region is *absorbing*. For such *monotone* problems the one-step-lookahead rule coincides with the full Bellman solution, so no multi-step reasoning can improve on it.

The value of the game. Under the optimal policy the number of draws is geometric with success probability $1 - t^* = \sqrt{2c}$, so $E[N] = 1/\sqrt{2c}$, and the accepted draw is uniform on $(t^*, 1)$ with mean $(1 + t^*)/2$. Therefore

$$E[\text{payoff}] = \frac{1 + t^*}{2} - \frac{c}{\sqrt{2c}} = 1 - \frac{\sqrt{2c}}{2} - \frac{\sqrt{2c}}{2} = 1 - \sqrt{2c} = t^*.$$

The value of the game equals the threshold itself — a tidy self-consistency: the price at which you are willing to stop is exactly what the opportunity is worth. For $c = 0.02$: $t^* = 0.8$ and the expected payoff is 0.8.

The two problems bracket the informational spectrum of optimal stopping: with ranks only, the solution is a threshold in *time* ($1/e$ of the horizon); with full values, a threshold in *value* ($1 - \sqrt{2c}$). Real trading problems — when to hit a bid, when to stop polling quotes — live between the two, and diagnosing which information you actually observe is the first step in choosing the right stopping framework.

Secretary Problem Summary

1. **The 37% rule:** observe and reject the first n/e candidates, then accept the first one better than all previously seen. Success probability $1/e$.
2. **Why $1/e$:** the optimal trade-off between learning (observing more) and acting (before the best passes) lies at exactly $1/e$.
3. **Sensitivity:** accepting the first candidate ($r = 0$) or being forced onto the last ($r = n - 1$) each yields success probability $1/n$; the optimum lies strictly between.
4. **Generalizations:** top- k acceptance (phased cutoffs), expected-rank minimization (the CMRS constant 3.8695), and the full-information variant (values, not just ranks).
5. **Optimal stopping is the unifying frame:** the secretary problem is a finite-horizon stopping problem solved by backward induction, only comparing the payoff of stopping now against the continuation value (the Bellman equation). The same machinery prices American options; the difference is informational. The secretary problem is *ordinal* (only ranks observed $\Rightarrow$ threshold in *time*, $\approx n/e$); American exercise is *cardinal* (payoff $(S_t - K)^+$ a function of price $\Rightarrow$ threshold in *price*, the early-exercise boundary). The secretary problem is optimal stopping stripped of all cardinal information, it is the timing decision in its purest form.

4 Urn Models and the Pólya Urn

4.1 The Classical Pólya Urn Scheme

Definition 4.1 (Pólya Urn). An urn initially contains a red balls and b black balls. At each step:

- (1) Draw a ball uniformly at random from the urn.
- (2) Return it, together with c additional balls of the same colour.

Write R_n and L_n for the numbers of red and black balls present after n draws (so $R_0 = a$, $L_0 = b$), and let

$$X_n = \frac{R_n}{R_n + L_n}$$

be the *urn composition* after n draws. Throughout this section $\mathcal{F}_n$ denotes the information generated by the first n draws.

The reinforcement c is taken to be a positive real parameter (integer values are the classical case; for non-integer c one reads the counts as weights). The degenerate and negative choices $c = 0$ and $c = -1$ are genuinely different schemes and are treated separately in Section 4.6.

Alongside the composition it will be useful to track the *drawn proportion*

$$\bar{X}_n = \frac{\#\{\text{red balls drawn in the first } n \text{ draws}\}}{n},$$

a distinct quantity that happens to share the same limit when $c > 0$, but not otherwise.

4.2 The Martingale Structure

Theorem 4.2 (The Composition Is a Martingale). $(X_n)_{n \geq 0}$ is a martingale with respect to $(\mathcal{F}_n)$.

Proof. Suppose $X_n = x$, so $R_n = x(R_n + L_n)$. A red is drawn with probability x and a black with probability $1 - x$; in either case the total becomes $R_n + L_n + c$. Hence

$$\begin{aligned} E[X_{n+1} \mid \mathcal{F}_n] &= x \cdot \frac{R_n + c}{R_n + L_n + c} + (1 - x) \cdot \frac{R_n}{R_n + L_n + c} \\ &= \frac{x(R_n + c) + (1 - x)R_n}{R_n + L_n + c} = \frac{R_n + xc}{R_n + L_n + c}. \end{aligned}$$

Substituting $R_n = x(R_n + L_n)$,

$$E[X_{n+1} \mid \mathcal{F}_n] = \frac{x(R_n + L_n) + xc}{R_n + L_n + c} = \frac{x(R_n + L_n + c)}{R_n + L_n + c} = x = X_n. \quad \square$$

Remark 4.3 (What This Does and Does Not Give). Since $X_n \in [0, 1]$, the martingale is bounded, hence uniformly integrable, and the Martingale Convergence Theorem yields $X_n \rightarrow X_\infty$ almost surely and in L^1 . This settles *that* a limit exists and that $E[X_\infty] = X_0 = a/(a + b)$, but it says nothing whatever about the law of X_∞ . Identifying that law requires a second, independent idea.

4.3 Exchangeability of the Draw Sequence

Let $\xi_1, \xi_2, \dots$ record the colours drawn, with $\xi_i = 1$ for red and $\xi_i = 0$ for black.

Proposition 4.4 (Exchangeability). *The sequence (ξ_i) is exchangeable: the probability of any finite colour pattern depends only on how many reds and blacks it contains, not on their order.*

Proof. A pattern with k reds and m blacks in a specified order has probability

$$\frac{\prod_{i=0}^{k-1} (a + ic) \prod_{j=0}^{m-1} (b + jc)}{\prod_{t=0}^{k+m-1} (a + b + tc)},$$

because at each step the denominator is the current total (which grows by c regardless of colour), while the numerator contributes $a + ic$ at the $(i + 1)$ -st red and $b + jc$ at the $(j + 1)$ -st black. Reordering the pattern permutes the factors of the numerator without changing them, and leaves the denominator untouched. $\square$

de Finetti's Representation

Corollary 4.5. *By de Finetti's theorem there exists a random variable $\Theta \in [0, 1]$ such that, conditionally on $\Theta = \theta$, the draws (ξ_i) are i.i.d. Bernoulli(θ). Moreover $\Theta = X_\infty$: applying the law of large numbers conditionally, the drawn proportion $\bar{X}_n \rightarrow \Theta$ almost surely, and $\bar{X}_n$ and X_n share the same limit when $c > 0$.*

The corollary supplies the bridge we need. For any k , conditioning on Θ and using conditional independence,

$$P(\xi_1 = \dots = \xi_k = 1) = E[\Theta^k] = E[X_\infty^k]. \quad (\dagger)$$

A question about an asymptotic random variable has become a question about a finite-sample probability — and that one is a two-line computation.

4.4 The Limiting Distribution

Limit of the Pólya Urn

Theorem 4.6. *As $n \rightarrow \infty$,*

$$X_n \longrightarrow X_\infty \sim \text{Beta}\left(\frac{a}{c}, \frac{b}{c}\right) \quad \text{almost surely.}$$

Proof. Existence of the limit is Theorem 4.2 together with the Martingale Convergence Theorem. It remains to identify the law, which we do by matching moments.

(a) *The probability of k consecutive reds.* Before the first draw the urn holds a red out of $a + b$; after j red draws it holds $a + jc$ out of $a + b + jc$. Multiplying the conditional probabilities,

$$P(\text{first } k \text{ draws all red}) = \prod_{j=0}^{k-1} \frac{a + jc}{a + b + jc}. \quad (*)$$

(b) *The moments of a Beta law.* For $Y \sim \text{Beta}(\alpha, \beta)$, repeated use of $\Gamma(z + 1) = z \Gamma(z)$ gives

$$E[Y^k] = \frac{\Gamma(\alpha + k) \Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\alpha + \beta + k)} = \prod_{j=0}^{k-1} \frac{\alpha + j}{\alpha + \beta + j}. \quad (**)$$

(c) *The match.* Dividing numerator and denominator of each factor of (*) by c ,

$$\prod_{j=0}^{k-1} \frac{a + jc}{a + b + jc} = \prod_{j=0}^{k-1} \frac{\frac{a}{c} + j}{\frac{a}{c} + \frac{b}{c} + j},$$

which by (**) is the k -th moment of $\text{Beta}(a/c, b/c)$. Combining with the bridge (†) of Section 4.3,

$$E[X_\infty^k] = E[Y^k] \quad \text{for every } k \geq 1, \quad Y \sim \text{Beta}\left(\frac{a}{c}, \frac{b}{c}\right).$$

(d) *Moments determine the law.* Both variables are supported on the compact interval $[0, 1]$, and a distribution on a compact interval is determined by its moment sequence (polynomials are dense in $C[0, 1]$ by Weierstrass). Hence $X_\infty \sim \text{Beta}(a/c, b/c)$. $\square$

Remark 4.7 (What Each Tool Contributed). The roles are worth separating. *Martingale convergence* supplied existence, and nothing more. *Exchangeability and de Finetti* supplied the bridge (†), converting a statement about a limiting random variable into one about finite-sample probabilities. The identification itself was then an elementary product computation — no measure theory, just the conditional probabilities of the urn. This is a recurring pattern: heavy theorems establish that an object exists and is well behaved, while its identity falls out of a concrete calculation.

4.5 Interpretation: The Reinforcement as a Dial

Where the Parameters a/c and b/c Come From

The division by c in step (c) is the whole content of the theorem, and it reads cleanly: *the Beta parameters count the initial composition in units of the reinforcement*. Starting with a red balls and adding c at a time means beginning with a/c “reinforcement units” of red. Absolute ball counts are irrelevant; only their ratio to the step size matters.

The Beta moments then quantify the urn’s behaviour. The mean recovers the martingale property,

$$E[X_\infty] = \frac{a/c}{a/c + b/c} = \frac{a}{a + b}$$

— reinforcement does not bias the urn on average, it only spreads the outcomes — while the variance, rewritten in the original parameters, makes the role of c explicit:

$$\text{Var}(X_\infty) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{abc}{(a + b)^2(a + b + c)}, \quad \alpha = \frac{a}{c}, \beta = \frac{b}{c}.$$

Recalling that $\text{Beta}(\alpha, \beta)$ concentrates for large parameters and pushes mass to the end-points for parameters below 1, three regimes emerge, each with its numerical signature.

- **c small relative to a, b** (gentle reinforcement): the parameters $a/c, b/c$ are large, the Beta concentrates near $a/(a + b)$, and the variance vanishes *linearly* in c . The long-run composition is nearly deterministic; early draws barely matter.
- **c large** (aggressive reinforcement): the parameters are small, the Beta becomes U-shaped, and the variance saturates at $ab/(a + b)^2$, which for $a = b$ is $1/4$ — the variance of a Bernoulli($1/2$). The reading is literal: with overwhelming reinforcement *the first draw decides the urn*, and X_∞ degenerates to a coin flip between 0 and 1. (Simulation with $a = b = 10, c = 500$: variance 0.241 against the predicted 0.240, with 96% of paths ending below 0.10 or above 0.90.)
- **$a = b = c$** : the parameters are both 1, so $X_\infty \sim \text{Uniform}(0, 1)$ — every limiting proportion equally likely. The variance formula agrees: $\frac{1}{4 \cdot 3} = \frac{1}{12}$. This is the classical and most striking special case.

Path dependence is therefore not a fixed feature of the Pólya urn but a dial: as c grows from small to large, the limit interpolates between a point mass at the initial proportion and a coin flip between the two extremes.

Remark 4.8 (The Urn Performs Bayesian Updating). The scheme is not merely analogous to Beta–Binomial conjugacy; it executes it. Take the prior $\text{Beta}(\alpha, \beta)$ with $\alpha = a/c, \beta = b/c$ — which by Corollary 4.5 is exactly the de Finetti mixing measure of the draw sequence. After observing k reds and m blacks, Bayesian updating gives the posterior $\text{Beta}(\alpha + k, \beta + m)$, and the urn’s composition at that moment is $(a + kc, b + mc)$ — the same parameters, measured in units of c . The two predictive probabilities coincide identically:

$$\underbrace{\frac{a + kc}{a + b + (k + m)c}}_{\text{urn draw probability}} = \underbrace{\frac{\alpha + k}{\alpha + \beta + k + m}}_{\text{Beta posterior predictive}}.$$

Adding c balls of the drawn colour is incrementing a Beta parameter by one. The urn is a physical realisation of conjugate updating, with reinforcement playing the role of evidence.

Remark 4.9 (Reinforcement Processes in the Wild). The Pólya urn is the simplest *reinforcement process*: outcomes that occur become more likely to recur. Models sharing this structure include

- opinion dynamics and social influence, where early adopters shape eventual consensus;
- preferential attachment in network growth, where well-connected nodes attract further connections;
- price formation, where trades that move the price attract attention and further trades.

In each case the qualitative signature is the same: the long-run state is random rather than determined, and early history is never washed out.

4.6 The Sign of c : Reinforcement, Independence, Exhaustion

The rule “draw, return, add c of the same colour” remains meaningful for $c = 0$ and $c = -1$, and the resulting family unifies three classical sampling schemes.

A caution about which proportion is being tracked. For $c > 0$ the composition X_n and the drawn proportion $\bar{X}_n$ converge to the same limit (Corollary 4.5), so the distinction is immaterial. For $c \leq 0$ it is essential, and the *drawn* proportion is the meaningful object:

- If $c = 0$ the composition never changes at all: $X_n \equiv a/(a+b)$ identically. Nothing converges because nothing moves; what obeys the law of large numbers is $\bar{X}_n$.
- If $c = -1$ the drawn ball is discarded, so the urn empties after $a + b$ draws and X_n becomes undefined ($0/0$) at the end — indeed just before the end a single ball remains and $X_n \in \{0, 1\}$. The quantity that is pinned is again $\bar{X}_n$: once every ball has been drawn, $\bar{X}_{a+b} = a/(a+b)$ *exactly*.

c	Scheme	Limit of the drawn proportion $\bar{X}_n$
$c > 0$	reinforcement (rich get richer)	random: Beta(a/c , b/c)
$c = 0$	sampling <i>with</i> replacement	deterministic: $a/(a+b)$, by the LLN
$c = -1$	sampling <i>without</i> replacement	deterministic and <i>forced</i> : exactly $a/(a+b)$

The three regimes differ in how strongly the past constrains the future. Positive c is *self-reinforcing*: early imbalances amplify and the limit stays random forever. Zero c is *memoryless*. Negative c is *self-correcting*: drawing a red makes red less likely next time, and the path is dragged back toward balance. For $c < 0$ the interesting question is therefore not what the final proportion is — it is fixed — but how the path *fluctuates* on the way there. That is where the Brownian bridge enters.

4.6.1 The Brownian Bridge

Definition 4.10 (Brownian Bridge). A *standard Brownian bridge* on $[0, 1]$ is the Gaussian process $(\mathcal{B}_t)_{t \in [0, 1]}$ with $\mathcal{B}_0 = \mathcal{B}_1 = 0$, $E[\mathcal{B}_t] = 0$, and

$$\text{Cov}(\mathcal{B}_s, \mathcal{B}_t) = \min(s, t) - st, \quad \text{so} \quad \text{Var}(\mathcal{B}_t) = t(1 - t).$$

Equivalently $\mathcal{B}_t = W_t - tW_1$ for a standard Brownian motion W : it is Brownian motion *conditioned to return to zero at time 1*.

A Path Pinned at Both Ends

Ordinary Brownian motion starts at 0 and wanders freely, its variance t growing without bound. A bridge is the same motion subject to the constraint of being back at 0 when the clock runs out, and the constraint has two visible consequences.

The variance is a parabola. $\text{Var}(\mathcal{B}_t) = t(1-t)$ vanishes at both endpoints and peaks at $t = 1/2$ with value $1/4$: uncertainty is greatest in the middle, far from both anchors, and is squeezed to zero at the ends.

The path is mean-reverting. Being high at time t leaves less room to return, so the bridge is pulled back toward zero with a drift $-\mathcal{B}_t/(1-t)$ that strengthens as the deadline approaches. A bridge does not wander; it is tethered.

4.6.2 The Card Deck as a Discrete Bridge

Take $c = -1$ with $a = b = N/2$: a deck of $N = 52$ cards, 26 red and 26 black, drawn without replacement. Track the running imbalance

$$S_k = (\text{reds drawn in the first } k) - (\text{blacks drawn in the first } k), \quad k = 0, 1, \dots, N,$$

so that $S_0 = 0$ and, crucially, $S_N = 0$: once the deck is exhausted the counts must balance. The path is pinned at both ends by construction.

Proposition 4.11 (Variance of the Imbalance). *For sampling without replacement from $N/2$ reds and $N/2$ blacks,*

$$E[S_k] = 0, \quad \text{Var}(S_k) = \frac{k(N-k)}{N-1}.$$

The factor $(N-k)$ is the finite-population correction: it encodes the exhaustion constraint and is exactly what drives the variance back to zero at $k = N$. For sampling *with* replacement the variance would be simply k , growing without bound.

Convergence to the Bridge

Theorem 4.12. *As the deck size $N \rightarrow \infty$, the rescaled imbalance process*

$$t \mapsto \frac{S_{\lfloor Nt \rfloor}}{\sqrt{N}}, \quad t \in [0, 1],$$

converges in distribution to a standard Brownian bridge.

The variance computation makes this plausible in one line: setting $k = Nt$ in Proposition 4.11,

$$\text{Var}\left(\frac{S_{Nt}}{\sqrt{N}}\right) = \frac{1}{N} \cdot \frac{Nt(N-Nt)}{N-1} \rightarrow t(1-t),$$

precisely the bridge variance of Definition 4.10. A simulation with $N = 52$ already gives 0.256 at $t = 1/2$ against the limiting 0.250, and the covariance matches too: 0.091 at $(s, t) = (0.3, 0.7)$ against $\min(s, t) - st = 0.090$.

Remark 4.13 (Why This Matters for the Card Problem). The red–black stopping problem asks when to halt a sequence of draws from a shuffled deck — a question about the *path* of S_k , not its endpoint. Theorem 4.12 says that path is, to good approximation, a Brownian bridge, so questions about maxima, crossings and hitting times for the deck can be answered with the corresponding bridge formulas, the pinning $\mathcal{B}_0 = \mathcal{B}_1 = 0$ standing in for the exhaustion constraint

$S_0 = S_N = 0$. Concretely, the mean-reverting drift $-\mathcal{B}_t/(1-t)$ is the continuous expression of a fact every card player knows: an excess of reds drawn early guarantees a compensating excess of blacks remaining, and the compensation grows more forceful as the deck thins.

Pólya Urn Key Facts

1. **Reinforcement scheme:** draw a ball, return it, add c more of the same colour. Success breeds success.
2. **Martingale structure:** the composition $X_n = R_n/(R_n + L_n)$ is a bounded martingale, hence converges a.s. and in L^1 with $E[X_\infty] = a/(a+b)$. This gives existence of a limit but nothing about its law.
3. **Exchangeability:** the draw sequence is exchangeable, so by de Finetti it is a mixture of i.i.d. Bernoulli sequences, and the mixing variable *is* the limit X_∞ . This converts $E[X_\infty^k]$ into $P(k \text{ reds in a row})$.
4. **Limit distribution:** $X_\infty \sim \text{Beta}(a/c, b/c)$, identified by matching moments: $P(\text{first } k \text{ red}) = \prod_{j < k} \frac{a+jc}{a+b+jc}$ is exactly the k -th Beta moment. The parameters *count the initial composition in units of the reinforcement*.
5. **Path dependence is a dial, not a switch:** early draws are amplified by all later reinforcement, so — unlike i.i.d. trials, where the LLN erases every finite history — the randomness is never washed out. How much survives is set by c :

$$\text{Var}(X_\infty) = \frac{abc}{(a+b)^2(a+b+c)},$$

vanishing linearly as c becomes small and saturating at $ab/(a+b)^2$ for large c , where the first draw alone decides the urn.

6. **The sign of c selects the scheme,** with the *drawn* proportion as the common object of study: $c > 0$ is reinforcement (random limit, Beta); $c = 0$ is sampling with replacement (composition frozen, drawn proportion deterministic by the LLN); $c = -1$ is sampling without replacement (self-correcting, drawn proportion forced to $a/(a+b)$, fluctuations converging to a Brownian bridge).
7. **The urn performs Bayesian updating:** with prior $\text{Beta}(a/c, b/c)$, observing k reds and m blacks yields posterior $\text{Beta}(a/c + k, b/c + m)$, and the urn's draw probability equals the posterior predictive identically. Adding c balls of the drawn colour *is* incrementing a Beta parameter by one, with the de Finetti mixing measure of item 3 playing the role of the prior.

5 The Egg Drop Problem

5.1 Problem Statement

You are given k identical eggs and a building with F floors. There exists a *critical floor* $f^* \in \{0, 1, \dots, F\}$ such that an egg breaks if and only if it is dropped from a floor strictly above f^* .

You may drop eggs sequentially, observing the outcome before choosing the next floor. A broken egg cannot be reused. The goal is to **identify f^* with certainty** using the **fewest drops in the worst case**.

Definition 5.1. Let $f(n, k)$ denote the maximum number of floors that can be *resolved* (i.e., f^* determined with certainty) using at most n drops and k eggs.

The answer to the original problem is the smallest n such that $f(n, k) \geq F$.

5.2 The Recursion

Lemma 5.2 (Fundamental Recursion). *For all $n \geq 1$ and $k \geq 1$,*

$$f(n, k) = f(n-1, k-1) + 1 + f(n-1, k),$$

with base cases $f(0, k) = 0$ for all $k \geq 0$, and $f(n, 0) = 0$ for all $n \geq 0$.

Proof. With n drops and k eggs, choose some floor m and drop an egg. Two outcomes:

- (i) **Egg breaks.** Now $f^* \in \{0, \dots, m-1\}$, and we have $n-1$ drops, $k-1$ eggs to resolve $m-1$ floors. Maximum: $m-1 = f(n-1, k-1)$.
- (ii) **Egg survives.** Now $f^* \in \{m, \dots, F\}$, and we have $n-1$ drops, k eggs to resolve floors above. Maximum: $f(n-1, k)$.

Setting $m = f(n-1, k-1) + 1$ balances both branches: $f(n, k) = f(n-1, k-1) + 1 + f(n-1, k)$. $\square$

5.3 The Closed-Form Solution

Main Theorem

Theorem 5.3. *For all $n \geq 0$ and $k \geq 1$,*

$$f(n, k) = \sum_{i=1}^k \binom{n}{i}$$

Proof. By strong induction on n .

Base case. $n = 0$: $f(0, k) = 0 = \sum_{i=1}^k \binom{0}{i}$.

Inductive step. By Lemma 5.2,

$$f(n, k) = \sum_{i=1}^{k-1} \binom{n-1}{i} + \sum_{i=1}^k \binom{n-1}{i} + 1.$$

Re-index the first sum: $\sum_{i=1}^{k-1} \binom{n-1}{i} = \sum_{j=2}^k \binom{n-1}{j-1}$.

Isolate the $i = 1$ term from the second sum: $\sum_{i=1}^k \binom{n-1}{i} = \binom{n-1}{1} + \sum_{i=2}^k \binom{n-1}{i}$.

Combine:

$$\begin{aligned} f(n, k) &= \sum_{i=2}^k \left[\binom{n-1}{i-1} + \binom{n-1}{i} \right] + (n-1) + 1 \\ &= \sum_{i=2}^k \binom{n}{i} + n = \sum_{i=1}^k \binom{n}{i}, \end{aligned}$$

using Pascal's identity. $\square$ $\square$

5.4 Application: Two Eggs and 100 Floors

Corollary 5.4. *With $k = 2$ eggs and $F = 100$ floors, the minimum number of drops in the worst case is $n = 14$.*

Proof. $f(n, 2) = n + n(n-1)/2 = n(n+1)/2$. Need $n(n+1)/2 \geq 100$: $n = 13$ gives 91, $n = 14$ gives 105. $\square$

Example 5.5 (Optimal Strategy for $n = 14$, $k = 2$). The first egg is dropped from floors whose spacing decreases by one at each step:

Attempt	Floor	Block size
1	14	14
2	27	13
3	39	12
4	50	11
5	60	10
6	69	9
7	77	8
8	84	7
9	90	6
10	95	5
11	99	4
12	100	1

If the first egg breaks on attempt j , we have used j drops with $14 - j$ remaining. The block below has $14 - j$ floors, which the second egg searches linearly. Total: $j + (14 - j) = 14$ in every case.

Why Binomial Coefficients?

Each sequence of n drops produces a binary string of outcomes: **B**reak or **S**urvive. With k eggs, at most k of the n outcomes can be **B**.

The number of distinct binary strings of length n with at most k ones is $\sum_{i=0}^k \binom{n}{i}$. Subtracting the single all-**S** string (which provides no boundary information) gives $\sum_{i=1}^k \binom{n}{i}$ distinguishable outcomes—one per identifiable critical floor.

The recursion mirrors Pascal's identity because each drop is a binary decision, just as each entry in Pascal's triangle is the sum of two entries above.

Egg Drop Essential Points

1. **Start with $k = 2$.** Decreasing-block strategy gives $n(n+1)/2 \geq F$, i.e., $n = 14$ for $F = 100$.
2. **General formula.** $f(n, k) = \sum_{i=1}^k \binom{n}{i}$. Reduces to $n(n+1)/2$ for $k = 2$.
3. **One-line intuition.** Each drop is binary; with k eggs we can have at most k breaks in n drops, giving $\sum \binom{n}{i}$ distinguishable outcome strings.
4. **Connection to Pascal's triangle.** The recursion reduces to Pascal's identity after re-indexing.

6 The Red–Black Card Stopping Problem

6.1 Problem Statement

A standard deck contains 26 red cards and 26 black cards. Cards are drawn sequentially without replacement. Each red card scores +1, each black card scores −1. The current score after k draws is

$$S_k = (\text{red drawn so far}) - (\text{black drawn so far}).$$

At any point you may *stop* and collect S_k as your final score. If you draw all 52 cards, $S_{52} = 0$.

Goal. Determine the stopping rule τ^* that maximises $E[S_\tau]$ and compute the optimal expected value $V^* = E[S_{\tau^*}]$.

6.2 Why This Is Not a Martingale Problem

Remark 6.1 (With Replacement: No Strategy Can Profit). Suppose first that each card is drawn *with* replacement, so every draw is red or black with probability $1/2$ independently of the past. Then (S_k) is a genuine martingale, and for any stopping time $\tau \leq 52$ the optional stopping theorem gives

$$E[S_\tau] = E[S_0] = 0.$$

No stopping rule — however clever, however adaptive — can extract positive expected value from a fair coin. If the card problem is to be interesting at all, the absence of replacement must break this argument somewhere.

Remark 6.2 (Without Replacement: The Martingale Breaks). It breaks at the very first step: (S_k) is *not* a martingale. Suppose after k draws the running score is S_k , and let r and b be the reds and blacks remaining in the deck, with $r + b = N - k$ (here $N = 52$). Since each colour started with $N/2$ copies,

$$r - b = \left(\frac{N}{2} - \#\text{reds seen}\right) - \left(\frac{N}{2} - \#\text{blacks seen}\right) = -S_k :$$

the deck holds the mirror image of the score. The next card is red with probability $r/(N - k)$ — not $1/2$ unless $S_k = 0$ — and

$$E[S_{k+1} \mid \mathcal{F}_k] = S_k + \frac{r - b}{N - k} = \boxed{S_k \left(1 - \frac{1}{N - k}\right)}.$$

The process is *mean-reverting*: a supermartingale whenever $S_k > 0$, a submartingale whenever $S_k < 0$, with a restoring pull $-S_k/(N - k)$ that strengthens as the deck thins — the discrete counterpart of the Brownian bridge drift $-\mathcal{B}_t/(1 - t)$ derived in the Pólya chapter (set $t = k/N$). A quick numerical check: conditional on $S_{10} = +4$, simulation gives $E[S_{11}] \approx 3.90$, matching $4(1 - \frac{1}{42}) \approx 3.905$ and refuting the martingale value 4.

Where the Value Comes From

Mean reversion cuts in both directions, and the two directions are not symmetric for a player who chooses when to stop.

When ahead ($S_k > 0$): the surplus of reds already drawn means the remaining deck is black-heavy, and the restoring drift erodes your lead. Waiting is costly — indeed waiting to the end is worthless, since $S_N = 0$ deterministically. This pushes you to **stop early when winning**.

When behind ($S_k < 0$): the same mechanism works for you. The deck now owes you reds: the remaining cards are red-heavy, and the drift pulls your deficit back toward zero. You are never forced to realise a loss — by waiting, in the worst case, until $S_N = 0$. This makes it optimal to **keep drawing when losing**.

Stop your gains before they revert; ride your losses back to par at no cost. The combination — an option exercised only on favourable paths, with a guaranteed floor on unfavourable ones — is why the game has strictly positive value: the backward induction of the next subsection yields $E[S_\tau^*] \approx 2.62$ for the full deck. Contrast this with replacement, where a losing streak promises nothing: there, past draws carry no information about future ones, and the value is zero. *Without replacement, the deck remembers — and its memory is an asset to whoever holds the timing option.*

Remark 6.3 (Consequence: Dynamic Programming, Not Martingales). Because (S_k) fails to be a martingale, no optional-stopping shortcut is available, in either direction: the theorem neither kills the game (as it did with replacement) nor solves it. The conditional drift depends on the full state — how far the score sits from zero and how many cards remain — so the optimal rule must be computed state by state, comparing the value of stopping against the expected value of continuing. That is a Bellman recursion, and it is the subject of the next subsection.

6.3 Dynamic Programming Formulation

Definition 6.4. Let $V(r, b)$ denote the maximum expected *additional* gain starting from a deck state with r red and b black cards remaining.

Proposition 6.5 (Bellman Equation). *For $r, b \geq 0$ with $r + b \geq 1$,*

$$V(r, b) = \max\left(0, \frac{r}{r+b}(1 + V(r-1, b)) + \frac{b}{r+b}(-1 + V(r, b-1))\right),$$

with $V(0, b) = 0$ for all $b \geq 0$ and $V(r, 0) = r$ for all $r \geq 0$.

Proof. The max reflects the choice between stopping (0 additional) and drawing. Drawing yields a red with probability $r/(r+b)$, contributing +1 plus continuation $V(r-1, b)$. Analogously for black.

Boundary $V(0, b) = 0$: drawing further can only decrease the score. Boundary $V(r, 0) = r$: all remaining cards are red. $\square$

Value of the Game

Theorem 6.6. *The optimal expected gain is*

$$V^* = V(26, 26) \approx 2.6245.$$

The value is obtained by tabulating the Bellman equation upwards from the boundary. No simple closed form is known.

6.4 Structure of the Optimal Policy

Proposition 6.7 (Threshold Structure). *The optimal stopping rule is a state-dependent threshold. Define $s(r, b) = b - r$ (the score after drawing all cards no longer in the deck). The optimal policy stops at state (r, b) iff*

$$s(r, b) \geq s^*(r, b),$$

where $s^*(r, b)$ is a threshold depending on the remaining deck composition.

Remark 6.8 (Why Not a Fixed Threshold?). A naive “stop at score $+k$ ” policy is suboptimal because the value of holding depends on how many cards remain. Late in the deck with few reds left, even a modest positive score should be locked in; early in the deck with many reds left, it may pay to keep drawing.

6.5 Fixed-Threshold Approximations

For intuition, consider the family of policies

$$\tau_k = \min\{j : S_j \geq k\},$$

stopping the first time the score reaches a fixed level k (taking $\tau_k = \infty$ if never reached, with payoff 0).

Proposition 6.9 (Reflection-Style Hitting Probability). *For $1 \leq k \leq 26$,*

$$P(\tau_k < \infty) = \frac{\binom{52}{26-k}}{\binom{52}{26}}.$$

The expected payoff is $E[S_{\tau_k}] = k \cdot P(\tau_k < \infty)$:

k	$P(\tau_k < \infty)$	$E[S_{\tau_k}]$
1	0.9630	0.963
2	0.8598	1.720
3	0.6890	2.067
4	0.5374	2.149
5	0.4199	2.099
6	0.3257	1.954
7	0.2511	1.758

The best fixed-threshold policy uses $k^* = 4$, achieving $E[S_{\tau_4}] \approx 2.149$. This is noticeably below the true optimum $V^* \approx 2.6245$.

Reasoning Hierarchy

1. **Spot the martingale.** (S_k) is a martingale with $E[S_{52}] = 0$. The naive answer would be 0.
2. **Identify the asymmetry.** Without replacement, the process *must* return to zero. Stopping at a positive score captures real value.
3. **Propose a fixed-threshold heuristic.** Compute $E[S_{\tau_k}]$ for small k and find $k^* = 4$ with value ≈ 2.15 .
4. **Recognise the true optimum.** The genuinely optimal rule is a state-dependent threshold, solvable by DP, yielding $V^* \approx 2.625$.

7 Pattern Hitting Times: Setup and Autocorrelation

7.1 The Problem

Fix a fair coin tossed independently forever, producing a sequence $X_1, X_2, X_3, \dots$ of i.i.d. symbols from $\{H, T\}$, each with probability $1/2$. Fix a pattern $P = p_1 p_2 \dots p_n$ of length n . Define the *first hitting time*

$$T_P = \min\{k \geq n : (X_{k-n+1}, \dots, X_k) = P\}.$$

We are interested in $E[T_P]$, the expected number of tosses until the pattern P first appears as a contiguous block.

Remark 7.1 (Why Hitting Times Are Not Equal Across Patterns). The naive intuition is that all patterns of length n should appear with the same expected time, since each occupies one of 2^n equally likely length- n windows. This is **false**. Two patterns of the same length can have very different expected hitting times depending on how they overlap with themselves. The patterns $P_1 = \text{HHH}$ and $P_2 = \text{HHT}$ both occur with window probability $1/8$, yet $E[T_{P_1}] = 14$ while $E[T_{P_2}] = 8$.

7.1.1 The Autocorrelation Polynomial

Definition 7.2 (Autocorrelation Vector). For a pattern $P = p_1 p_2 \dots p_n$ of length n , the *autocorrelation vector* is the binary vector $c(P) = (c_0, c_1, \dots, c_{n-1})$ where

$$c_i = \begin{cases} 1 & \text{if } p_{i+1} p_{i+2} \dots p_n = p_1 p_2 \dots p_{n-i}, \\ 0 & \text{otherwise.} \end{cases}$$

In words, $c_i = 1$ iff the suffix of P of length $n - i$ equals the prefix of P of the same length.

By convention $c_0 = 1$ (the full pattern matches itself).

Definition 7.3 (Conway Leading Number). The *leading number* (correlation polynomial evaluated at 2) of P is

$$L(P, P) = \sum_{i=0}^{n-1} c_i \cdot 2^{n-i}.$$

Example 7.4 (Worked Calculations). For each pattern of length 3 over $\{H, T\}$:

Pattern P	Vector $c(P)$	$L(P, P)$	$E[T_P]$
HHH	(1, 1, 1)	$2^3 + 2^2 + 2^1 = 14$	14
HHT	(1, 0, 0)	$2^3 = 8$	8
HTH	(1, 0, 1)	$2^3 + 2^1 = 10$	10
HTT	(1, 0, 0)	$2^3 = 8$	8
THH	(1, 0, 0)	$2^3 = 8$	8
THT	(1, 0, 1)	$2^3 + 2^1 = 10$	10
TTH	(1, 0, 0)	$2^3 = 8$	8
TTT	(1, 1, 1)	14	14

The equality of $L(P, P)$ and $E[T_P]$ in every row is the content of the next theorem.

7.2 The Guibas–Odlyzko Theorem

Guibas–Odlyzko

Theorem 7.5. *For a fair coin and any pattern P of length n ,*

$$E[T_P] = L(P, P) = \sum_{i=0}^{n-1} c_i(P) \cdot 2^{n-i}$$

where $c(P)$ is the autocorrelation vector of Definition 7.2.

We present two proofs. The first is via probability generating functions; the second is the elegant “infinite casino” argument due to Li (1980).

7.2.1 Proof I: Probability Generating Functions

Throughout, fix the pattern P of length n , write $T = T_P$, and abbreviate $f_k = P(T = k)$, $q_k = P(T > k)$.

Definition 7.6 (The Three Generating Functions).

$$\begin{aligned} F(z) &:= \sum_{k \geq 0} f_k z^k = E[z^T], & (\text{distribution of } T) \\ S(z) &:= \sum_{k \geq 0} q_k z^k, & (\text{survival probabilities}) \\ A(z) &:= \sum_{i=0}^{n-1} c_i z^i. & (\text{autocorrelation polynomial}) \end{aligned}$$

Two standing facts. First, $S(z) = \frac{1-F(z)}{1-z}$ for $|z| < 1$, by summing $q_k = 1 - \sum_{m \leq k} f_m$. Second, and crucially for the endgame, the tail-sum formula gives

$$S(1) = \sum_{k \geq 0} P(T > k) = E[T], \quad F(1) = P(T < \infty) = 1,$$

both series converging since $E[T] < \infty$ (the pattern appears eventually: disjoint blocks of n tosses each spell P with probability 2^{-n}).

Lemma 7.7 (Append-the-Pattern Identity). *For every $k \geq 0$,*

$$q_k \cdot 2^{-n} = \sum_{j=1}^n f_{k+j} c_{n-j} 2^{-(n-j)}.$$

Proof. Consider the event $E_k = \{T > k\} \cap \{\text{tosses } k+1, \dots, k+n \text{ spell } P\}$. Since $\{T > k\}$ depends only on the first k tosses, independence gives $P(E_k) = q_k \cdot 2^{-n}$: this is the left side.

Now compute $P(E_k)$ a second way. On E_k the pattern certainly occurs ending at position $k+n$, so the first occurrence satisfies $k < T \leq k+n$; partition by $T = k+j$, $j = 1, \dots, n$. Suppose the first occurrence ends at $k+j$. Its last j letters occupy positions $k+1, \dots, k+j$ — which, on E_k , hold the *first* j letters of P . So the event $\{T = k+j\}$ is compatible with E_k only if the length- j suffix of P equals its length- j prefix, i.e. only if $c_{n-j} = 1$. When it is compatible, completing E_k requires the remaining tosses $k+j+1, \dots, k+n$ of the block to match the last $n-j$ letters of P : these are free future tosses, contributing $2^{-(n-j)}$ independently. Hence $P(E_k \cap \{T = k+j\}) = f_{k+j} c_{n-j} 2^{-(n-j)}$, and summing over j gives the right side. $\square$

Lemma 7.8 (Master Equation). *For $z \in [0, 1]$,*

$$S(z) z^n = 2^n A\left(\frac{z}{2}\right) F(z).$$

Proof. Multiply Lemma 7.7 by z^k and sum over $k \geq 0$. The left side gives $2^{-n}S(z)$. On the right, for each fixed j ,

$$\sum_{k \geq 0} f_{k+j} z^k = z^{-j} \sum_{m \geq j} f_m z^m = z^{-j} F(z),$$

where the missing terms $m < j$ vanish because $f_m = 0$ for $m < n$ and $j \leq n$. Therefore

$$2^{-n}S(z) = F(z) \sum_{j=1}^n c_{n-j} 2^{j-n} z^{-j} \stackrel{i=n-j}{=} F(z) z^{-n} \sum_{i=0}^{n-1} c_i \left(\frac{z}{2}\right)^i = F(z) z^{-n} A\left(\frac{z}{2}\right),$$

which rearranges to the claim. $\square$

Proof of Theorem 7.5. Set $z = 1$ in the master equation. Since $S(1) = E[T]$ and $F(1) = 1$,

$$E[T_P] = 2^n A\left(\frac{1}{2}\right) = \sum_{i=0}^{n-1} c_i 2^{n-i}. \quad \square$$

Reading the Master Equation

Each generating function carries one kind of information — F says *when the pattern arrives*, S says *how long it has stayed away*, A encodes the *geometry of the pattern's self-overlaps* — and the master equation is the bookkeeping identity that ties the three together. It is nothing more than Lemma 7.7 summed over all starting times: one counting argument, dressed in power series.

The final formula deserves a second look, because each term has a meaning:

$$E[T] = \sum_{i: c_i=1} 2^{n-i},$$

one contribution of 2^ℓ for every self-overlap of length $\ell = n - i$. These are exactly the stakes of the surviving gamblers in the casino proof of Section 7.4: the gambler aligned with an overlap of length ℓ holds 2^ℓ at the stopping time. The two proofs — generating functions here, optional stopping there — agree not merely in their total but *term by term*: what the martingale argument reads as surviving stakes, the analytic argument reads as the coefficients of $2^n A(1/2)$.

7.2.2 Proof II: The Infinite Casino

Setup of the Casino Game

At every time step $t = 1, 2, 3, \dots$, a fresh *gambler* walks into the casino with one dollar. Each gambler, upon entering at time t , immediately bets on the event that the next n tosses (positions $t, t+1, \dots, t+n-1$) form precisely the pattern P .

The gambler bets sequentially, symbol by symbol, with a “double-or-nothing” rule:

- Bet $\$w$ on the next symbol being p_1 . If correct, the gambler now holds $\$2w$. If wrong, the gambler leaves with $\$0$.
- Continue: bet the entire current pile on the next symbol matching the next character of P .

A gambler “wins” (leaves with money) if and only if her entry time t is such that the next n symbols spell out P , in which case she leaves with $\$2^n$.

Lemma 7.9 (Casino Fairness). *Let W_t denote the total wealth in the hands of all gamblers currently inside the casino at time t , and let $D_t = t - W_t$ denote the casino’s net profit (dollars paid in by entering gamblers minus dollars still held by them). Then (D_t) is a martingale with $D_0 = 0$.*

Proof. Each gambler’s wealth process is a fair game (double-or-nothing on a fair coin), so its expectation is preserved at each step. Summing over all gamblers preserves the martingale property, and subtracting the deterministic entries t keeps it a martingale. $\square$

Lemma 7.10 (Wealth at Time T). *At time T , the only gamblers still holding money are those whose entry time s is such that the symbols $X_s X_{s+1} \dots X_T$ form a prefix of P . Such a gambler entered at time $s = T - (n - i) + 1$ for some i with $c_i = 1$, and holds $\$2^{n-i}$. Hence*

$$W_T = \sum_{i=0}^{n-1} c_i \cdot 2^{n-i} = L(P, P).$$

Proof. A gambler who entered at time s has been betting for $T - s + 1 = j$ rounds. She still holds money iff her bets on positions $s, s+1, \dots, s+j-1$ have all been correct, i.e., iff $X_s X_{s+1} \dots X_T = p_1 p_2 \dots p_j$.

At time T , the last n symbols equal P . For the gambler entering at s to be still alive, her matched prefix $p_1 \dots p_j$ must coincide with the suffix $X_s \dots X_T$ of the pattern. Writing $j = n - i$, this is exactly the condition $c_i = 1$.

Such a gambler holds $\$2^j = \2^{n-i} . Summing over all i with $c_i = 1$ gives $W_T = L(P, P)$. $\square$

Proof of Theorem 7.5 (Casino Argument). Apply the Optional Stopping Theorem to the martingale $D_t = t - W_t$ at the bounded stopping time T :

$$E[D_T] = D_0 = 0 \implies E[T] = E[W_T].$$

By Lemma 7.10, $W_T = L(P, P)$ deterministically. Hence $E[T_P] = L(P, P)$. $\square$ $\square$

Why This Proof Is Beautiful

The casino is a fair game: each gambler pays \$1 and either wins $\$2^n$ or loses everything. Over many time steps, the casino's average inflow equals its average outflow. The **moment of payout** is exactly when the pattern appears. At that moment, the gamblers who happen to be "aligned with the pattern" all cash out simultaneously, and the amounts they cash out are governed by the self-overlap structure of P .

In short: $E[T_P]$ is large when P has lots of self-overlaps, because many gamblers cash out simultaneously when the pattern finally appears, and the casino must have collected their winnings over a long time.

7.3 Conway's Trick: Pattern Races

7.3.1 The Penney Ante Game

The Game

Two players choose patterns A and B of equal length. A fair coin is tossed repeatedly. The first player whose pattern appears as a contiguous block in the toss sequence wins.

Surprising fact (Penney, 1969). For every pattern A of length $n \geq 3$, there exists another pattern B of the same length such that $P(B \text{ beats } A) > 1/2$.

This makes the game **non-transitive**: there is no “best” pattern.

Definition 7.11 (Cross-Correlation). For patterns $A = a_1 \cdots a_n$ and $B = b_1 \cdots b_n$ of equal length, the *cross-correlation vector* $c^{AB} = (c_0^{AB}, c_1^{AB}, \dots, c_{n-1}^{AB})$ is

$$c_i^{AB} = \begin{cases} 1 & \text{if } a_{i+1}a_{i+2} \cdots a_n = b_1b_2 \cdots b_{n-i}, \\ 0 & \text{otherwise.} \end{cases}$$

The corresponding Conway number is

$$L(A, B) = \sum_{i=0}^{n-1} c_i^{AB} 2^{n-i}.$$

Note that $L(A, A)$ recovers the autocorrelation leading number, and in general $L(A, B) \neq L(B, A)$.

Conway's Race Formula

Theorem 7.12. Let A and B be patterns of the same length, and let T_{AB} be the first time either appears in a fair-coin toss sequence. The probability that A appears first is

$$P(A \prec B) = \frac{L(B, B) - L(B, A)}{[L(A, A) - L(A, B)] + [L(B, B) - L(B, A)]}$$

Proof via the Casino Argument. Run the casino game with **both** betting schemes operating simultaneously: at each time step, two gamblers enter, one betting on A and one betting on B . The game halts at T_{AB} .

Let W_T^A, W_T^B denote the total wealth of A - and B -betting gamblers at time $T = T_{AB}$. By the same analysis as Lemma 7.10:

- If A wins: $W_T^A = L(A, A)$, $W_T^B = L(B, A)$.
- If B wins: $W_T^A = L(A, B)$, $W_T^B = L(B, B)$.

Apply OST to the martingale $D_t = 2t - W_t^A - W_t^B$:

$$E[W_T^A + W_T^B] = 2E[T_{AB}].$$

Write $p = P(A \prec B)$ and $q = 1 - p$. Then

$$\begin{aligned} E[W_T^A] &= p L(A, A) + q L(A, B), \\ E[W_T^B] &= p L(B, A) + q L(B, B). \end{aligned}$$

Conway's identity in standard form is

$$p [L(A, A) - L(A, B)] = q [L(B, B) - L(B, A)],$$

and combined with $p + q = 1$:

$$p = \frac{L(B, B) - L(B, A)}{[L(A, A) - L(A, B)] + [L(B, B) - L(B, A)]}. \quad \square$$

Example 7.13 (HTH vs HHT). Compute $P(\text{HTH} \prec \text{HHT})$.

Step 1: Autocorrelations.

- HTH: suffix H matches prefix H ($c_2 = 1$); suffix TH does not match prefix HT ($c_1 = 0$); $c_0 = 1$. $c(\text{HTH}) = (1, 0, 1)$, $L(\text{HTH}, \text{HTH}) = 8 + 2 = 10$.
- HHT: no nontrivial matches. $c(\text{HHT}) = (1, 0, 0)$, $L(\text{HHT}, \text{HHT}) = 8$.

Step 2: Cross-correlations.

- $L(\text{HTH}, \text{HHT})$: suffixes of HTH vs prefixes of HHT. Only length-1 matches (H = H). $L(\text{HTH}, \text{HHT}) = 2$.
- $L(\text{HHT}, \text{HTH})$: suffix HT matches prefix HT ($i = 1$, $2^2 = 4$). $L(\text{HHT}, \text{HTH}) = 4$.

Step 3: Apply Conway's formula.

$$P(\text{HTH} \prec \text{HHT}) = \frac{8 - 4}{(10 - 2) + (8 - 4)} = \frac{4}{12} = \frac{1}{3}.$$

So **HHT beats HTH with probability 2/3**. Both patterns have the same single-window probability of 1/8, yet one dominates the other in head-to-head racing.

Proposition 7.14 (Expected First-Hit Time of Pair). *For two patterns A and B,*

$$E[T_{AB}] = \frac{[L(A, A) - L(A, B)][L(B, B) - L(B, A)]}{[L(A, A) - L(A, B)] + [L(B, B) - L(B, A)]}.$$

Example 7.15 (Expected Time for HTH vs HHT). Using Example 7.13:

$$E[T_{AB}] = \frac{8 \cdot 4}{8 + 4} = \frac{32}{12} = \frac{8}{3}.$$

On average $8/3 \approx 2.67$ tosses are needed for either pattern to first appear. Compare with the individual hitting times $E[T_{\text{HTH}}] = 10$ and $E[T_{\text{HHT}}] = 8$.

7.3.2 Penney's Strategy

Proposition 7.16 (Penney's Strategy). *Given that the first player chooses a pattern $A = a_1 a_2 a_3 \cdots a_n$ of length $n \geq 3$, the second player wins more than half the time by choosing*

$$B = \bar{a}_2 a_1 a_2 \cdots a_{n-1},$$

where $\bar{a}_2$ denotes the complement of a_2 .

Example 7.17 (Penney's Strategy in Action). • Opponent picks HHH. Penney response: THH. Penney wins with probability 7/8.

- Opponent picks HTH. Penney response: HHT. Penney wins with probability 2/3.
- Opponent picks TTT. Penney response: HTT. Penney wins with probability 7/8.

Why Penney's Trick Works

By placing $\bar{a}_2$ at the start of B , Penney ensures that whenever the opponent's pattern A is one symbol away from completion, the previous $n - 1$ symbols already form a prefix of B . Effectively, B "intercepts" A just before A can complete.

7.3.3 Multi-Pattern Races

Theorem 7.18 (Multi-Pattern Conway Formula). *Suppose m patterns $P^{(1)}, \dots, P^{(m)}$ of equal length race. Let T^* be the first hit time and let $p_k = P(\text{winner is } P^{(k)})$. The probabilities satisfy the linear system*

$$\sum_{k=1}^m p_k L(P^{(j)}, P^{(k)}) = E[T^*], \quad j = 1, \dots, m,$$

with normalization $\sum_k p_k = 1$.

Sketch. Run the casino with m parallel betting schemes; at each step m gamblers enter. The casino's net flow $D_t = mt - \sum_k W_t^{(k)}$ is a martingale; OST at T^* gives $mE[T^*] = \sum_k E[W_{T^*}^{(k)}]$. Using the suffix-prefix matching argument patternwise yields the linear system. $\square$

Remark 7.19 (Solving the System). The system has $m + 1$ equations and $m + 1$ unknowns $(p_1, \dots, p_m, E[T^*])$. It is linear in the Conway matrix $[L(P^{(j)}, P^{(k)})]$ and solvable by standard linear algebra. For $m = 2$ this recovers Theorem 7.12.

Pattern Hitting Times: Essential Points

1. **Equal-window patterns can have unequal hitting times.** Self-overlap structure is what matters.
2. **Conway's leading number.** $L(P, P) = \sum c_i \cdot 2^{n-i}$ with $c_i = 1$ iff suffix-prefix overlap of length $n - i$ holds.
3. **Guibas–Odlyzko theorem.** $E[T_P] = L(P, P)$, provable via the infinite casino martingale.
4. **Conway's race formula.** $P(A \prec B) = [L(B, B) - L(B, A)] / \{[L(A, A) - L(A, B)] + [L(B, B) - L(B, A)]\}$.
5. **Penney's strategy.** Given $A = a_1 a_2 \dots a_n$, respond with $B = \bar{a}_2 a_1 a_2 \dots a_{n-1}$ to win more than half the time.
6. **Non-transitivity.** No globally best pattern; the game has rock-paper-scissors structure.

7.4 ABRACADABRA: Pattern Hitting Times via Martingales

The generating-function machinery developed above is the systematic route to pattern hitting times. There is, however, a second proof technique — a martingale argument of striking elegance — that produces the expected hitting time in three lines. We present it through its most famous incarnation.

Example 7.20 (The ABRACADABRA Problem). A monkey types i.i.d. uniform letters on a 26-key typewriter, one letter per second. What is the expected number of seconds until the word ABRACADABRA first appears?

The answer, which we derive below, is

$$E[T] = 26^{11} + 26^4 + 26 \approx 3.67 \times 10^{15},$$

about 116 million years. The striking feature is not the size of the number but its *shape*: why three terms, and why these exponents?

7.4.1 The Casino of Fair Gamblers

Imagine a casino that offers fair bets on the monkey's letters. Just before each second $t = 1, 2, 3, \dots$, a new gambler walks in with 1 euro and adopts the following strategy, keyed to the target word $w = w_1 w_2 \cdots w_{11} = \text{ABRACADABRA}$:

- She bets her 1 euro that the next letter is $w_1 = \text{A}$. The bet pays fair odds: if she wins she receives 26 euros (probability $1/26$); if she loses she leaves with nothing.
- If she wins, she immediately bets her entire 26 euros that the following letter is $w_2 = \text{B}$, receiving 26^2 if right.
- She continues letting everything ride on successive letters of w . A gambler who survives k letters holds 26^k euros; the first wrong letter wipes her out.

Every individual bet is fair (expected value zero), so the casino's aggregate net position — total euros paid in by gamblers minus total euros currently held by gamblers — is a **martingale**.

7.4.2 Stopping at the Moment of Victory

Let T be the first time ABRACADABRA has just been completed. Stop the game at T and take stock.

Money paid in. One gambler entered per second, so exactly T euros have been paid in.

Money held by gamblers. Almost every gambler has been wiped out. The survivors are precisely those whose betting sequence is currently *aligned with a suffix of the typed text that matches a prefix of w* :

- The gambler who entered 11 seconds ago has seen the full word ABRACADABRA unfold exactly as she bet. She holds 26^{11} .
- The gambler who entered 4 seconds ago has so far seen ABRA — the last four letters of the completed word — which are exactly the first four letters of w . She is alive and holds 26^4 .
- The gambler who entered 1 second ago has seen A — the final letter — which is the first letter of w . She holds 26.
- Every other gambler is broke: for any other entry time, the letters observed fail to match the corresponding prefix of w .

The surviving stakes correspond exactly to the *self-overlaps* of ABRACADABRA: the lengths $L \in \{1, 4, 11\}$ for which the length- L prefix equals the length- L suffix (A, ABRA, and the word itself).

Optional stopping. The net casino position is a martingale started at 0, and T is a stopping time with $E[T] < \infty$ (the pattern appears eventually with geometric-type tail bounds), so the optional stopping theorem applies:

$$0 = E[\text{net position at } T] = E\left[\underbrace{26^{11} + 26^4 + 26}_{\text{held by survivors}} - \underbrace{T}_{\text{paid in}}\right].$$

Rearranging,

$E[T] = 26^{11} + 26^4 + 26.$

Hitting Time of a General Pattern

Theorem 7.21. Let $w = w_1 \cdots w_m$ be a pattern over an alphabet of size A , typed as i.i.d. uniform letters. Then

$$E[T_w] = \sum_{\substack{1 \leq L \leq m \\ w_1 \cdots w_L = w_{m-L+1} \cdots w_m}} A^L,$$

the sum running over all lengths L at which a prefix of w coincides with a suffix of w (including $L = m$, the word itself).

The overlap set is exactly the *correlation set* of the pattern from the Guibas–Odlyzko theory developed earlier in this section: the martingale argument re-derives, by an entirely different route, the fact that hitting times are governed by the pattern’s self-correlation structure.

Example 7.22 (Sanity Checks on a Two-Letter Alphabet). For coin flips ($A = 2$):

- $w = \text{HH}$: overlaps at $L = 1, 2$, so $E[T] = 2 + 4 = 6$.
- $w = \text{HT}$: only the trivial overlap $L = 2$, so $E[T] = 4$.
- $w = \text{HTH}$: overlaps at $L = 1, 3$, so $E[T] = 2 + 8 = 10$.

These recover the classical values (and the familiar asymmetry $E[T_{\text{HH}}] = 6 > 4 = E[T_{\text{HT}}]$: patterns that overlap themselves take *longer* to appear, because failed attempts destroy accumulated progress).

Why Self-Overlap Slows a Pattern Down

The martingale computation makes the role of overlaps tangible. Each self-overlap of w is a surviving gambler at time T — extra money the casino must have collected, hence extra expected time. Equivalently: when a self-overlapping pattern fails midway, part of the partial progress can be *reused* (after typing ABRACADABR A... a failure can still leave you holding A or ABRA as live progress), and fair pricing of that reusable progress is precisely the extra terms $26^4 + 26$. Non-overlapping patterns restart from zero after every failure, which — counterintuitively at first — makes them *faster* to hit on average.

Remark 7.23 (Two Proofs, One Structure). We now possess two independent derivations of pattern hitting times: the Guibas–Odlyzko generating-function approach (systematic, yields full distributions and competition probabilities as in Penney’s game) and the gamblers’ martingale (three lines, yields the expectation with maximal transparency). The correlation polynomial of the former and the surviving gamblers of the latter encode the same combinatorial object — the self-overlap set of the pattern. When only $E[T]$ is required, the martingale route is unbeatable; when distributions or head-to-head pattern races are required, the generating-function machinery earns its keep.

Remark 7.24 (Trading Connection: Fair Games and Optional Stopping). Beyond its combinatorial charm, the argument is a template for a reasoning pattern ubiquitous in quantitative finance: *price a complicated random quantity by constructing a fair game (martingale) and applying optional stopping*. This is, in miniature, the logic of risk-neutral pricing — the value of a derivative is the stake that makes the hedged position a fair game. The gambler construction also illustrates a subtler point relevant to strategy design: sequences of individually fair bets can be assembled so that, at a cleverly chosen stopping time, the *bookkeeping identity itself* reveals a nontrivial expectation. No single bet has edge; the information is extracted purely from the stopping structure.

8 Order Statistics: Maximum, Minimum, and Their Symmetries

Many probability problems that look laborious collapse into two-line solutions once the basic order-statistics toolkit is in hand. The expected maximum and minimum of n samples from a uniform distribution, the pointwise identities relating them to sums and absolute differences, and a clean rule of thumb bridging the discrete and continuous cases are the entire content of this section. Their value lies in reach: a handful of elementary facts about maxima and minima dissolves a whole class of problems that appear, at first sight, to require separate ad hoc calculations.

8.1 The Pointwise Identities

Before any probability enters the picture, we record two elementary algebraic identities that hold for any pair of real numbers x, y .

Proposition 8.1 (Sum and Difference Identities). *For any two real numbers $x, y \in \mathbb{R}$:*

$$\max(x, y) + \min(x, y) = x + y,$$

$$\max(x, y) - \min(x, y) = |x - y|.$$

Proof. Either $x \leq y$ or $y \leq x$. In the first case, $\max(x, y) = y$ and $\min(x, y) = x$, giving sum $x + y$ and difference $y - x = |x - y|$. The second case is symmetric. $\square$

Taking expectations (which is linear), these identities immediately yield:

Corollary 8.2 (Sum and Difference of Expectations). *For any pair of random variables X, Y on a common probability space:*

$$E[\max(X, Y)] + E[\min(X, Y)] = E[X] + E[Y],$$

$$E[\max(X, Y)] - E[\min(X, Y)] = E|X - Y|.$$

Why This Is So Useful

The two identities together act as a system: *the sum of the expectations gives the total of the variables, and the difference gives the expected absolute spread.* In practice this means that if you can compute *either* $E[\max]$ or $E[\min]$, the other follows immediately from the sum identity, and the expected absolute difference $E|X - Y|$ follows for free. The quantity $E|X - Y|$ looks hard on its own: the absolute value resists naive expansion, and computing it directly means splitting into the cases $X > Y$ and $X < Y$ and summing each separately. But via Corollary 8.2 the problem reduces to $E[\max] - E[\min]$, which is purely a question about order statistics with no absolute value in sight. A term that obstructs direct calculation has been traded for two that do not.

8.2 The Tail-Sum Formula

The central computational tool for $E[\max]$ and $E[\min]$ is the *tail-sum formula*, which converts an expectation into an integral (or sum) of survival probabilities.

Lemma 8.3 (Tail-Sum Formula). *Let Z be a non-negative random variable.*

- If Z is continuous with values in $[0, M]$:

$$E[Z] = \int_0^M P(Z > t) dt.$$

- If Z is discrete with values in $\{1, 2, \dots, M\}$:

$$E[Z] = \sum_{k=1}^M P(Z \geq k).$$

Proof. For the continuous case, swap the order of integration:

$$\int_0^M P(Z > t) dt = \int_0^M \int_t^M f_Z(z) dz dt = \int_0^M f_Z(z) \int_0^z dt dz = \int_0^M z f_Z(z) dz = E[Z].$$

For the discrete case:

$$\sum_{k=1}^M P(Z \geq k) = \sum_{k=1}^M \sum_{j=k}^M P(Z = j) = \sum_{j=1}^M j P(Z = j) = E[Z]. \quad \square$$

The reason this formula is the central tool: *the survival probabilities of maxima and minima have explicit product forms*, while their distributions do not.

8.3 Expected Maximum and Minimum: Continuous Uniform

Continuous Uniform Order Statistics

Theorem 8.4. Let $X_1, \dots, X_n$ be i.i.d. $\text{Uniform}(0, S)$. Then

$$E[\max(X_1, \dots, X_n)] = \frac{nS}{n+1}, \quad E[\min(X_1, \dots, X_n)] = \frac{S}{n+1}.$$

Proof. For $t \in [0, S]$:

$$P(\max > t) = 1 - P(\text{all } X_i \leq t) = 1 - \left(\frac{t}{S}\right)^n.$$

By the tail-sum formula,

$$E[\max] = \int_0^S \left[1 - \left(\frac{t}{S}\right)^n\right] dt = S - \frac{S}{n+1} = \frac{nS}{n+1}.$$

Similarly,

$$P(\min > t) = P(\text{all } X_i > t) = \left(\frac{S-t}{S}\right)^n,$$

so

$$E[\min] = \int_0^S \left(\frac{S-t}{S}\right)^n dt = \frac{S}{n+1}. \quad \square$$

Remark 8.5 (A Built-In Sanity Check). The two formulas satisfy

$$E[\max] + E[\min] = \frac{nS + S}{n+1} = S,$$

which is also $E[X_1] + E[X_n] = 2 \cdot S/2$ via Corollary 8.2 applied to the case $n = 2$. More generally, for $\text{Uniform}(a, b)$:

$$E[\max] + E[\min] = a + b,$$

which is twice the mean. Any answer violating this identity is wrong.

The “Equispaced Sample” Mental Model

The formula $E[\max] = nS/(n+1)$ has a striking interpretation. Imagine throwing n darts uniformly into the interval $[0, S]$, then ordering them as $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$. Adding the two endpoints 0 and S as “virtual” order statistics gives $n+2$ points partitioning $[0, S]$ into $n+1$ sub-intervals.

Key fact: By symmetry, the expected length of each sub-interval is exactly $S/(n+1)$. Therefore

$$E[X_{(k)}] = \frac{kS}{n+1},$$

i.e. the k -th order statistic sits, on average, at the $k/(n+1)$ fraction of the interval.

This generalises Theorem 8.4: setting $k = n$ gives $E[\max] = nS/(n+1)$, and $k = 1$ gives $E[\min] = S/(n+1)$. Internalising this picture lets you write down the expected position of *any* order statistic in one second.

8.4 Expected Maximum and Minimum: Discrete Uniform

Discrete Uniform Order Statistics

Theorem 8.6. Let $X_1, \dots, X_n$ be i.i.d. uniform on $\{1, 2, \dots, S\}$. Then

$$E[\max] = \sum_{k=1}^S \left[1 - \left(\frac{k-1}{S} \right)^n \right], \quad E[\min] = \sum_{k=1}^S \left(\frac{S-k+1}{S} \right)^n.$$

Proof. For $k \in \{1, \dots, S\}$:

$$P(\max \geq k) = 1 - P(\text{all } X_i \leq k-1) = 1 - \left(\frac{k-1}{S} \right)^n,$$

$$P(\min \geq k) = P(\text{all } X_i \geq k) = \left(\frac{S-k+1}{S} \right)^n.$$

The result follows from the discrete tail-sum formula. □

Example 8.7 (Two Dice: Computing $E[\max]$ Directly). For $S = 6$, $n = 2$:

$$E[\max] = \sum_{k=1}^6 \left[1 - \left(\frac{k-1}{6} \right)^2 \right] = 6 - \frac{0+1+4+9+16+25}{36} = 6 - \frac{55}{36} = \frac{161}{36} \approx 4.47.$$

Similarly $E[\min] = 91/36 \approx 2.53$.

8.5 The Discrete Sum Identity

The continuous formula $E[\max] + E[\min] = S$ has a discrete counterpart that is even cleaner.

Discrete Sum Identity

Theorem 8.8. For $X_1, \dots, X_n$ i.i.d. uniform on $\{1, \dots, S\}$ and **any** $n \geq 1$:

$$E[\max] + E[\min] = S + 1.$$

Proof. Substitute $j = S - k + 1$ in the formula for $E[\min]$ of Theorem 8.6. When k ranges over $\{1, \dots, S\}$, j ranges over $\{S, S-1, \dots, 1\}$, i.e. the same set in reverse:

$$E[\min] = \sum_{k=1}^S \left(\frac{S-k+1}{S} \right)^n = \sum_{j=1}^S \left(\frac{j}{S} \right)^n.$$

For $E[\max]$:

$$E[\max] = \sum_{k=1}^S 1 - \sum_{k=1}^S \left(\frac{k-1}{S} \right)^n = S - \sum_{j=0}^{S-1} \left(\frac{j}{S} \right)^n.$$

Adding:

$$E[\max] + E[\min] = S - \sum_{j=0}^{S-1} \left(\frac{j}{S} \right)^n + \sum_{j=1}^S \left(\frac{j}{S} \right)^n.$$

The two sums overlap on $j = 1, 2, \dots, S-1$ and cancel there, leaving

$$- \left(\frac{0}{S} \right)^n + \left(\frac{S}{S} \right)^n = 0 + 1 = 1.$$

Therefore $E[\max] + E[\min] = S + 1$. □

The Mirror Symmetry

The identity $E[\max] + E[\min] = S + 1$ reflects a deep symmetry of the discrete uniform distribution: the map $k \mapsto S - k + 1$ is a bijection on $\{1, \dots, S\}$ that preserves the uniform distribution and swaps maxima with minima.

Concretely: if X_i is uniform on $\{1, \dots, S\}$, then $Y_i := S - X_i + 1$ is also uniform on $\{1, \dots, S\}$, and

$$\max(Y_1, \dots, Y_n) = S + 1 - \min(X_1, \dots, X_n).$$

Taking expectations: $E[\max] = S + 1 - E[\min]$, which gives the identity.

The continuous version, $E[\max] + E[\min] = S$, has the same mirror symmetry but without the “+1”: the continuous reflection is $x \mapsto S - x$, taking $[0, S]$ to itself. The discrete “+1” is the consequence of reflecting around the midpoint $(S + 1)/2$ instead of $S/2$.

8.6 The Rule of Thumb: Approximating Discrete by Continuous

The exact formulas in Theorem 8.6 involve sums of n -th powers, which are awkward to compute mentally. This “Rule of Thumb” gives you a quick estimation.

Rule of Thumb

Theorem 8.9. For $X_1, \dots, X_n$ i.i.d. uniform on $\{1, \dots, S\}$, the following approximations hold:

Crude version (treating dice as $U(0, S)$):

$$E[\max] \approx \frac{nS}{n+1}, \quad E[\min] \approx \frac{S}{n+1}.$$

Refined version (with continuity correction, treating dice as $U(\frac{1}{2}, S + \frac{1}{2})$):

$$E[\max] \approx \frac{1}{2} + \frac{nS}{n+1}, \quad E[\min] \approx \frac{1}{2} + \frac{S}{n+1}.$$

Proof. The crude version comes from Theorem 8.4 applied to $\text{Uniform}(0, S)$. The refined version comes from treating the discrete distribution as $\text{Uniform}(a, b)$ with $a = 1/2$ and $b = S + 1/2$ (the natural “continuity correction” that puts each integer at the centre of an interval of length one), and noting that for $\text{Uniform}(a, b)$:

$$E[\max] = a + (b - a) \cdot \frac{n}{n + 1}, \quad E[\min] = a + (b - a) \cdot \frac{1}{n + 1}.$$

With $a = 1/2$ and $b - a = S$, this gives the refined formulas. $\square$

Example 8.10 (How Good Are These for Two Dice?). With $S = 6$, $n = 2$: the exact values are $E[\max] = 161/36 \approx 4.472$ and $E[\min] = 91/36 \approx 2.528$.

	Crude $U(0, S)$		Refined $U(\frac{1}{2}, S + \frac{1}{2})$	
	approx	error	approx	error
$E[\max]$	4.000	−0.472	4.500	+0.028
$E[\min]$	2.000	−0.528	2.500	−0.028
sum	6.000	−1.000	7.000	0.000

The crude version is off by roughly $\frac{1}{2}$ in each direction; the refined version is off by less than 3%. The refined version also **automatically satisfies the sum identity** $E[\max] + E[\min] = S + 1 = 7$, while the crude version misses it by 1.

Which Approximation to Use

In a fast-paced interview where you need a quick number, the crude version $E[\max] \approx nS/(n + 1)$ is fine. It captures the right scaling and gives a usable estimate.

For accuracy or written work, use the refined version with the $+1/2$ correction. It is essentially exact in practice and automatically satisfies the discrete sum identity.

The pattern to remember: *anything you’d compute by treating a die as a continuous uniform should be shifted by $1/2$ to account for the fact that the die’s values are centred at integers, not at the midpoints of unit intervals starting at zero.* This continuity correction appears in many places (normal approximations to binomials, sums of dice, etc.) and is worth internalising as a reflex.

8.7 Application: Solving $E|X - Y|$ for Dice in One Line

We now revisit the expected absolute difference of two dice, this time solving it almost without calculation.

Example 8.11 ($E|X - Y|$ via Order Statistics). Let X, Y be independent uniform on $\{1, \dots, S\}$.

By Corollary 8.2 and Theorem 8.8:

$$E[\max] + E[\min] = S + 1, \quad E[\max] - E[\min] = E|X - Y|.$$

Solving the system:

$$E[\max] = \frac{S + 1 + E|X - Y|}{2}, \quad E[\min] = \frac{S + 1 - E|X - Y|}{2}.$$

To find $E|X - Y|$, compute either $E[\max]$ or $E[\min]$ exactly. We computed $E[\max] = (4n - 1)(n + 1)/(6n)$ earlier (or by direct summation with $n = 2$):

$$E[\max] = \sum_{k=1}^S \left[1 - \left(\frac{k-1}{S} \right)^2 \right] = S - \frac{(S-1)S(2S-1)}{6S^2} = \frac{(4S-1)(S+1)}{6S}.$$

Then

$$E|X - Y| = 2E[\max] - (S+1) = \frac{(4S-1)(S+1)}{3S} - (S+1) = \frac{(S+1)(4S-1-3S)}{3S} = \frac{(S+1)(S-1)}{3S} = \frac{S^2-1}{3S}.$$

For $S = 6$: $E|X - Y| = 35/18 \approx 1.94$. For $S = 30$: $E|X - Y| = 899/90 \approx 9.99$.

Remark 8.12 (Why This Solution Is Better). Compare with the direct combinatorial approach that summed $|i - j|$ over all S^2 pairs. Both give the same answer, but the order-statistics route:

- Avoids the absolute value entirely.
- Generalises immediately to other distributions: for any pair with known $E[\max]$, the same identity gives $E|X - Y|$.
- Extends to min / max of more than two variables without modification.

This is the typical pattern: a single algebraic identity (Proposition 8.1) converts a hard problem into an easy one.

8.8 Example of verification: The Same Problem by Direct Summation

The previous subsection obtained $E|X - Y| = \frac{S^2-1}{3S}$ in a few lines, using only the max-min identity and the discrete sum identity. It is worth carrying out the direct combinatorial computation as well: it confirms the result independently, and it makes visible exactly how much labour the order-statistics machinery absorbed.

Proposition 8.13 (Direct Summation). *Let X, Y be i.i.d. uniform on $\{1, \dots, S\}$. Then*

$$E|X - Y| = \frac{S^2 - 1}{3S},$$

in agreement with the order-statistics derivation.

Proof. By definition,

$$E|X - Y| = \frac{1}{S^2} \sum_{i=1}^S \sum_{j=1}^S |i - j|.$$

Group the double sum by the gap $d = |i - j|$. For each $d \in \{1, \dots, S-1\}$ there are exactly $2(S-d)$ ordered pairs (i, j) at distance d — the $S-d$ pairs with $j = i + d$ and the $S-d$ with $j = i - d$. The diagonal $d = 0$ contributes S pairs but nothing to the sum. Hence

$$\sum_{i,j} |i - j| = 2 \sum_{d=1}^{S-1} d(S-d).$$

Using $\sum_{d=1}^{S-1} d = \frac{(S-1)S}{2}$ and $\sum_{d=1}^{S-1} d^2 = \frac{(S-1)S(2S-1)}{6}$,

$$\sum_{d=1}^{S-1} d(S-d) = \frac{(S-1)S}{6} [3S - (2S-1)] = \frac{S(S-1)(S+1)}{6}.$$

Therefore $\sum_{i,j} |i - j| = \frac{S(S^2-1)}{3}$, and dividing by S^2 gives the claim. $\square$

Remark 8.14 (What the Machinery Saved). The direct route needs the gap decomposition, the counting $2(S - d)$, both closed forms for $\sum d$ and $\sum d^2$, and a cancellation. The order-statistics route needed one algebraic identity and one tail sum. Both are elementary; only one is fast enough to carry out under interview conditions, and only one survives a change of question — replace $|X - Y|$ by the range of n dice and the direct sum becomes intractable, while $E[\max] - E[\min]$ does not change at all.

Example 8.15 (Numerical Values).

S	$E X - Y $ exact	decimal
2	$1/2$	0.500
6	$35/18$	1.944
10	$33/10$	3.300
20	$133/20$	6.650
30	$899/90$	9.989

For two ordinary dice ($S = 6$) the expected gap is $35/18 \approx 1.94$: a little under two pips.

Asymptotic Sanity Check

Rewriting the closed form,

$$E|X - Y| = \frac{S}{3} - \frac{1}{3S} \xrightarrow{S \rightarrow \infty} \frac{S}{3}.$$

This is exactly the continuous answer. For U, V i.i.d. $\text{Unif}(0, 1)$ the same max–min identity with the continuous formulas of Theorem 8.4 gives

$$E|U - V| = E[\max] - E[\min] = \frac{2}{3} - \frac{1}{3} = \frac{1}{3},$$

and rescaling to $\text{Unif}(0, S)$ multiplies by S . The discrete correction $-\frac{1}{3S}$ is the quantisation penalty: dice cannot land between pips, which trims the expected gap slightly. Note that it is a *relative* correction of order $1/S^2$, so already for $S = 30$ the continuous approximation $S/3 = 10$ is within 0.2% of the exact 9.989.

8.9 Application: The Expected Median of Three Dice

We close with an application that showcases how the symmetry arguments developed above collapse a seemingly computational problem into a one-line answer.

Example 8.16 (Expected Median of Three Dice). Three fair dice are rolled. What is the expected value of the median (the middle value after ordering)?

Denote the ordered outcomes $X_{(1)} \leq X_{(2)} \leq X_{(3)}$, so the median is $X_{(2)}$. We give three solutions, in increasing order of elegance.

8.9.1 Method 1: Direct Computation

The distribution of the median follows from its CDF. Writing $p = k/6$ for the probability a single die is $\leq k$,

$$P(X_{(2)} \leq k) = P(\text{Bin}(3, p) \geq 2) = 3p^2(1 - p) + p^3 = 3p^2 - 2p^3.$$

Differencing gives $P(X_{(2)} = k)$, and summing $k P(X_{(2)} = k)$ yields the distribution

$$\left(\frac{2}{27}, \frac{5}{27}, \frac{13}{54}, \frac{13}{54}, \frac{5}{27}, \frac{2}{27} \right) \quad \text{for } k = 1, \dots, 6,$$

and hence $E[X_{(2)}] = 7/2$. Correct, but laborious — and it obscures *why* the answer is exactly the single-die mean.

8.9.2 Method 2: The Sum Identity

Ordering preserves the total, so

$$X_{(1)} + X_{(2)} + X_{(3)} = X_1 + X_2 + X_3,$$

and taking expectations,

$$E[X_{(1)}] + E[X_{(2)}] + E[X_{(3)}] = 3 E[X_1] = 3 \cdot \frac{7}{2} = \frac{21}{2}.$$

By the discrete sum identity (Theorem 8.8), $E[X_{(1)}] + E[X_{(3)}] = E[\min] + E[\max] = S + 1 = 7$. Subtracting,

$$E[X_{(2)}] = \frac{21}{2} - 7 = \frac{7}{2}.$$

This is faster and reuses the machinery we already built, but the cleanest argument dispenses with arithmetic entirely.

8.9.3 Method 3: The Reflection Symmetry

Recall the reflection $x \mapsto S + 1 - x = 7 - x$, which is a distribution-preserving bijection of $\{1, \dots, 6\}$ (the same map that powered the sum identity). Set $Y_i := 7 - X_i$. Then (Y_1, Y_2, Y_3) has the same joint distribution as (X_1, X_2, X_3) , and reflection reverses order:

$$Y_{(1)} = 7 - X_{(3)}, \quad Y_{(2)} = 7 - X_{(2)}, \quad Y_{(3)} = 7 - X_{(1)}.$$

In particular $Y_{(2)} = 7 - X_{(2)}$ has the same distribution as $X_{(2)}$, so

$$X_{(2)} \stackrel{d}{=} 7 - X_{(2)}.$$

Taking expectations, $E[X_{(2)}] = 7 - E[X_{(2)}]$, whence

$$E[X_{(2)}] = \frac{7}{2}.$$

Why This Argument Is the “Right” One

Methods 1 and 2 compute the answer; Method 3 *explains* it. The reflection $X \stackrel{d}{=} 7 - X$ shows that the *entire distribution* of the median is symmetric about $7/2$, not merely that its mean happens to land there. The numerical distribution confirms this: $P(X_{(2)} = k) = P(X_{(2)} = 7 - k)$ for every k , so the histogram is a mirror image about 3.5 . A symmetric distribution has its mean at the centre of symmetry, and the answer follows with no computation.

Expected Median, General Case

Theorem 8.17. For n (odd) independent dice uniform on $\{1, \dots, S\}$, the median $X_{((n+1)/2)}$ has a distribution symmetric about $(S + 1)/2$, and therefore

$$E[\text{median}] = \frac{S + 1}{2}.$$

Proof. The reflection $X_i \mapsto S + 1 - X_i$ preserves the uniform distribution on $\{1, \dots, S\}$ and reverses the order statistics, mapping the median to $S + 1 - (\text{median})$. Hence the median satisfies $X_{((n+1)/2)} \stackrel{d}{=} (S + 1) - X_{((n+1)/2)}$, a distribution symmetric about $(S + 1)/2$. Its mean is the centre of symmetry, $(S + 1)/2$. $\square$

Remark 8.18 (The General Lesson). The expected median of any odd number of uniform dice equals the single-die mean $(S+1)/2$ — independently of how many dice are rolled. Rolling more dice sharpens the median's distribution around the centre (the variance shrinks), but never shifts its mean, because symmetry about $(S+1)/2$ is preserved for every odd n .

The broader principle worth internalising: *for any symmetric distribution, the mean, median, and mode all coincide at the centre of symmetry*. Detecting a reflection symmetry — here $x \mapsto S+1-x$ — reduces an entire class of order-statistic expectation problems to identifying the centre. This is often the fastest route in an interview: before computing, ask whether the distribution has a symmetry that pins the answer down for free.

8.10 Application: Auction Revenue and the Equispaced-Sample Formula

The order-statistic means $E[U_{(k)}] = k/(n+1)$ answer a central question of auction theory in two lines.

Let n bidders hold i.i.d. $\text{Unif}(0,1)$ private valuations. In a *second-price* (Vickrey) auction, bidding one's true valuation is a dominant strategy and the winner pays the *second*-highest bid, so the seller's expected revenue is

$$E[U_{(n-1)}] = \frac{n-1}{n+1}.$$

In a *first-price* auction the winner pays her own bid, so bidding truthfully guarantees zero surplus and every bidder shades downward. The symmetric equilibrium bidding *function* — the strategy mapping a valuation v to a bid — is

$$b(v) = \frac{n-1}{n} v,$$

i.e. each bidder offers the expected highest rival valuation *conditional on winning*: given that all $n-1$ rivals fall below v , their maximum is the top of $n-1$ uniforms on $[0, v]$, with mean $\frac{n-1}{n}v$. (Shading vanishes as n grows: competition erodes the surplus.) The winner is the bidder with valuation $U_{(n)}$, so expected revenue is

$$E[b(U_{(n)})] = \frac{n-1}{n} \cdot \frac{n}{n+1} = \frac{n-1}{n+1}.$$

The two formats yield *identical* expected revenue, although the payment rules differ entirely and the second valuation is never actually paid in the first-price auction. This is the simplest instance of the **Revenue Equivalence Theorem** (Vickrey, Myerson): under independent private values and risk-neutral bidders, all standard auction formats raise the same expected revenue. What changes across formats is *who bears the shading*, not what the seller collects — and the whole computation reduces to knowing where the order statistics of a uniform sample sit.

8.11 Where Order Statistics Appear in Markets

Remark 8.19 (Extremes, Quantiles, and Ranges). Order statistics enter whenever one cares about *extreme or ranked realisations* of a sample rather than its average. Three instances:

- **Range-based volatility estimation.** The daily high and low are the max and min of the price path, and their spread carries far more information about intraday variance than the close-to-close return alone. The Parkinson estimator builds on $E[(\max - \min)^2]$ for Brownian motion over $[0, T]$, achieving roughly five times the efficiency of squared returns; the continuous limits of the identities in this section are the discrete shadow of that calculation.

- **Empirical quantiles and Value-at-Risk.** The k -th order statistic of an i.i.d. sample is the empirical quantile: by the equispaced-sample formula, $E[U_{(k)}] = k/(n+1)$, so $X_{(k)}$ sits, on average, at the $k/(n+1)$ quantile of the underlying law. A historical 99% VaR computed from $n = 500$ daily returns is nothing but the 5th order statistic of the losses — and the formula explains both why the estimator is centred correctly and why it is noisy: near the tail, consecutive order statistics are spaced $\sim 1/(n+1)$ apart in probability but far apart in *value*, since few observations populate the region.
- **Auction mechanics.** Expected revenue in standard auction formats reduces entirely to the order-statistic means $E[U_{(k)}] = k/(n+1)$; this was developed in the auction-revenue application above, where both the first- and second-price formats yield $E[U_{(n-1)}] = \frac{n-1}{n+1}$.

The pattern across all three: replace a question about an awkward functional of a sample — a range, a tail, a winning bid — by a question about where ranked uniforms sit, and the equispaced-sample picture answers it.

Key Takeaways

Order Statistics: Essentials

1. **Pointwise identities.** $\max + \min = x + y$ and $\max - \min = |x - y|$. Taking expectations: $E[\max] + E[\min] = E[X] + E[Y]$ and $E[\max] - E[\min] = E|X - Y|$.
2. **Tail-sum formula.** $E[Z] = \int_0^M P(Z > t) dt$ (continuous) or $E[Z] = \sum_{k=1}^M P(Z \geq k)$ (discrete). The workhorse for computing extremes.
3. **Continuous uniform.** $E[\max] = nS/(n+1)$, $E[\min] = S/(n+1)$, $E[X_{(k)}] = kS/(n+1)$. Mnemonic: *n darts divide $[0, S]$ into $n+1$ equal expected sub-intervals.*
4. **Discrete uniform sum identity.** For dice on $\{1, \dots, S\}$, any $n \geq 1$: $E[\max] + E[\min] = S + 1$.
5. **Rule of thumb.** Crude: $E[\max] \approx nS/(n+1)$, $E[\min] \approx S/(n+1)$. Refined (with continuity correction): $E[\max] \approx 1/2 + nS/(n+1)$, $E[\min] \approx 1/2 + S/(n+1)$.
6. **Master technique.** When a problem asks for $E|X - Y|$, $E[\text{range}]$, or anything involving absolute values of differences, immediately rewrite in terms of $E[\max]$ and $E[\min]$ and compute those.
7. **Trading hooks.** Parkinson volatility estimator, VAR and second-price auction revenue.

9 Size-Biasing: When Random Sampling Lies

9.1 The Waiting Time Paradox

Buses arrive at a stop according to a process in which the times between consecutive buses are i.i.d. random variables with mean μ . You arrive at the stop at a “random” instant, independent of the bus process. Two natural questions:

- What is the expected time you wait until the next bus?
- What is the expected length of the interval *between buses* that your arrival falls into?

The naive answer to (a) is “half the mean interval,” $\mu/2$: you land somewhere in the middle of a typical gap, so on average you wait half of it. The naive answer to (b) is “the mean interval,” μ : you fall into a typical gap.

Both naive answers are wrong. For buses with exponentially distributed inter-arrival times of mean μ , the expected waiting time is μ , *not* $\mu/2$, and the expected length of the captured interval is 2μ , *not* μ . You wait, on average, a full mean inter-arrival time — as long as if you had started your own timer at the previous bus.

The Waiting Time Paradox

Theorem 9.1. *Let inter-arrival times be i.i.d. with mean μ and second moment $E[X^2]$. For an observer arriving at a random instant:*

$$E[\text{captured interval}] = \frac{E[X^2]}{\mu}, \quad E[\text{waiting time}] = \frac{E[X^2]}{2\mu}.$$

For the exponential distribution, $E[X^2] = 2\mu^2$, giving captured interval 2μ and waiting time μ . For a degenerate (deterministic) distribution, $E[X^2] = \mu^2$, giving captured interval μ and waiting time $\mu/2$ — the naive answer, recovered only when there is no randomness.

Why the Naive Answer Fails

The error in the naive reasoning is subtle: it assumes that the interval you land in is a *typical* interval. It is not. **A random arrival is more likely to land in a long interval than a short one**, simply because long intervals occupy more of the timeline.

Concretely: if half the gaps are 1 minute and half are 10 minutes, then although the gaps are 50/50 by count, they are not 50/50 by *duration*. The 10-minute gaps account for 10/11 of the total time, so a random arrival lands in a 10-minute gap with probability 10/11, not 1/2. The interval you sample is biased toward large values. This is the phenomenon of *size-biasing* (or length-biasing), and it is the subject of this entire section.

9.2 The Length-Biased Distribution

The resolution of the paradox rests on a single concept. When we sample an interval by “landing in it at a random time,” we are not sampling from the interval-length distribution f ; we are sampling from a *length-biased* version of it.

Length-Biased Density

Theorem 9.2. *Let intervals have density $f(x)$ with mean $\mu = E[X]$. The interval captured by an observer arriving at a uniformly random time has the length-biased density*

$$\tilde{f}(x) = \frac{x f(x)}{\mu}.$$

Proof. Consider a long timeline of total length L , containing $N \approx L/\mu$ intervals. Among these, the number with length in $[x, x + dx]$ is $Nf(x) dx$. The *total length* occupied by such intervals is $x \cdot Nf(x) dx$. An observer arriving at a random time lands in an interval of length $[x, x + dx]$ with probability proportional to the total length such intervals occupy:

$$P(\text{captured length} \in [x, x + dx]) = \frac{x \cdot Nf(x) dx}{L} = \frac{x f(x) dx}{\mu},$$

using $L = N\mu$. This is the claimed density. $\square$

Remark 9.3 (Mean of the Length-Biased Distribution). The expected captured interval is

$$E[\tilde{X}] = \int_0^\infty x \tilde{f}(x) dx = \int_0^\infty \frac{x^2 f(x)}{\mu} dx = \frac{E[X^2]}{\mu}.$$

Since $E[X^2] = \text{Var}(X) + \mu^2 \geq \mu^2$, we always have $E[\tilde{X}] \geq \mu$, with equality if and only if X is deterministic. *The captured interval is always at least as long as a typical interval, and strictly longer whenever there is any variability.* The ratio

$$\frac{E[\tilde{X}]}{\mu} = \frac{E[X^2]}{\mu^2} = 1 + \frac{\text{Var}(X)}{\mu^2} = 1 + \text{CV}^2$$

is governed by the squared coefficient of variation $\text{CV} = \sigma/\mu$. The more variable the intervals, the stronger the bias.

The Waiting Time from Length-Biasing

Once you have landed in an interval of length $\tilde{X}$, your position within it is uniform (by symmetry, conditional on the captured length). So your expected wait, given the captured length $\tilde{X} = \ell$, is $\ell/2$. Unconditioning:

$$E[\text{waiting time}] = E\left[\frac{\tilde{X}}{2}\right] = \frac{1}{2} \cdot \frac{E[X^2]}{\mu} = \frac{E[X^2]}{2\mu},$$

which is Theorem 9.1. The paradox is now transparent: you wait half of a *length-biased* interval, not half of a typical one.

9.3 Renewal Theory: The Density of Points

The waiting time paradox is the visible face of a deeper structure: *renewal theory*, which governs the long-run behaviour of processes built from i.i.d. increments.

Definition 9.4 (Renewal Process). Let $X_1, X_2, \dots$ be i.i.d. positive random variables with mean μ . The *renewal epochs* are the partial sums $S_0 = 0$, $S_n = X_1 + \dots + X_n$. The points $\{S_0, S_1, S_2, \dots\}$ form a renewal process on $[0, \infty)$.

The central quantity is the probability that the process visits a given point. For an integer-valued process, define

$$u_s = P(S_n = s \text{ for some } n).$$

Blackwell's Renewal Theorem, Lattice Case

Theorem 9.5. *For a renewal process with integer increments whose greatest common divisor is 1, and mean μ :*

$$\lim_{s \rightarrow \infty} u_s = \frac{1}{\mu}.$$

Heuristic derivation. Over a long interval $[0, N]$, the process places approximately N/μ renewal epochs (since consecutive epochs are spaced μ apart on average). The expected number of epochs in $[0, N]$ is therefore $\approx N/\mu$. On the other hand, this expected count equals $\sum_{s=0}^N u_s \approx N \cdot u_\infty$ once u_s has stabilised at its limit u_∞ . Equating, $N \cdot u_\infty \approx N/\mu$, so $u_\infty = 1/\mu$.

A rigorous proof proceeds via the generating function $U(z) = \sum_s u_s z^s = (1 - G(z))^{-1}$, where $G(z) = E[z^X]$ is the increment generating function, and analyses the dominant pole at $z = 1$, where $\text{Res}_{z=1} U(z) = -1/G'(1) = -1/\mu$. See Feller, *An Introduction to Probability Theory*, Vol. I, Ch. XIII, for the complete argument. $\square$

Remark 9.6 (The Recurrence Determines Everything but the Level). The visitation probabilities satisfy

$$u_s = \sum_k P(X = k) u_{s-k},$$

because to reach s , the last step was some k and before it the process stood at $s - k$. Substituting a constant $u_s \equiv u_\infty$ satisfies this recurrence for *any* u_∞ — the recurrence alone does not fix the level. It is the global conservation argument (density of points = $1/\mu$) that pins down $u_\infty = 1/\mu$. This is the discrete shadow of the length-biasing we saw for continuous intervals: both express the same “ $1/\mu$ points per unit length” principle.

Remark 9.7 (Connection to Length-Biasing). The renewal theorem and the length-biased distribution are two views of one phenomenon. The renewal theorem says renewal epochs occur with density $1/\mu$; the length-biasing result says a random sampling point lands in an interval of expected length $E[X^2]/\mu$. Both quantify how a “randomly chosen location” interacts with an i.i.d. sequence, and both feature the harmonic weighting by $1/\mu$. The continuous version (residual life) has the *equilibrium density* $h(y) = (1 - F(y))/\mu$, whose mean is exactly $E[X^2]/(2\mu)$ — the waiting time.

9.4 The Generalised Dice Problem

We now apply renewal theory to a clean discrete puzzle whose answer is initially surprising.

The Last Roll Before Crossing a Threshold

Theorem 9.8. *A fair n -sided die (faces $1, \dots, n$) is rolled repeatedly, accumulating the total, until the total reaches or exceeds a large threshold M . The probability that the final roll shows the value k is*

$$P(\text{final roll} = k) = \frac{2k}{n(n+1)}, \quad k = 1, \dots, n.$$

For the standard die ($n = 6$): $P(\text{final} = k) = k/21$, so the most likely final roll is 6, with probability $6/21 = 2/7 \approx 28.6\%$.

Proof. The accumulated total is a renewal process with increments uniform on $\{1, \dots, n\}$, mean $\mu = (n+1)/2$. Let T be the first index with $S_T \geq M$. The final roll equals k precisely when the total just before the final roll, S_{T-1} , lies in $\{M-k, M-k+1, \dots, M-1\}$ — a set of k states — and the final roll is k :

$$P(\text{final} = k) = \sum_{s=M-k}^{M-1} P(S_{T-1} = s) \cdot \frac{1}{n}.$$

For M large, each state s near M is a renewal epoch with probability $u_s = 1/\mu = 2/(n+1)$ by Theorem 9.5. Hence

$$P(\text{final} = k) = \frac{1}{n} \cdot k \cdot \frac{2}{n+1} = \frac{2k}{n(n+1)}. \quad \square$$

The Dice Problem Is Size-Biasing in Disguise

Why is a large final roll more likely? Because a roll of k can terminate the game from k distinct pre-roll states (namely $M-k, \dots, M-1$), whereas a roll of 1 can terminate only from the single state $M-1$. Large rolls have *more opportunities* to be the terminating roll — exactly the size-biasing principle. The terminating roll is sampled proportionally to its value, just as the captured bus interval is sampled proportionally to its length.

The unifying statement: *when an event is selected by “whatever happens to cross a boundary,” larger crossings are over-represented in proportion to their size.*

Example 9.9 (Exact Values for the Standard Die, $M = 1000$). Direct computation of the visitation probabilities via the recurrence $u_s = \frac{1}{6}(u_{s-1} + \dots + u_{s-6})$ confirms $u_s = 2/7$ exactly for all s in $\{994, \dots, 999\}$ (the convergence to the limit is geometrically fast, and is already exact to machine precision far below $s = 1000$). The resulting distribution:

k	$P(\text{final} = k)$	decimal
1	1/21	0.0476
2	2/21	0.0952
3	3/21	0.1429
4	4/21	0.1905
5	5/21	0.2381
6	6/21	0.2857

9.5 A Gallery of Size-Biasing Paradoxes

The same mechanism — sampling proportional to size — generates a family of paradoxes across wildly different domains. Each is the formula $E[X^2]/E[X]$ in disguise.

Remark 9.10 (The Class Size Paradox). A university reports a mean class size of, say, 30. Yet students, when surveyed, report an average class size far higher — often double. Both are correct. The administrator averages over *classes*; the student averages over *enrolments*. Since a large class contains more students, a randomly chosen student is more likely to sit in a large class. The student-experienced mean is the size-biased mean

$$\frac{E[C^2]}{E[C]} = E[C] (1 + \text{CV}^2),$$

where C is the class-size distribution. With realistic class-size variability, the inflation factor is easily 2 or more.

Remark 9.11 (The Family Size Paradox). In a population where families have on average 2 children, ask a randomly chosen *child* how many children are in their family. The answer averages more than 2, because children are over-represented in large families. The child-experienced family size is again $E[N^2]/E[N]$, where N is the number of children per family. The same arithmetic underlies why your friends seem to come from bigger families than you would guess from birth statistics.

Remark 9.12 (The Friendship Paradox). Perhaps the most striking instance: *on average, your friends have more friends than you do*. In any social network, the mean number of friends of a randomly chosen *person* is $E[d]$ (the mean degree), but the mean number of friends of a randomly chosen *friend* (i.e. a random endpoint of a random edge) is

$$\frac{E[d^2]}{E[d]} = E[d] (1 + CV^2) \geq E[d].$$

The reason: high-degree individuals appear in many people’s friend lists, so they are over-sampled when you look at “a random friend.” For heavy-tailed social networks, where degree variance is enormous, this effect is dramatic — empirically, a random neighbour can have several times the average degree. The friendship paradox has real applications: it underpins efficient strategies for early detection of epidemics (monitor random friends, not random people, because friends are more central) and for identifying influential nodes in a network.

The One Formula Behind Them All

Every paradox in this gallery is the statement

$$\underbrace{\frac{E[X^2]}{E[X]}}_{\text{size-biased mean}} = \underbrace{E[X]}_{\text{naïve mean}} \cdot (1 + CV^2) \geq E[X].$$

Buses (intervals sampled by time), dice (rolls sampled by crossing opportunity), classes (sizes sampled by students), families (sizes sampled by children), networks (degrees sampled by edges): in each case a quantity is sampled with weight proportional to its size, inflating the observed mean by the factor $1 + CV^2$. *Whenever you sample “by encounter” rather than “by unit,” you must correct for size-biasing.*

9.6 Applications to Trading

Size-biasing is not a mathematical curiosity; it is a recurring source of error — and occasionally of edge — in quantitative finance.

Remark 9.13 (Microstructure: The Trade You Observe Is Not Typical). Suppose trades in an instrument arrive with i.i.d. inter-trade times of mean μ . If you begin observing the tape at a random instant, the inter-trade gap you land in has expected length $E[X^2]/\mu$, not μ — you are more likely to arrive during a quiet stretch than during a burst, because quiet stretches occupy more clock time.

This has a concrete consequence for latency-sensitive strategies: *the time-until-next-trade experienced by an order resting in the book is the residual life $E[X^2]/(2\mu)$, which exceeds $\mu/2$ whenever trade timing is variable*. Naively estimating fill times as “half the average inter-trade gap” systematically underestimates how long a passive order will wait. Trade arrivals are famously bursty (high CV), so the underestimate can be severe.

Remark 9.14 (Volatility Clustering and Hawkes Processes). Market events — large prints, volatility spikes, liquidity withdrawals — cluster in time. A standard model is the *Hawkes*

process, a self-exciting point process in which each event raises the intensity of further events. Hawkes processes are, locally, renewal-like, and the same size-biasing correction applies: if you sample the market at a random time, you are disproportionately likely to land inside a cluster, because clusters — being periods of high event density — are over-represented by the size-biasing of *duration weighted by intensity*.

The practical lesson: intensity estimates conditioned on “a randomly chosen observation time” are biased high relative to the unconditional event rate. Estimators of realised event intensity must account for this, or they will overstate the baseline arrival rate of shocks.

Remark 9.15 (Position Duration and Portfolio Sampling). Consider a book of positions, each held for a random duration with distribution f . If a risk manager audits the book at a random instant and asks “how long has this position been open, and how long will it remain open,” the answer is governed by length-biasing: the position *captured* by a random audit has expected total duration $E[X^2]/\mu$, longer than the mean holding period μ .

Long-held positions are over-represented in any snapshot of open positions. A study of “currently open positions” that naively treats them as a representative sample of all positions ever taken will overweight slow, long-duration strategies and underweight fast, short-duration ones — a size-biasing artefact, not a real feature of the trading process.

Remark 9.16 (Time-to-Next-Default from a Random Observation). In credit modelling, defaults in a pool arrive with some inter-arrival law. The time from a randomly chosen observation date to the next default is a residual-life quantity, with expectation $E[X^2]/(2\mu)$ rather than $\mu/2$. For heavy-tailed default clustering (defaults cluster in crises), the residual-life inflation is large: conditional on observing “in a calm period,” the expected wait to the next default is much longer than half the average inter-default time — precisely because calm periods are long and thus over-sampled by a random observation date.

Key Takeaways

Size-Biasing and Renewal Theory

1. **The master formula.** Sampling a quantity X with weight proportional to its size yields the length-biased mean $E[X^2]/E[X] = E[X](1 + CV^2) \geq E[X]$. The inflation is governed by the squared coefficient of variation.
2. **Waiting time paradox.** An observer arriving at a random time waits $E[X^2]/(2\mu)$, not $\mu/2$. For exponential inter-arrivals, this is a full mean μ , not $\mu/2$. The captured interval has mean $E[X^2]/\mu = 2\mu$.
3. **Length-biased density.** The interval captured by a random observer has density $\tilde{f}(x) = xf(x)/\mu$. The residual (waiting) time has the equilibrium density $h(y) = (1 - F(y))/\mu$, with mean $E[X^2]/(2\mu)$.
4. **Renewal theorem.** For integer increments of mean μ , the visitation probability satisfies $u_s \rightarrow 1/\mu$. Renewal epochs occur with density $1/\mu$ per unit length.
5. **Generalised dice problem.** The final roll before crossing a large threshold has distribution $P(\text{final} = k) = 2k/(n(n+1))$. For $n = 6$: $k/21$, most likely 6 at $2/7$. The large roll wins because it has more crossing opportunities — size-biasing again.
6. **The paradox gallery.** Class sizes (sampled by students), family sizes (sampled by children), and network degrees (sampled by edges: the friendship paradox) are all the same $E[X^2]/E[X]$ inflation.
7. **Trading applications.** (a) Passive fill times exceed half the average inter-trade gap. (b) Intensity estimates conditioned on a random observation time are biased high (Hawkes clustering). (c) Snapshots of open positions over-represent long-duration strategies. (d) Residual time to next default from a random date exceeds $\mu/2$.
8. **The one-sentence summary.** Whenever you sample “by encounter” rather than “by unit,” correct for size-biasing, or your estimates will be systematically inflated by $1 + CV^2$.

10 Random Walks and Gambler's Ruin

10.1 The Symmetric Random Walk on $\{0, 1, \dots, N\}$

Consider a particle that starts at position $k \in \{0, 1, \dots, N\}$ and at each time step moves $+1$ or -1 with equal probability $1/2$, independently of the past. The walk continues until the particle hits either 0 or N , at which point it stops.

This is the classical *gambler's ruin* setup: a gambler with initial fortune k plays a fair game against a casino with fortune $N - k$, and we ask for the probability of going broke (hitting 0) versus winning the entire pot (hitting N).

Definition 10.1. Let p_k denote the probability that the walk, starting at position k , hits N before 0 . Let τ_k denote the number of steps until the walk hits the boundary $\{0, N\}$, and let $T_k = E[\tau_k]$.

10.2 Hitting Probability: The Linear Solution

Hitting Probability

Theorem 10.2. For the symmetric random walk on $\{0, 1, \dots, N\}$,

$$p_k = \frac{k}{N}$$

Proof: Recurrence and Boundary Conditions. Condition on the first step. From position k (with $1 \leq k \leq N - 1$), the walk moves to $k + 1$ with probability $1/2$ or to $k - 1$ with probability $1/2$. By the strong Markov property:

$$p_k = \frac{1}{2} p_{k+1} + \frac{1}{2} p_{k-1}.$$

Rearranging,

$$p_{k+1} - p_k = p_k - p_{k-1},$$

which says that the sequence (p_k) has constant first differences—it is an *arithmetic progression*. Equivalently, p_k is a linear function of k :

$$p_k = \alpha + \beta k$$

for some constants α, β .

Applying the boundary conditions $p_0 = 0$ (already at 0 , cannot win) and $p_N = 1$ (already at N , has won) yields $\alpha = 0$ and $\beta = 1/N$. Hence

$$p_k = \frac{k}{N}.$$

□

Why the Hitting Probability Is Linear

The recurrence $p_k = \frac{1}{2} p_{k+1} + \frac{1}{2} p_{k-1}$ is the *discrete harmonic equation*. In the continuous analogue (Brownian motion), the corresponding equation is $\frac{1}{2} u''(x) = 0$, whose general solution is the linear function $u(x) = \alpha + \beta x$.

The boundary conditions “pin” the line: $p_0 = 0$ and $p_N = 1$ give the unique line passing through $(0, 0)$ and $(N, 1)$. Hence $p_k = k/N$.

This is a discrete instance of the maximum principle: harmonic functions are uniquely determined by their boundary values.

Alternative Proof via Optional Stopping. The position S_t of the walk at time t is itself a martingale (each step has mean zero conditional on the past). Apply the Optional Stopping Theorem at the bounded stopping time τ_k :

$$E[S_{\tau_k}] = S_0 = k.$$

But $S_{\tau_k} \in \{0, N\}$, so

$$E[S_{\tau_k}] = 0 \cdot (1 - p_k) + N \cdot p_k = Np_k.$$

Equating: $Np_k = k$, hence $p_k = k/N$. □

Remark 10.3 (Two Equivalent Worldviews). The recurrence proof is constructive: it builds the solution from the local dynamics. The martingale proof is structural: it exploits a conservation law (mean position is preserved). Both yield the same answer because the random walk is *both* locally harmonic and a martingale—these are two sides of the same coin.

10.3 Expected Hitting Time: The Quadratic Solution

Expected Hitting Time

Theorem 10.4. For the symmetric random walk on $\{0, 1, \dots, N\}$ starting at position k ,

$$T_k = E[\tau_k] = k(N - k)$$

Proof: Second-Difference Equation. Condition on the first step. Since each step adds exactly one unit to the time count,

$$T_k = 1 + \frac{1}{2} T_{k+1} + \frac{1}{2} T_{k-1}, \quad 1 \leq k \leq N - 1,$$

with boundary conditions $T_0 = 0$ and $T_N = 0$ (no steps needed if already at the boundary).

Rearranging,

$$T_{k+1} - 2T_k + T_{k-1} = -2.$$

The left-hand side is the *discrete second derivative* of T at k . This is a discrete Poisson equation: $\Delta^2 T = -2$ with zero boundary conditions.

General solution. The equation $T_{k+1} - 2T_k + T_{k-1} = -2$ has:

- Homogeneous solutions: any linear function $T_k^{\text{hom}} = \alpha + \beta k$ satisfies $T_{k+1}^{\text{hom}} - 2T_k^{\text{hom}} + T_{k-1}^{\text{hom}} = 0$.
- A particular solution: try $T_k^{\text{part}} = -k^2$. Then $T_{k+1}^{\text{part}} - 2T_k^{\text{part}} + T_{k-1}^{\text{part}} = -(k+1)^2 + 2k^2 - (k-1)^2 = -2$. ✓

General solution:

$$T_k = -k^2 + \alpha + \beta k.$$

Applying boundary conditions $T_0 = 0$ and $T_N = 0$:

$$\begin{aligned} T_0 = \alpha &= 0, \\ T_N = -N^2 + \beta N &= 0 \implies \beta = N. \end{aligned}$$

Therefore

$$T_k = -k^2 + Nk = k(N - k). \quad \square$$

The Discrete Poisson Equation Interpretation

The equation $T_{k+1} - 2T_k + T_{k-1} = -2$ is the discrete analogue of the continuous Poisson equation $\frac{1}{2}u''(x) = -1$ with $u(0) = u(N) = 0$, whose solution is $u(x) = x(N - x)$.

The -1 on the right-hand side comes from the *constant rate of time accumulation*: each step adds exactly one unit of time. In the continuous case, this would be the “source term” for hitting time in Brownian motion.

The fact that T_k is quadratic in k (whereas p_k was linear) reflects that hitting times accumulate *twice as fast* as displacements in random walks—a quantitative version of the diffusive scaling $T \sim k^2$ for symmetric walks.

Example 10.5 (Sanity Checks). • $N = 2, k = 1$: $T_1 = 1 \cdot 1 = 1$. Starting at the midpoint with $N = 2$, one step always hits a boundary.

- $N = 10, k = 5$: $T_5 = 5 \cdot 5 = 25$ steps from the midpoint of a length-10 interval.
- $N = 100, k = 50$: $T_{50} = 2500$, growing as $N^2/4$ from the midpoint—diffusive scaling.

10.4 Conditional Hitting Time

Theorem 10.6 (Conditional Hitting Time). *For the symmetric random walk on $\{0, 1, \dots, N\}$ starting at k , the expected time to hit N conditional on hitting N before 0 is*

$$E[\tau_k \mid \text{hit } N \text{ first}] = \frac{1}{3}(N^2 - k^2).$$

Proof Sketch. Let $V_k = E[\tau_k \mid \mathcal{H}_{\text{hit } N}]$ be the “unnormalized” conditional expectation. Condition on the first step:

$$V_k = \frac{1}{2}(V_{k+1} + p_{k+1}) + \frac{1}{2}(V_{k-1} + p_{k-1}),$$

where the $+p$ terms come from the one additional step taken on the way. Using $p_k = k/N$ and simplifying yields a second-order linear recurrence with cubic particular solution. After solving with boundary conditions $V_0 = 0$ and $V_N = 0$:

$$V_k = \frac{k(N^2 - k^2)}{3N},$$

and dividing by $p_k = k/N$:

$$E[\tau_k \mid \text{hit } N] = \frac{V_k}{p_k} = \frac{N^2 - k^2}{3}. \quad \square$$

Remark 10.7 (Surprising Asymmetry). Note that $E[\tau_k \mid \text{hit } N] = (N^2 - k^2)/3$ depends on k through $N^2 - k^2$, not through $k(N - k)$ as the unconditional time does. In particular, starting close to 0 but conditioning on hitting N first gives a much *longer* expected time than the unconditional case, because the conditioning excludes the quick paths that go directly to 0.

10.5 The Asymmetric Random Walk

Now suppose each step moves $+1$ with probability p and -1 with probability $q = 1 - p$, where $p \neq 1/2$. Let $r = q/p$ denote the ratio of probabilities (with $r \neq 1$).

Asymmetric Hitting Probability

Theorem 10.8. *For the asymmetric random walk on $\{0, 1, \dots, N\}$ with step distribution $P(+1) = p, P(-1) = q = 1 - p$, the probability of hitting N before 0 starting from k is*

$$p_k = \frac{1 - r^k}{1 - r^N}, \quad r = \frac{q}{p}.$$

Proof: Geometric Recurrence. Condition on the first step:

$$p_k = p \cdot p_{k+1} + q \cdot p_{k-1}, \quad 1 \leq k \leq N-1.$$

Rearranging,

$$p(p_{k+1} - p_k) = q(p_k - p_{k-1}).$$

Setting $d_k = p_{k+1} - p_k$, this gives the geometric recurrence

$$d_k = r \cdot d_{k-1}, \quad r = q/p,$$

hence $d_k = r^k d_0$.

Therefore

$$p_k = p_0 + \sum_{j=0}^{k-1} d_j = 0 + d_0 \sum_{j=0}^{k-1} r^j = d_0 \cdot \frac{1 - r^k}{1 - r}.$$

Applying $p_N = 1$:

$$1 = d_0 \cdot \frac{1 - r^N}{1 - r} \implies d_0 = \frac{1 - r}{1 - r^N}.$$

Substituting back:

$$p_k = \frac{1 - r^k}{1 - r^N}.$$

□

Why Geometric, Not Linear?

In the symmetric case, the recurrence preserves *differences* (arithmetic progression). In the asymmetric case, the recurrence preserves *ratios of differences* (geometric progression). Equivalently: in the symmetric case, the position S_t is a martingale. In the asymmetric case, the position is no longer a martingale, but the *geometric process* $M_t = r^{S_t}$ is a martingale. This explains why the answer involves r^k .

When $p > 1/2$ ($r < 1$), the walk drifts toward N , and $p_k \rightarrow 1$ as $N \rightarrow \infty$ for fixed k .

When $p < 1/2$ ($r > 1$), the walk drifts toward 0, and $p_k \rightarrow 0$ as $N \rightarrow \infty$ for fixed k (the walk almost surely reaches 0 first).

Proof via Geometric Martingale. Define $M_t = r^{S_t}$. Then

$$E[M_{t+1} | \mathcal{F}_t] = p \cdot r^{S_t+1} + q \cdot r^{S_t-1} = r^{S_t}(pr + qr^{-1}).$$

Computing $pr + qr^{-1} = p \cdot \frac{q}{p} + q \cdot \frac{p}{q} = q + p = 1$, we see that $M_t = r^{S_t}$ is a martingale.

Apply optional stopping at τ_k :

$$E[M_{\tau_k}] = M_0 = r^k.$$

Since $S_{\tau_k} \in \{0, N\}$,

$$E[M_{\tau_k}] = (1 - p_k) \cdot r^0 + p_k \cdot r^N = (1 - p_k) + p_k r^N.$$

Setting equal: $(1 - p_k) + p_k r^N = r^k$, so

$$p_k = \frac{1 - r^k}{1 - r^N}.$$

□

Asymmetric Expected Hitting Time

Theorem 10.9. For the asymmetric random walk with $p \neq 1/2$ and $r = q/p$,

$$T_k = \frac{1}{q - p} \left(k - N \cdot \frac{1 - r^k}{1 - r^N} \right).$$

Proof. The recurrence is

$$T_k = 1 + pT_{k+1} + qT_{k-1}, \quad T_0 = T_N = 0.$$

The corresponding homogeneous equation $pT_{k+1} - T_k + qT_{k-1} = 0$ has characteristic equation $p\lambda^2 - \lambda + q = 0$ with roots $\lambda = 1, r$. So the homogeneous solution is $\alpha + \beta r^k$.

For a particular solution, try $T_k^{\text{part}} = k/(q-p)$. Substituting into $T_k = 1 + pT_{k+1} + qT_{k-1}$:

$$\frac{k}{q-p} = 1 + p \cdot \frac{k+1}{q-p} + q \cdot \frac{k-1}{q-p} = 1 + \frac{(p+q)k + (p-q)}{q-p} = 1 + \frac{k}{q-p} - 1 = \frac{k}{q-p}. \quad \checkmark$$

Therefore the general solution is

$$T_k = \frac{k}{q-p} + \alpha + \beta r^k.$$

Boundary conditions:

$$T_0 = 0 : \quad \alpha + \beta = 0,$$

$$T_N = 0 : \quad \frac{N}{q-p} + \alpha + \beta r^N = 0.$$

Solving:

$$\beta = -\alpha, \quad \frac{N}{q-p} + \alpha(1 - r^N) = 0, \quad \alpha = -\frac{N}{(q-p)(1-r^N)}.$$

Substituting back:

$$T_k = \frac{k}{q-p} - \frac{N}{(q-p)(1-r^N)}(1 - r^k) = \frac{1}{q-p} \left(k - N \frac{1-r^k}{1-r^N} \right). \quad \square$$

Remark 10.10 (Limit as $p \rightarrow 1/2$). The formula appears singular at $p = 1/2$ (division by $q - p = 0$), but taking the limit using L'Hôpital's rule on the ratio $(1 - r^k)/(1 - r^N)$ as $r \rightarrow 1$ recovers the symmetric formula $T_k = k(N - k)$. This is reassuring: the symmetric case is the limit of the asymmetric one.

10.6 Worked Examples

Example 10.11 (The Drunkard's Walk Home). A drunkard stands at position k between his home (position 0) and the river (position N). He stumbles left or right with equal probability at each step. How worried should his family be?

By Theorem 10.2, the probability he falls in the river is k/N . If he starts halfway ($k = N/2$), this is exactly $1/2$. His expected walking time is $T_k = k(N - k)$ steps—quadratic in the distances.

Example 10.12 (Gambler's Ruin Against a House Edge). A gambler with \$100 plays a game with $p = 0.49$ per round against a casino with \$10,000 ($N = 10,100$). Then $r = 51/49 \approx 1.0408$, and

$$p_{100} = \frac{1 - r^{100}}{1 - r^{10100}} \approx \frac{1 - 56.4}{1 - \text{enormous}} \approx 0.$$

Essentially zero chance of winning. Even a tiny edge, compounded over many rounds, drives the gambler's ruin probability to 1.

Example 10.13 (Coin-Flip Until 3 Heads Ahead). Flip a fair coin and track $S_t = \#H - \#T$. Expected flips until $|S_t| = 3$?

Translating to standard form ($k = 3, N = 6$): $T_3 = 3 \cdot 3 = 9$.

Example 10.14 (Conditional Times). For $k = 1, N = 6$ (start at 1, hit ± 5):

$$T_1 = 1 \cdot 5 = 5 \quad (\text{unconditional}),$$

$$E[\tau_1 \mid \text{hit 6 first}] = \frac{36-1}{3} = \frac{35}{3} \approx 11.67.$$

Conditioning on the unlikely event makes the expected time much longer.

Random Walks: Essential Facts

1. **Symmetric hitting probability.** $p_k = k/N$ (linear, harmonic equation).
2. **Symmetric hitting time.** $T_k = k(N - k)$ (quadratic, discrete Poisson equation).
3. **Asymmetric hitting probability.** $p_k = (1 - r^k)/(1 - r^N)$, with $r = q/p$ (geometric).
4. **Connection to martingales.** In the symmetric case, S_t itself is a martingale. In the asymmetric case, r^{S_t} is a martingale.
5. **Two views of the same problem.** Recurrence + boundary conditions $\Leftrightarrow$ martingale + optional stopping. Use whichever is faster.
6. **Continuous analogues.** $\frac{1}{2}u'' = 0$ for hitting probability; $\frac{1}{2}u'' = -1$ for hitting time. These are the Brownian motion versions with the same functional form as their discrete counterparts.

11 The Reflection Principle and the Ballot Problem

Before diving into the individual exercises of this Part, we establish a single combinatorial tool that recurs across several of them: the *reflection principle*. It is the master device for counting lattice paths constrained to stay on one side of a barrier, and it is the engine behind the Catalan first-passage distribution, the ballot problem, the arcsine law, and the casino argument for Guibas–Odlyzko pattern-hitting times. Establishing it once in a canonical form makes every subsequent application a matter of citing a lemma rather than reproving a bijection.

11.1 Lattice Paths and the Setup

Definition 11.1 (Lattice Path). A *lattice path* of length n from a to b is a sequence $(S_0, S_1, \dots, S_n)$ with $S_0 = a$, $S_n = b$, and $S_{i+1} - S_i \in \{+1, -1\}$ for each i . Each such path corresponds to a sequence of n symbols $X_1, \dots, X_n \in \{+1, -1\}$ via $S_k = a + \sum_{i \leq k} X_i$.

Proposition 11.2 (Path Count Between Endpoints). *The number of lattice paths of length n from a to b is $\binom{n}{(n+b-a)/2}$ when $n + b - a$ is even and $|b - a| \leq n$, and zero otherwise.*

Proof. A path from a to b in n steps has u up-steps and $d = n - u$ down-steps with $u - d = b - a$, hence $u = (n + b - a)/2$. The path is determined by the positions of the up-steps among the n steps, giving $\binom{n}{u}$ choices. Parity and bound constraints are necessary for u to be a non-negative integer at most n . $\square$

11.2 The Reflection Principle

Reflection Principle

Theorem 11.3. *Fix integers a, b and an integer barrier c with $c > a$ and $c > b$ (the barrier lies strictly above both endpoints). The number of lattice paths of length n from a to b that touch or cross the level c at some intermediate step equals the total number of lattice paths of length n from $2c - a$ to b :*

$$\#\{\text{paths } a \rightarrow b \text{ touching } c\} = \#\{\text{paths } (2c - a) \rightarrow b\}.$$

By symmetry, the same statement holds with all inequalities reversed (barrier strictly below both endpoints, reflecting a to $2c - a$ below).

Proof: An Explicit Bijection. We construct an explicit bijection between paths in the two sets. Let

$$\mathcal{P}_{\text{touch}} = \{\text{paths } a \rightarrow b \text{ of length } n \text{ touching } c\}, \quad \mathcal{P}_{\text{reflected}} = \{\text{paths } (2c - a) \rightarrow b \text{ of length } n\}.$$

The map $\Phi : \mathcal{P}_{\text{touch}} \rightarrow \mathcal{P}_{\text{reflected}}$. Given a path $\pi \in \mathcal{P}_{\text{touch}}$, let $\tau(\pi)$ be the *first* time step at which the path touches the barrier c (it exists by hypothesis). Define $\Phi(\pi)$ as the path obtained by reflecting the portion of π before τ across the horizontal line at height c :

$$\Phi(\pi)_k = \begin{cases} 2c - \pi_k & \text{if } k \leq \tau(\pi), \\ \pi_k & \text{if } k > \tau(\pi). \end{cases}$$

Geometric picture: take a mirror placed at height c and reflect the first part of the path (from the start up to the first touch) across it. The reflected portion starts at $2c - a$ and ends at $2c - c = c$, which is exactly where the unreflected portion begins. The two halves glue into a single valid lattice path because the reflection preserves ± 1 step sizes and the join point matches.

The image $\Phi(\pi)$ starts at $2c - a$ (the reflection of a) and ends at $\pi_n = b$. Its length is still n . Hence $\Phi(\pi) \in \mathcal{P}_{\text{reflected}}$.

The inverse map $\Psi : \mathcal{P}_{\text{reflected}} \rightarrow \mathcal{P}_{\text{touch}}$. Given any path ρ from $2c - a$ to b of length n : since $2c - a > c$ (because $c > a$) and $b < c$, the path ρ starts *above* the barrier and ends *below* it. By continuity of ± 1 steps, ρ must cross the level c at some step. Let $\tau'(\rho)$ be the first time ρ touches c . Define

$$\Psi(\rho)_k = \begin{cases} 2c - \rho_k & \text{if } k \leq \tau'(\rho), \\ \rho_k & \text{if } k > \tau'(\rho). \end{cases}$$

The image $\Psi(\rho)$ starts at a , ends at b , and touches c at step $\tau'(\rho)$. Hence $\Psi(\rho) \in \mathcal{P}_{\text{touch}}$.

Φ and Ψ are mutual inverses. Both maps reflect the path before the first touch of c , and the first-touch time is preserved under reflection of that initial segment (since reflection across c maps crossings to crossings). Hence $\Phi \circ \Psi$ and $\Psi \circ \Phi$ are both the identity.

Therefore $|\mathcal{P}_{\text{touch}}| = |\mathcal{P}_{\text{reflected}}|$. □

The One-Line Summary

Reflecting a path before its first contact with the barrier swaps “paths from a that touch c ” with “paths from $2c - a$.” The bijection works because the first touch is well-defined, the reflection is an involution on that initial segment, and the rest of the path is unchanged.

The power of the principle is that it converts a constrained count (paths from a that touch a barrier) into an unconstrained count (paths from a reflected starting point) that is computed by an ordinary binomial coefficient. Every application that follows is an instance of this single move.

Corollary 11.4 (Paths Avoiding a Barrier). *The number of lattice paths from a to b of length n that never touch the level c (with $c > a, b$) is*

$$\binom{n}{(n+b-a)/2} - \binom{n}{(n+b-(2c-a))/2} = \binom{n}{(n+b-a)/2} - \binom{n}{(n+b+a-2c)/2}.$$

Proof. Subtract the touching paths (counted by Theorem 11.3) from the total paths (counted by Proposition 11.2). □

11.3 The Ballot Problem

The most celebrated immediate consequence of the reflection principle is Bertrand’s *ballot problem* (1887), whose combinatorial proof by reflection is due to Désiré André the same year.

The Question

In an election, candidate A receives p votes and candidate B receives q votes, with $p > q$. The $n = p + q$ ballots are counted one at a time in a uniformly random order. What is the probability that A is *strictly* ahead of B throughout the entire count?

Bertrand’s Ballot Problem

Theorem 11.5. *The probability that candidate A leads strictly throughout the count is*

$$P(A \text{ always strictly ahead}) = \frac{p-q}{p+q}.$$

Proof. Encode each ballot for A as +1 and each ballot for B as -1, and let S_k be the running tally after the first k ballots. The total count is a uniformly random lattice path of length $n = p + q$ from $S_0 = 0$ to $S_n = p - q$.

A is strictly ahead throughout the count if and only if $S_k > 0$ for all $k \geq 1$. The first step must therefore be +1 (otherwise $S_1 = -1 < 0$). Conditional on the first step being +1, we need the remaining path $(S_1, S_2, \dots, S_n)$, which starts at $S_1 = +1$ and ends at $S_n = p - q$, to stay *strictly positive* throughout, i.e. never to touch the level 0.

By Corollary 11.4 applied with starting point $a = 1$, endpoint $b = p - q$, length $n - 1$, and barrier $c = 0$ (below both endpoints, using the symmetric form of the reflection principle):

$$\#\{\text{paths } 1 \rightarrow (p - q) \text{ never touching } 0\} = \binom{n-1}{p-1} - \binom{n-1}{p}.$$

The first term counts all paths from 1 to $p - q$ of length $n - 1$ (which have $p - 1$ up-steps and q down-steps). The second counts those touching 0, obtained by reflecting the start 1 to -1: paths from -1 to $p - q$ of length $n - 1$ have p up-steps and $q - 1$ down-steps.

The total number of orderings of p A-ballots and q B-ballots is $\binom{n}{p}$. The number of orderings starting with an A and having strictly positive tally afterwards is therefore $\binom{n-1}{p-1} - \binom{n-1}{p}$. The probability we want is

$$P = \frac{\binom{n-1}{p-1} - \binom{n-1}{p}}{\binom{n}{p}}.$$

Using $\binom{n-1}{p-1} = \frac{p}{n} \binom{n}{p}$ and $\binom{n-1}{p} = \frac{q}{n} \binom{n}{p}$:

$$P = \frac{(p/n) - (q/n)}{1} \binom{n}{p} / \binom{n}{p} = \frac{p - q}{p + q}. \quad \square$$

Example 11.6 (Numerical Cases). • $p = 3, q = 2$: probability = $1/5 = 20\%$.

- $p = 7, q = 3$: probability = $4/10 = 40\%$.
- $p = 100, q = 1$: probability = $99/101 \approx 98\%$.

Even when A wins overwhelmingly, the probability of *leading throughout* is bounded by $(p - q)/(p + q)$, not by 1. The proportion of orderings where A leads continuously equals the relative margin of victory.

Corollary 11.7 (Weak Lead Version). *The probability that A is weakly ahead (i.e. $S_k \geq 0$ for all k , allowing ties) is*

$$P(\text{A always weakly ahead}) = \frac{p - q + 1}{p + 1}.$$

Sketch. Apply the same reflection argument with the barrier shifted by one unit. Equivalently, conditioning on first vote being A is no longer required; the count is allowed to touch 0 but not go below. $\square$

11.4 First-Passage Probabilities for the Random Walk

The reflection principle gives, almost for free, the distribution of the first time a simple symmetric random walk reaches a target level. This result is used directly in the Catalan/ $\pi/4$ first-crossing exercise later in this Part.

Corollary 11.8 (First-Passage to +1). *For a simple symmetric random walk (S_n) started at $S_0 = 0$, the probability that the first passage to +1 occurs at time $n = 2k + 1$ is*

$$P(\tau_{+1} = 2k + 1) = \frac{C_k}{2^{2k+1}}, \quad C_k = \frac{1}{k+1} \binom{2k}{k}.$$

Proof. The first passage to $+1$ at step $2k + 1$ requires the path to start at 0, end at $+1$ at step $2k + 1$, and stay ≤ 0 at all earlier steps. Conditioning on the first step:

- If the first step is $+1$ (probability $\frac{1}{2}$), the walk is at $+1$ immediately and $\tau_{+1} = 1$, so this corresponds to $k = 0$ and contributes $\frac{1}{2} = C_0/2^1$.
- If the first step is -1 (probability $\frac{1}{2}$), the walk is at -1 and must reach $+1$ in $2k$ further steps without first revisiting $+1$. We count paths from -1 to $+1$ of length $2k$ that stay ≤ 0 except at the final step.

For the second case, by Corollary 11.4 applied with $a = -1$, $b = +1$, length $2k$, and barrier $c = 0$ (the path must stay strictly below 0 until the final arrival at $+1$; equivalently, the path from -1 to 0 of length $2k - 1$ stays ≤ -1 except at the very end): the count of such restricted paths equals

$$\frac{1}{2k+1} \binom{2k+1}{k}.$$

Combining with the probability $\frac{1}{2} \cdot 2^{-2k}$ of the specific tail and using the identity $\frac{1}{2k+1} \binom{2k+1}{k} = C_k$,

$$P(\tau_{+1} = 2k + 1) = \frac{C_k}{2^{2k+1}}. \quad \square$$

Remark 11.9 (Recurrence and Infinite Mean). Summing using the Catalan generating function $\sum_k C_k x^k = (1 - \sqrt{1 - 4x})/(2x)$ at $x = 1/4$:

$$\sum_{k=0}^{\infty} \frac{C_k}{2^{2k+1}} = \frac{1}{2} \sum_{k=0}^{\infty} C_k 4^{-k} = \frac{1}{2} \cdot 2 = 1,$$

confirming that the walk reaches $+1$ almost surely. However, $\sum_k (2k + 1) \cdot C_k / 2^{2k+1}$ diverges (the tail decays as $k^{-3/2}$, so k times the tail is not summable), so $E[\tau_{+1}] = \infty$. This is the classic “recurrent but with infinite mean hitting time” phenomenon of one-dimensional simple random walks.

11.5 Connection to Brownian Motion

Remark 11.10 (The Continuous Analogue). The reflection principle has a continuous-time counterpart for Brownian motion (Proposition ?? in Part II): for standard Brownian motion $(W_t)_{t \geq 0}$ and any $a > 0$,

$$P\left(\max_{0 \leq s \leq t} W_s \geq a\right) = 2 P(W_t \geq a).$$

The proof is structurally identical to the discrete case: the path of W that first hits level a at some time $\tau \leq t$ is reflected across a for $s \in [\tau, t]$, producing a path that ends at $2a - W_t$. This bijection between $\{W_t \geq a\}$ and the reflected event yields the factor of 2.

The discrete reflection principle above is the natural finite counterpart, and the two results converge under the diffusive scaling $S_{[nt]}/\sqrt{n} \Rightarrow W_t$ (Donsker’s invariance principle). When we apply the discrete version to first-passage calculations of simple random walks, we are working with the same combinatorial content that, in the continuous limit, becomes the Brownian reflection identity.

Key Takeaways

Reflection Principle Essentials

1. **Statement.** The number of paths from a to b touching a barrier c (with $c > a, b$) equals the number of unconstrained paths from $2c - a$ to b .
2. **Proof mechanism.** An explicit bijection: reflect the path before its first contact with the barrier across the barrier. The bijection is an involution.
3. **Ballot problem.** $P(\text{A always strictly ahead}) = (p - q)/(p + q)$ when A gets p votes and B gets $q < p$, with ballots counted in random order.
4. **First-passage distribution.** $P(\tau_{+1} = 2k + 1) = C_k/2^{2k+1}$ for the symmetric walk, where C_k is the k -th Catalan number.
5. **Recurrence with infinite mean.** The simple symmetric walk hits $+1$ a.s. but with $E[\tau_{+1}] = \infty$.
6. **Continuous analogue.** Brownian reflection: $P(\max_{s \leq t} W_s \geq a) = 2P(W_t \geq a)$. The discrete and continuous statements share the same combinatorial DNA, linked by Donsker's theorem.
7. **What it is for.** Reflection is the universal tool for converting barrier-constrained path counts into ordinary binomial counts. Whenever a problem involves “paths that stay above/below some level until time n ,” reach for reflection first.

12 First Crossing of the Majority: Catalan Numbers and $\pi/4$

12.1 Problem Statement

A fair coin is tossed repeatedly. After each toss, compute the running proportion of heads

$$r_n = \frac{H_n}{n},$$

where H_n is the number of heads and n the total number of tosses so far. Continue tossing as long as $r_n \leq \frac{1}{2}$; in particular, if $r_n = \frac{1}{2}$ exactly, you must continue. Stop at the first instant $r_n > \frac{1}{2}$, and record the final proportion $a = r_n$ at that stopping time.

Goal. Compute $E[a]$, the expected final proportion over many independent rounds.

12.2 Reduction to a First-Passage Problem

The first observation is that the proportion condition is really a statement about a *simple symmetric random walk*. Let $T_n = n - H_n$ be the number of tails and define the imbalance

$$S_n = H_n - T_n = 2H_n - n.$$

Each toss changes S_n by $+1$ (heads) or -1 (tails), independently and with probability $\frac{1}{2}$ each, so (S_n) is exactly a simple symmetric random walk started at $S_0 = 0$.

Proposition 12.1 (Translation of the Stopping Rule). *The running proportion satisfies*

$$r_n > \frac{1}{2} \iff S_n > 0, \quad r_n = \frac{1}{2} \iff S_n = 0.$$

Consequently, the game stops at the first time the walk (S_n) becomes strictly positive. Since the steps are ± 1 , the first strictly positive value of S_n is necessarily $+1$. Hence the stopping time is

$$\tau = \min\{n \geq 1 : S_n = +1\},$$

the *first-passage time to $+1$* .

Proof. From $S_n = 2H_n - n$ we get $r_n = H_n/n = \frac{1}{2} + S_n/(2n)$, so $r_n > \frac{1}{2} \iff S_n > 0$ and $r_n = \frac{1}{2} \iff S_n = 0$. A walk with ± 1 steps cannot jump over $+1$, so the first time it is positive it equals exactly $+1$. $\square$

Why This Reduction Is the Whole Game

The seemingly analytic condition “proportion exceeds one half” is, after the change of variables $S_n = 2H_n - n$, the purely combinatorial condition “the walk reaches $+1$.” Everything that follows—Catalan numbers, the value $\pi/4$, the convergence analysis—flows from this single translation. Recognising a proportion threshold as a random-walk level crossing is a recurring move: the same idea reappears in the ballot problem, in the red–black card stopping problem, and in the Guibas–Odlyzko casino argument.

12.3 The Shape of the Stopping State

Lemma 12.2 (Where the Walk Stops). *If the first passage to $+1$ occurs at time $\tau = n$, then n is odd, say $n = 2k + 1$ with $k \geq 0$, and at that moment*

$$H_n = k + 1, \quad T_n = k, \quad a = r_n = \frac{k + 1}{2k + 1}.$$

Proof. $S_n = +1$ means $H_n - T_n = 1$. Together with $H_n + T_n = n$ this gives $H_n = (n+1)/2$ and $T_n = (n-1)/2$, which forces n odd. Writing $n = 2k+1$ yields $H_n = k+1$, $T_n = k$, and $a = H_n/n = (k+1)/(2k+1)$. $\square$

Remark 12.3 (The First Few Stopping Values).

$$a_0 = \frac{1}{1} = 1, \quad a_1 = \frac{2}{3}, \quad a_2 = \frac{3}{5}, \quad a_3 = \frac{4}{7}, \quad \dots, \quad a_k = \frac{k+1}{2k+1} \xrightarrow[k \rightarrow \infty]{} \frac{1}{2}.$$

The very first toss already ends the game with probability $\frac{1}{2}$ (a head on toss 1 gives $a = 1$). Longer games end with a ever closer to $\frac{1}{2}$ from above.

12.4 The Analytic Toolbox: Two Generating Functions

Before computing the first-passage law we assemble the two power series that drive this section — and not only this one: the arcsine-law section reuses both, so they are proved here once, in full.

Definition 12.4 (Catalan Numbers). The k -th *Catalan number* is

$$C_k = \frac{1}{k+1} \binom{2k}{k} = \binom{2k}{k} - \binom{2k}{k+1},$$

with $C_0 = 1$, $C_1 = 1$, $C_2 = 2$, $C_3 = 5$, $C_4 = 14, \dots$

Lemma 12.5 (Central Binomial Generating Function).

$$C(x) := \sum_{n \geq 0} \binom{2n}{n} x^n = \frac{1}{\sqrt{1-4x}}, \quad |x| < \frac{1}{4}.$$

Proof. Newton's generalised binomial theorem gives, for any real α , $(1-z)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} (-z)^n$ with $\binom{\alpha}{n} = \alpha(\alpha-1) \cdots (\alpha-n+1)/n!$. Take $\alpha = -\frac{1}{2}$ and compute the coefficient:

$$\binom{-1/2}{n} = \frac{(-\frac{1}{2})(-\frac{3}{2}) \cdots (-\frac{2n-1}{2})}{n!} = (-1)^n \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n n!} = (-1)^n \frac{(2n-1)!!}{2^n n!}.$$

Multiplying and dividing the double factorial by the even numbers, $(2n-1)!! = (2n)!/(2^n n!)$, so

$$\binom{-1/2}{n} (-1)^n = \frac{(2n)!}{2^n n! \cdot 2^n n!} = \frac{1}{4^n} \binom{2n}{n}.$$

Therefore $(1-z)^{-1/2} = \sum_n \binom{2n}{n} (z/4)^n$, and setting $z = 4x$ gives the claim. *Numerical check:* $\binom{-1/2}{n} (-1)^n = 1, \frac{1}{2}, \frac{3}{8}, \frac{5}{16}, \frac{35}{128}$ for $n = 0, \dots, 4$, matching $\binom{2n}{n}/4^n$ exactly. $\square$

Lemma 12.6 (Catalan Generating Function).

$$\mathcal{C}(x) := \sum_{m \geq 0} C_m x^m = \frac{1 - \sqrt{1-4x}}{2x}, \quad |x| \leq \frac{1}{4}.$$

Proof. The first-return decomposition of non-negative (Dyck) paths gives the quadratic relation $\mathcal{C} = 1 + x\mathcal{C}^2$: a path is either empty, or an up-step, a shifted Dyck path, a down-step, and another Dyck path. Solving for $\mathcal{C}$ and choosing the branch analytic at $x = 0$ (the “−” root, since $\mathcal{C}(0) = C_0 = 1$) yields the stated form. $\square$

12.5 The First-Passage Distribution: Catalan Numbers Appear

First-Passage Probabilities

Theorem 12.7. *For the simple symmetric random walk, the probability that the first passage to +1 occurs exactly at time $n = 2k + 1$ is*

$$p_k = P(\tau = 2k + 1) = \frac{C_k}{2^{2k+1}}.$$

Proof. Let $\Phi(z) = E[z^\tau] = \sum_{n \geq 1} P(\tau = n)z^n$ be the generating function of the first-passage time, and condition on the first step.

If the first step is a head (probability $\frac{1}{2}$), then $\tau = 1$: contribution $\frac{z}{2}$.

If it is a tail (probability $\frac{1}{2}$), the walk sits at -1 and must reach $+1$. By the strong Markov property this splits into two *independent* stages, each a fresh copy of the original problem: first reach 0 from -1 (a first passage one level up), then reach $+1$ from 0 (another). Each stage contributes an independent copy of τ , so this branch contributes $\frac{z}{2} \Phi(z)^2$ — one factor z for the initial tail, one Φ per stage. Hence

$$\Phi(z) = \frac{z}{2} + \frac{z}{2} \Phi(z)^2,$$

the same quadratic as in Lemma 12.6. Solving and taking the branch with $\Phi(0) = 0$,

$$\Phi(z) = \frac{1 - \sqrt{1 - z^2}}{z} = \frac{z}{2} \cdot \frac{1 - \sqrt{1 - 4(z^2/4)}}{2(z^2/4)} = \frac{z}{2} \mathcal{C}\left(\frac{z^2}{4}\right) = \sum_{k \geq 0} \frac{C_k}{2^{2k+1}} z^{2k+1},$$

using Lemma 12.6 in the last two steps. Reading off the coefficient of z^{2k+1} gives the theorem. *Numerical check:* the series begins $\frac{z}{2} + \frac{z^3}{8} + \frac{z^5}{16} + \frac{5z^7}{128} + \frac{7z^9}{256}$, matching $C_k/2^{2k+1}$ for $C_k = 1, 1, 2, 5, 14$. $\square$

Remark 12.8 (The Combinatorial Reading). The same probabilities admit a path-counting derivation: the number of first-passage paths to $+1$ of length $2k + 1$ is

$$\frac{1}{2k+1} \binom{2k+1}{k},$$

by the ballot/cycle argument of the reflection section (Theorem 11.3), and this equals C_k by direct expansion:

$$\frac{1}{2k+1} \binom{2k+1}{k} = \frac{(2k+1)!}{(2k+1)k!(k+1)!} = \frac{(2k)!}{k!(k+1)!} = \frac{1}{k+1} \binom{2k}{k} = C_k.$$

The renewal proof above and the counting proof agree, as they must; the first generalises more readily (it is the one that survives for a biased coin below), while the second explains *why* a Catalan number: it is a binomial count with a barrier constraint imposed.

Why Catalan and Not Just Binomial

A binomial coefficient $\binom{2k+1}{k}$ counts *all* paths ending at $+1$. The Catalan number is what remains after we impose the **first-passage constraint**: the walk may not have reached $+1$ before the final step. The reflection principle removes exactly the “early-crossing” paths, and the surviving count is the Catalan number.

This is the same combinatorial object that counts balanced parentheses, monotone lattice paths below the diagonal, and binary trees. Its appearance here is not a coincidence: all of these are “stay-below-a-barrier-until-the-end” counts, and that is precisely the first-passage condition.

Proposition 12.9 (The Game Ends Almost Surely). *The stopping probabilities sum to one:*

$$\sum_{k=0}^{\infty} p_k = \sum_{k=0}^{\infty} \frac{C_k}{2^{2k+1}} = 1.$$

Proof. Immediate from the proof of Theorem 12.7: setting $z = 1$ in $\Phi(z) = (1 - \sqrt{1 - z^2})/z$ gives $\Phi(1) = 1$. Equivalently, by Lemma 12.6 at $x = \frac{1}{4}$: $\sum_k C_k 4^{-k} = \mathcal{C}(\frac{1}{4}) = 2$, and dividing by 2 gives the claim. $\square$

Remark 12.10 (A Recurrent Walk with Infinite Patience). That the probabilities sum to 1 is the statement that the symmetric walk hits +1 in finite time almost surely (recurrence of the one-dimensional simple random walk). Note, however, that $E[\tau] = \infty$: the same walk that *certainly* reaches +1 takes, on average, infinitely long to do so. This is the classic warning attached to the Optional Stopping Theorem (see the OST failure example in Part II), and it is the reason we must compute $E[a]$ directly rather than through a naive martingale shortcut.

12.6 Computing the Expected Final Proportion

Main Result

Theorem 12.11. *The expected final proportion is*

$$E[a] = \sum_{k=0}^{\infty} \frac{k+1}{2k+1} \cdot \frac{C_k}{2^{2k+1}} = \frac{\pi}{4} \approx 0.785398.$$

Proof. By Lemma 12.2 and Theorem 12.7,

$$E[a] = \sum_{k=0}^{\infty} a_k p_k = \sum_{k=0}^{\infty} \frac{k+1}{2k+1} \cdot \frac{C_k}{2^{2k+1}}.$$

Step 1: the $(k+1)$'s cancel. Using $C_k = \frac{1}{k+1} \binom{2k}{k}$,

$$\frac{k+1}{2k+1} \cdot \frac{C_k}{2^{2k+1}} = \frac{1}{2k+1} \binom{2k}{k} \frac{1}{2^{2k+1}},$$

so that

$$E[a] = \frac{1}{2} \sum_{k=0}^{\infty} \frac{1}{2k+1} \binom{2k}{k} \frac{1}{4^k}.$$

This cancellation is the pivotal simplification: the Catalan structure of the probabilities and the shape of the payoff a_k conspire to leave a pure central-binomial series with a $1/(2k+1)$ weight — and a $1/(2k+1)$ weight is the signature of a term-by-term *integration*.

Step 2: integrate the central binomial series. By Lemma 12.5 with $x = t^2$,

$$\frac{1}{\sqrt{1-4t^2}} = \sum_{k=0}^{\infty} \binom{2k}{k} t^{2k},$$

and integrating term by term from 0 to u (justified by uniform convergence on $|t| \leq u < \frac{1}{2}$),

$$\int_0^u \frac{dt}{\sqrt{1-4t^2}} = \sum_{k=0}^{\infty} \binom{2k}{k} \frac{u^{2k+1}}{2k+1}.$$

The left side is elementary: substituting $t = \frac{1}{2} \sin \theta$,

$$\int_0^u \frac{dt}{\sqrt{1-4t^2}} = \frac{1}{2} \arcsin(2u).$$

Step 3: evaluate at $u = \frac{1}{2}$. With $u = \frac{1}{2}$ we have $u^{2k+1} = \frac{1}{2} \cdot 4^{-k}$, so

$$\frac{1}{2} \sum_{k=0}^{\infty} \frac{1}{2k+1} \binom{2k}{k} \frac{1}{4^k} = \frac{1}{2} \arcsin(1) = \frac{\pi}{4}.$$

The left side is exactly $E[a]$ by Step 1, completing the proof. (The evaluation at the boundary of convergence is legitimate by Abel's theorem, since the coefficient series converges there — its terms decay like $k^{-3/2}$, as quantified below.) $\square$

Why π From Coin Tosses?

The appearance of π is startling but structural. It enters through the *arcsine*, which is the antiderivative of $1/\sqrt{1-4t^2}$ — and that square root is precisely the central-binomial kernel of Lemma 12.5. Whenever first-passage or return-to-origin counts are summed against a $1/(2k+1)$ weight, an arcsin appears, and evaluation at the natural boundary produces $\arcsin(1) = \pi/2$.

The same kernel governs the celebrated **arcsine law** for the fraction of time a random walk spends positive — developed in full in the next section, which reuses Lemmas 12.5 and 12.6 proved here. The present problem and the arcsine law are cousins: one weights the kernel by a stopping payoff, the other by an occupation count, and both hand back π .

12.7 Convergence and the Tail

Remark 12.12 (The Series Converges, but Slowly). By Stirling's approximation, $C_k \sim \frac{4^k}{\sqrt{\pi} k^{3/2}}$, so the stopping probabilities decay as

$$p_k = \frac{C_k}{2^{2k+1}} \sim \frac{1}{2\sqrt{\pi} k^{3/2}}.$$

The contribution to $E[a]$ from term k behaves like

$$a_k p_k \sim \frac{1}{2} \cdot \frac{1}{2\sqrt{\pi} k^{3/2}} = \frac{1}{4\sqrt{\pi}} k^{-3/2},$$

which is summable (the exponent $3/2 > 1$), confirming $E[a] < \infty$. The decay is only polynomial, however: truncating at $k = K$ leaves a tail of order $K^{-1/2}$ — numerically, summing two million terms still leaves an error of 2×10^{-4} . This is why the closed-form arcsine evaluation is the right tool rather than brute summation.

Remark 12.13 (Contrast With $E[\tau] = \infty$) It is worth pausing on the contrast: the *stopping time* τ has infinite mean (its distribution decays as $k^{-3/2}$, and $\sum k \cdot k^{-3/2} = \sum k^{-1/2}$ diverges), yet the *stopping value* a has finite mean $\pi/4$. The reason is that the late-stopping events, though heavy enough to make $E[\tau]$ diverge, carry proportions $a_k \rightarrow \frac{1}{2}$ that are bounded, so they contribute a summable amount to $E[a]$. Finiteness of a functional of a stopping time does not follow from finiteness of the stopping time itself.

12.8 Generalizations and Connections

Remark 12.14 (The Biased Coin: An Exact Formula). Replace the fair coin by one with $P(\text{heads}) = p \geq \frac{1}{2}$ and $q = 1 - p$. The renewal proof of Theorem 12.7 survives verbatim with $\Phi(z) = pz + qz\Phi(z)^2$, giving $P(\tau = 2k + 1) = C_k p^{k+1} q^k$, and the game still ends almost surely (for $p < \frac{1}{2}$ it ends only with probability p/q). The same cancellation and integration as in Theorem 12.11 then yield the closed form

$$E[a] = p \frac{\arcsin(2\sqrt{pq})}{2\sqrt{pq}} \quad (p \geq \tfrac{1}{2}),$$

which contains the fair case: at $p = \frac{1}{2}$, $2\sqrt{pq} = 1$ and the formula reads $\frac{1}{2} \arcsin(1) = \pi/4$. *Numerical check by simulation (4×10^5 rounds each):*

p	formula	simulation
0.50	0.7854	0.7860
0.55	0.8129	0.8129
0.60	0.8386	0.8385
0.70	0.8854	0.8857

The value $\pi/4$ is thus the $p = \frac{1}{2}$ slice of a one-parameter family of arcsine evaluations — the special role of the fair coin is that it pushes the evaluation point to the boundary $\arcsin(1)$, where the answer becomes a rational multiple of π .

Remark 12.15 (Where This Sits in the Toolkit). This exercise is a compact tour of several techniques developed elsewhere in these notes:

- **Random-walk reduction** of a proportion condition (Part II, random walks).
- **First-return/renewal decomposition**, the same quadratic-equation mechanism that generates the Catalan numbers themselves.
- **Generating functions** (Lemmas 12.5 and 12.6), proved here and reused wholesale by the arcsine-law section.
- **Care with infinite-mean stopping times**, echoing the Optional Stopping Theorem failure modes.

Key Takeaways

First Crossing of the Majority

1. **Reduce to a walk.** $r_n > \frac{1}{2} \iff S_n = 2H_n - n > 0$. The game is the first passage of a simple symmetric walk to $+1$.
2. **Stopping shape.** Stop at $n = 2k + 1$ with $k + 1$ heads, value $a_k = \frac{k + 1}{2k + 1}$.
3. **Two generating functions, proved here and reused later.** $\sum \binom{2n}{n} x^n = 1/\sqrt{1 - 4x}$ (Newton's binomial series) and $\sum C_m x^m = (1 - \sqrt{1 - 4x})/(2x)$ (first return). The arcsine-law section builds on both.
4. **Catalan first-passage law.** $P(\tau = 2k + 1) = \frac{C_k}{2^{2k+1}}$, from the renewal equation $\Phi = \frac{z}{2} + \frac{z}{2}\Phi^2$ — the walk's own restatement of the Catalan recursion.
5. **Total probability one.** $\Phi(1) = 1$: the walk reaches $+1$ a.s., yet $E[\tau] = \infty$.
6. **The answer.** The $(k + 1)$ in a_k cancels the $1/(k + 1)$ in C_k ; the surviving $1/(2k + 1)$ weight signals an integration of the central-binomial series, and

$$E[a] = \frac{1}{2} \arcsin(1) = \frac{\pi}{4}.$$

7. **The biased family.** $E[a] = p \arcsin(2\sqrt{pq})/(2\sqrt{pq})$ for $p \geq \frac{1}{2}$; the fair coin is the boundary case where the arcsine hits 1 and π appears.
8. **Finite value, infinite time.** $E[a] = \pi/4$ is finite even though $E[\tau] = \infty$, because $a_k \rightarrow \frac{1}{2}$ is bounded while the time tail is heavy.

13 The Arcsine Law for the Time Spent Positive

13.1 Problem Statement

Toss a fair coin $2n$ times and let $S_k = H_k - T_k$ be the running imbalance after k tosses, a simple symmetric random walk with $S_0 = 0$. Call the i -th step *positive* if the segment $[i-1, i]$ of the walk lies weakly above zero, that is, if

$$\max(S_{i-1}, S_i) > 0,$$

and define

$$L_{2n} = \#\{i \in \{1, \dots, 2n\} : \max(S_{i-1}, S_i) > 0\},$$

the total number of positive steps, so that $L_{2n}/(2n)$ is the *fraction of time* spent on the positive side.

Question. What is the distribution of $L_{2n}/(2n)$ as $n \rightarrow \infty$?

Naive intuition suggests concentration around $1/2$: by symmetry the walk should spend roughly half its time above zero. The truth is strikingly different — the answer is maximised at the *extremes*.

13.2 Strategy of the Proof

The argument is a generating-function computation, and it is worth stating the plan before the machinery arrives, because every step exists to serve a single final move.

- (1) **Split every path in two** at its last visit to zero: a *returning prefix* followed by a *non-returning tail* (Lemma 13.6).
- (2) **Count each piece separately.** The prefix is governed by Sparre Andersen's theorem, which says its positive-step count is *uniform* (Lemma 13.4); the tail is trivial, being either entirely positive or entirely negative (Lemma 13.3).
- (3) **Encode both counts in a bivariate generating function** — one variable for length, one for positive steps — so that concatenation becomes *multiplication*.
- (4) **Multiply and watch a cancellation:** the two complicated factors collapse into the product of two central-binomial series.
- (5) **Read off a coefficient**, which factorises into $u_{2k}u_{2n-2k}$ — the theorem.

What the Two Variables Track

The whole computation hinges on a bookkeeping convention that should be fixed in the reader's mind before any algebra:

x marks *half the length*, y marks *half the number of positive steps*.

A path of length $2n$ with $2k$ positive steps contributes the monomial $x^n y^k$. So the master generating function

$$A(x, y) = \sum_{n \geq 0} \sum_{k=0}^n \alpha_{2n, 2k} x^n y^k, \quad \alpha_{2n, 2k} := \#\{\text{such paths}\},$$

holds the entire answer: extracting $[x^n y^k]$ recovers exactly the count we want.

Micro-example ($n = 1$). The four paths of length 2 are UU, UD, DU, DD. The first two never go below zero, so both their steps are positive ($k = 1$); the last two never rise above zero, so neither step is ($k = 0$). Hence $\alpha_{2,0} = \alpha_{2,2} = 2$, and the x^1 part of A is $2 + 2y$. The theorem predicts $\binom{0}{0} \binom{2}{1} = 2$ and $\binom{2}{1} \binom{0}{0} = 2$. ✓

13.3 Tools Recalled from the First-Crossing Section

Two power series drive the computation, both established in the first-crossing section and reused here without change: the central binomial generating function (Lemma 12.5),

$$\sum_{n \geq 0} \binom{2n}{n} x^n = \frac{1}{\sqrt{1-4x}},$$

proved there via Newton's binomial series, and the Catalan generating function (Lemma 12.6), $\sum_m C_m x^m = (1 - \sqrt{1-4x})/(2x)$, proved by the first-return decomposition.

Remark 13.1 (Reading the Kernel Probabilistically). Dividing the central binomial coefficients by 4^n turns them into the return-to-origin probabilities of Lemma 13.2: $\sum_n u_{2n} x^n = 1/\sqrt{1-x}$. The square root is therefore not an accident of algebra — it is the analytic signature of *returns to the origin*, and every appearance of $\sqrt{1-4x}$ below should be read as “a walk coming home.” In the first-crossing section the same kernel was weighted by a stopping payoff and produced $\pi/4$; here it will be weighted by an occupation count and produce the arcsine density.

13.4 Two Combinatorial Lemmas

Lemma 13.2 (Return-to-Origin Count). *The number of paths of length $2m$ from 0 to 0 is $\binom{2m}{m}$; equivalently the return probability is*

$$u_{2m} := P(S_{2m} = 0) = \frac{1}{4^m} \binom{2m}{m} \sim \frac{1}{\sqrt{\pi m}} \quad (m \rightarrow \infty),$$

the asymptotic following from Stirling's formula.

Proof. Such a path has m up-steps and m down-steps; the number of arrangements is $\binom{2m}{m}$. □

Lemma 13.3 (Non-Returning Path Count). *The number of paths of length $2m \geq 2$ starting at 0 that never return to 0 at times $1, \dots, 2m$ equals the number returning at time $2m$:*

$$\#\{\text{non-returning paths of length } 2m\} = \binom{2m}{m}.$$

Moreover exactly half are strictly positive throughout and half strictly negative, each side numbering $\frac{1}{2}\binom{2m}{m}$.

Proof of the sign split. Negating every step is an involution of the set of non-returning paths that exchanges the strictly positive with the strictly negative ones, hence a bijection between the two classes. $\square$

The count itself is classical; it follows from the reflection principle (Theorem 11.3) by a bijection equating non-returning with returning paths. See Feller, *An Introduction to Probability Theory and Its Applications*, Vol. I, Section III.3, for the explicit construction. *Numerical check:* for $m = 1, \dots, 6$ the non-returning counts are 2, 6, 20, 70, 252, $924 = \binom{2m}{m}$, splitting as 1+1, 3+3, 10+10, 35+35, 126+126, 462+462.

Lemma 13.4 (Sparre Andersen: Uniformity of Positive Steps). *Among the $\binom{2m}{m}$ paths of length $2m$ from 0 to 0, the number having exactly $2j$ positive steps is*

$$N_{2m}(2j) = \frac{1}{m+1} \binom{2m}{m} = C_m \quad \text{for every } j \in \{0, 1, \dots, m\}.$$

*That is, conditional on returning to zero at time $2m$, the number of positive steps is **uniformly distributed** on $\{0, 2, \dots, 2m\}$.*

The proof is the *cycle lemma* of Dvoretzky and Motzkin: the $2m$ cyclic shifts of a returning path realise each possible positive-step count exactly once, after accounting for the action of the cyclic group. The result is due to Sparre Andersen (1953/54); complete proofs appear in Feller, Vol. I, Section III.4, and Spitzer, *Principles of Random Walk*, Section II.4. *Numerical check:* for $m = 3$ the six possible counts $\{0, 2, 4, 6\}$ are attained by 5, 5, 5, 5 paths, and $C_3 = 5$; for $m = 4$, by 14 each, and $C_4 = 14$.

Why Uniformity Is the Surprise

Sparre Andersen is the engine of the whole result, and it is already counterintuitive on its own. Among paths that return to zero — the “balanced” paths, which one might expect to divide their time evenly — *every* split of time is equally likely: spending all $2m$ steps positive is exactly as common as splitting 50–50. The U-shape of the final answer is not created by the tail decomposition; it is already latent here, in the flatness of a distribution that intuition insists should be peaked.

13.5 The Prefix Generating Function

Lemma 13.5 (Bivariate Generating Function of Returning Paths). *With $N_{2m}(2j)$ as in Lemma 13.4,*

$$B(x, y) := \sum_{m \geq 0} \sum_{j=0}^m N_{2m}(2j) x^m y^j = \frac{1}{2x(1-y)} \left[\sqrt{1-4xy} - \sqrt{1-4x} \right].$$

Proof. By Lemma 13.4 the inner coefficient $N_{2m}(2j) = C_m$ does not depend on j , so the inner sum is a finite geometric series:

$$B(x, y) = \sum_{m \geq 0} C_m x^m (1 + y + \dots + y^m) = \sum_{m \geq 0} C_m x^m \frac{1 - y^{m+1}}{1 - y}.$$

Split the numerator and recognise two copies of the Catalan series (Lemma 12.6), the second evaluated at xy :

$$B(x, y) = \frac{1}{1-y} \left[\mathcal{C}(x) - y \mathcal{C}(xy) \right] = \frac{1}{1-y} \left[\frac{1 - \sqrt{1-4x}}{2x} - y \cdot \frac{1 - \sqrt{1-4xy}}{2xy} \right].$$

The factor y cancels in the second term, leaving a common denominator $2x$:

$$B(x, y) = \frac{1}{2x(1-y)} \left[(1 - \sqrt{1-4x}) - (1 - \sqrt{1-4xy}) \right] = \frac{1}{2x(1-y)} \left[\sqrt{1-4xy} - \sqrt{1-4x} \right]. \quad \square$$

13.6 The Last-Zero Decomposition

Lemma 13.6 (Last-Zero Bijection). *Every path σ of length $2n$ starting at 0 decomposes uniquely as $\sigma = \pi \parallel \tau$, where π is a path of length $2m$ from 0 to 0 (for a unique $m \in \{0, \dots, n\}$), τ is a path of length $2n - 2m$ starting at 0 that never returns to 0, and $\parallel$ is concatenation. The index $2m$ is the last time σ visits zero.*

Proof. Let $2m$ be the largest index with $S_{2m} = 0$; it exists since $S_0 = 0$, and it is unique by maximality. Set $\pi = \sigma|_{[0, 2m]}$ and $\tau = \sigma|_{[2m, 2n]}$ shifted to start at 0. Then π returns to zero by construction and τ never does, again by maximality of $2m$. Conversely any such pair concatenates to a path with last zero at $2m$. $\square$

Remark 13.7 (Why This Decomposition and Not Another). The last-zero split is chosen because it makes the two pieces *independently countable*: the prefix is an arbitrary returning path (counted by Sparre Andersen) and the tail an arbitrary non-returning one (counted by Lemma 13.3), with no interaction between them. Since the positive-step counts of the two pieces simply add, and lengths add, the bivariate generating function of the concatenation is the *product* of the two — which is precisely why generating functions are the right language here.

13.7 The Main Theorem

Lévy's Discrete Arcsine Law

Theorem 13.8. *For a simple symmetric random walk of length $2n$ started at 0,*

$$P(L_{2n} = 2k) = u_{2k} u_{2n-2k}, \quad k = 0, 1, \dots, n,$$

with $u_{2j} = \binom{2j}{j}/4^j$ as in Lemma 13.2.

Proof. Write $\alpha_{2n, 2k}$ for the number of paths of length $2n$ with exactly $2k$ positive steps; it suffices to prove

$$\alpha_{2n, 2k} = \binom{2k}{k} \binom{2(n-k)}{n-k},$$

since dividing by $2^{2n} = 4^k \cdot 4^{n-k}$ then gives the theorem.

Step 1: turn concatenation into multiplication. By Lemma 13.6 each path splits uniquely into a returning prefix and a non-returning tail, with lengths and positive-step counts *adding*. In the bookkeeping $x \leftrightarrow$ half-length, $y \leftrightarrow$ half-positive-steps, adding exponents is multiplying monomials, so

$$A(x, y) = B(x, y) \cdot T(x, y),$$

where B is the prefix series of Lemma 13.5 and

$$T(x, y) := \sum_{r, p} M_{2r}(2p) x^r y^p, \quad M_{2r}(2p) := \#\{\text{non-returning paths of length } 2r \text{ with } 2p \text{ positive steps}\}.$$

Step 2: the tail series is elementary. By Lemma 13.3 a non-returning path of length $2r \geq 2$ is either strictly positive — all $2r$ steps positive, so $p = r$ — or strictly negative, with $p = 0$;

and the two classes are equinumerous. Hence

$$M_{2r}(2p) = \begin{cases} \frac{1}{2} \binom{2r}{r}, & p = 0 \text{ or } p = r, \\ 0, & \text{otherwise,} \end{cases} \quad r \geq 1,$$

with $M_0(0) = 1$ for the empty tail. Summing, and using Lemma 12.5 twice (once at x , once at xy),

$$T(x, y) = 1 + \sum_{r \geq 1} \frac{1}{2} \binom{2r}{r} (x^r + (xy)^r) = \frac{1}{2} [C(x) + C(xy)] = \frac{1}{2} \left[\frac{1}{\sqrt{1-4x}} + \frac{1}{\sqrt{1-4xy}} \right].$$

Step 3: assemble. With B from Lemma 13.5,

$$A(x, y) = \underbrace{\frac{1}{2x(1-y)} [\sqrt{1-4xy} - \sqrt{1-4x}]}_{B(x,y)} \cdot \underbrace{\frac{1}{2} \left[\frac{1}{\sqrt{1-4x}} + \frac{1}{\sqrt{1-4xy}} \right]}_{T(x,y)}.$$

Step 4: the cancellation. Set $a = \sqrt{1-4x}$ and $b = \sqrt{1-4xy}$. The bracketed factors are $(b-a)$ and $(\frac{1}{a} + \frac{1}{b})$, whose product telescopes:

$$(b-a) \left(\frac{1}{a} + \frac{1}{b} \right) = \frac{(b-a)(a+b)}{ab} = \frac{b^2 - a^2}{ab} = \frac{(1-4xy) - (1-4x)}{\sqrt{(1-4x)(1-4xy)}} = \frac{4x(1-y)}{\sqrt{(1-4x)(1-4xy)}}.$$

The prefactor $\frac{1}{2x(1-y)} \cdot \frac{1}{2}$ cancels the numerator $4x(1-y)$ exactly, leaving

$$\boxed{A(x, y) = \frac{1}{\sqrt{(1-4x)(1-4xy)}} = \frac{1}{\sqrt{1-4x}} \cdot \frac{1}{\sqrt{1-4xy}}.}$$

Every trace of the Catalan numbers and of the factor $(1-y)$ has vanished: the joint generating function is simply a product of two central-binomial series.

Step 5: extract the coefficient. Expand each factor by Lemma 12.5, using *different* dummy indices since they will be matched separately:

$$\frac{1}{\sqrt{1-4xy}} = \sum_{a \geq 0} \binom{2a}{a} x^a y^a, \quad \frac{1}{\sqrt{1-4x}} = \sum_{b \geq 0} \binom{2b}{b} x^b,$$

so that

$$A(x, y) = \sum_{a \geq 0} \sum_{b \geq 0} \binom{2a}{a} \binom{2b}{b} x^{a+b} y^a.$$

Now hunt for $x^n y^k$, and note the crucial asymmetry: *y occurs in the first factor only*. Matching powers of y therefore pins the first index outright, $a = k$, with no freedom remaining; only then does matching x force $b = n - k$. Exactly one term survives:

$$[x^n y^k] A(x, y) = \binom{2k}{k} \binom{2(n-k)}{n-k} = \alpha_{2n, 2k},$$

whence

$$P(L_{2n} = 2k) = \frac{\alpha_{2n, 2k}}{2^{2n}} = \frac{\binom{2k}{k}}{4^k} \cdot \frac{\binom{2(n-k)}{n-k}}{4^{n-k}} = u_{2k} u_{2n-2k}. \quad \square$$

Why the Answer Factorises

Multiplying two ordinary one-variable series produces a *convolution*: $[x^n]$ of a product is $\sum_{a+b=n} c_a d_b$, a sum over every way of splitting n . Such sums are usually unwieldy, and a closed form is the exception.

Here the sum collapses to a *single term*, because the second variable y acts as a *tag* recording how much of the total came from the first factor. Demanding y^k is demanding “the first factor contributed exactly k ,” which discards every other term of the convolution. This is the entire reason the answer is a clean product $u_{2k}u_{2n-2k}$ rather than a sum — and the bivariate bookkeeping of Step 1 was set up precisely so that this collapse would occur.

Combinatorially the two factors are not interchangeable: $1/\sqrt{1-4xy}$ counts the *positive-time* part (length $2k$, all of it positive), $1/\sqrt{1-4x}$ the *non-positive* part (length $2(n-k)$, none of it positive). The theorem therefore says that a path with $2k$ positive steps is built by choosing these two blocks *independently* — which is also why the answer is symmetric in $k \leftrightarrow n-k$.

Remark 13.9 (Sanity Checks on $A(x, y)$). Two specialisations confirm the identity. Setting $y = 1$ ignores the positive-step tag and should recover all paths: $A(x, 1) = 1/(1-4x)$, whose x^n coefficient is $4^n = 2^{2n}$. ✓ Setting $y = 0$ keeps only $k = 0$: $A(x, 0) = 1/\sqrt{1-4x}$, giving $\alpha_{2n,0} = \binom{2n}{n}$, the paths that are never positive. ✓

13.8 The Continuous Limit: The Arcsine Distribution

Arcsine Law, Continuous Form

Theorem 13.10. As $n \rightarrow \infty$, $\frac{L_{2n}}{2n} \Rightarrow X$, where X has the arcsine distribution on $[0, 1]$:

$$f_X(x) = \frac{1}{\pi\sqrt{x(1-x)}}, \quad F_X(x) = \frac{2}{\pi} \arcsin(\sqrt{x}).$$

Proof. By Lemma 13.2, $u_{2j} \sim 1/\sqrt{\pi j}$. Put $k = \lfloor nx \rfloor$ with $x \in (0, 1)$ fixed; then

$$P(L_{2n} = 2k) = u_{2k}u_{2n-2k} \sim \frac{1}{\sqrt{\pi k}} \cdot \frac{1}{\sqrt{\pi(n-k)}} = \frac{1}{\pi\sqrt{k(n-k)}} \sim \frac{1}{\pi n\sqrt{x(1-x)}},$$

using $\sqrt{k(n-k)} \approx n\sqrt{x(1-x)}$. The factor $1/n$ is exactly the grid spacing of $\{0, 1/n, \dots, 1\}$, so these point masses form a Riemann sum for the density $1/(\pi\sqrt{x(1-x)})$ and

$$P\left(\frac{L_{2n}}{2n} \leq x\right) \rightarrow \int_0^x \frac{du}{\pi\sqrt{u(1-u)}}.$$

The substitution $u = \sin^2 \theta$ gives $du = 2 \sin \theta \cos \theta d\theta$ and $\sqrt{u(1-u)} = \sin \theta \cos \theta$, so the integrand reduces to $2 d\theta/\pi$ and the integral is $\frac{2}{\pi} \arcsin \sqrt{x}$. $\square$

Remark 13.11 (Particular Values of the CDF). Evaluating $F_X(x) = \frac{2}{\pi} \arcsin \sqrt{x}$ at three points shows how the mass distributes:

- $F_X(1/2) = \frac{2}{\pi} \arcsin(1/\sqrt{2}) = \frac{2}{\pi} \cdot \frac{\pi}{4} = \frac{1}{2}$ — the median is at $1/2$, by symmetry.
- $F_X(0.1) \approx 0.205$: a 20% chance of spending less than 10% of the time positive.
- $F_X(0.01) \approx 0.064$: a 6.4% chance of spending less than 1% of the time positive.

Substantial mass sits near the endpoints 0 and 1, and the distribution is comparatively flat in the middle.

13.9 The Counterintuitive Shape

Why a U-Shape, Not a Bell Curve

The density $f_X(x) = 1/(\pi\sqrt{x(1-x)})$ is **minimised** at $x = 1/2$, where $f_X(1/2) = 2/\pi \approx 0.637$, and **diverges** at $x = 0$ and $x = 1$. The modal outcome of “fraction of time positive” is therefore at the *extremes*.

Mechanism. Once a walk drifts away from zero, the time to return scales like the *square* of the displacement (diffusive scaling $T \sim \text{distance}^2$), so long excursions are common and a single one can dominate the total. The walker does not balance her time between the two sides; she gets *stuck* on one and stays there.

Note where this was already visible: Sparre Andersen (Lemma 13.4) says that even among balanced paths — those returning to zero — every time-split is equally likely. The U-shape then arises because the non-returning tail contributes *all-or-nothing*: it is entirely positive or entirely negative, pushing mass outward.

Example 13.12 (Numerical Illustration for $n = 8$). With $2n = 16$ tosses, $P(L_{16} = 2k) = u_{2k}u_{16-2k}$:

k	$L_{16} = 2k$	$P(L_{16} = 2k)$
0	0 (none positive)	0.1964
1	2	0.1047
2	4	0.0846
3	6	0.0769
4	8 (exactly half)	0.0748
5	10	0.0769
6	12	0.0846
7	14	0.1047
8	16 (all positive)	0.1964

Each extreme event is $0.1964/0.0748 \approx 2.6$ times more likely than the balanced one, and the U-shape is already unmistakable at $n = 8$. It sharpens as n grows, the middle flattening toward $2/\pi$ while the ends diverge.

13.10 Why π Appears

The Same Kernel as the Catalan Problem

The arcsine law and the first-crossing-of-the-majority problem ($\pi/4$) share one analytic source: both are built from central binomial coefficients, whose generating function is $1/\sqrt{1-4x}$ (Lemma 12.5), and both reduce in the limit to integrals of a square-root kernel,

$$\int \frac{dx}{\sqrt{x(1-x)}} = 2 \arcsin \sqrt{x}, \quad \int_0^1 \frac{dx}{\sqrt{x(1-x)}} = \pi.$$

The chain is now fully visible in this section: Stirling sends $u_{2m} \rightarrow 1/\sqrt{\pi m}$ (Lemma 13.2), the theorem multiplies two such factors, and the resulting $1/\sqrt{x(1-x)}$ integrates to an arcsine — with π entering through $\arcsin(1) = \pi/2$. The factorisation $A(x, y) = 1/\sqrt{(1-4x)(1-4xy)}$ of Step 4 is the algebraic form of the same fact: two central-binomial kernels, hence two $\sqrt{\pi}$'s, hence π .

That is why coin tosses produce π : the kernel is the kernel of first-passage and return phenomena, and only the weighting differs from problem to problem.

13.11 Applications to Trading

Remark 13.13 (Reading P&L Paths). A cumulative P&L modelled as a driftless random walk spends, under the arcsine law, most of its time on *one side* of breakeven. Reading such paths as though they ought to oscillate symmetrically around zero systematically misjudges typical behaviour: a zero-edge strategy will produce track records that *look* trending — long stretches above or below water — with no drift whatsoever. Conversely, a genuinely trending P&L is harder to distinguish from arcsine noise than intuition suggests, particularly over short horizons.

Remark 13.14 (Performance Evaluation Horizons). A manager with zero true edge will, over any finite horizon, show a record dominated by extended up or down streaks rather than balanced oscillation. The practical consequence: *the fraction of time an equity curve spent above its high-water mark is not a reliable signal of edge* unless the sample is enormous, since that fraction is arcsine-distributed and can land arbitrarily close to 0 or 1 by chance alone.

Remark 13.15 (Time Since the Last Crossing). For a walk observed over $[0, T]$, the *last* visit to zero before T is itself arcsine-distributed on $[0, T]$ — the same law, a different functional. This bears directly on statements like “the spread has not touched fair value in months.” Under the arcsine law long absences are *common*, not anomalous, and a spread that has stayed away for a long time is **not** thereby more likely to revert soon. The naive “it is due for a reversal” intuition inverts the truth.

Remark 13.16 (Timing of the Maximum, and Drawdowns). A companion arcsine law, also due to Lévy, governs the *location* of the running maximum:

$$\frac{\tau_{\max}}{T} \implies \text{Arcsine}(0, 1), \quad \tau_{\max} = \arg \max_{t \in [0, T]} S_t.$$

The maximum is therefore most likely to occur near the *beginning* or the *end* of the window, rarely in the middle. For drawdown analysis this means a strategy's worst peak-to-trough episode is most likely to be a recent or an initial event; risk frameworks that assume extremes fall uniformly through the sample understate both new-strategy and end-of-sample risk.

13.12 Connection to Brownian Motion

Remark 13.17 (The Continuous Analogue). Lévy’s classical arcsine theorem states that for standard Brownian motion the occupation fraction

$$\frac{1}{t} \int_0^t \mathbb{1}_{\{W_s > 0\}} ds$$

is *exactly* arcsine-distributed on $[0, 1]$ for every $t > 0$ — a stronger statement than the discrete result, which holds only asymptotically. The discrete law converges to it under Donsker scaling, and the two share the same combinatorial source: the proof in continuous time replaces the last-zero decomposition by the strong Markov property at the last zero, and the generating-function cancellation by Brownian scaling invariance.

Key Takeaways

The Arcsine Law

1. **Discrete form.** $P(L_{2n} = 2k) = u_{2k}u_{2n-2k}$ with $u_{2j} = \binom{2j}{j}/4^j$ — a product, not a sum.
2. **Continuous limit.** $L_{2n}/(2n) \Rightarrow X$ with density $1/(\pi\sqrt{x(1-x)})$ on $(0, 1)$ and CDF $\frac{2}{\pi} \arcsin \sqrt{x}$.
3. **The shocking shape.** The density is U-shaped: spending almost all the time on one side is far more likely than an even split. At $n = 8$ the extremes are already 2.6 times likelier than the balanced outcome.
4. **Sparre Andersen is the engine.** Among paths returning to zero, the positive-step count is *uniform*, each value attained by exactly C_m paths. The flatness of this conditional law is where the U-shape originates.
5. **Proof technique.** Last-zero decomposition $\rightarrow$ product of bivariate generating functions $\rightarrow$ a cancellation collapsing everything to $A(x, y) = 1/\sqrt{(1-4x)(1-4xy)} \rightarrow$ coefficient extraction.
6. **Why the answer factorises.** A one-variable product gives a convolution; the tag variable y , present in only one factor, selects a single term of it. That is precisely why the answer is $u_{2k}u_{2n-2k}$ and not a sum.
7. **Why π .** $1/\sqrt{1-4x}$ is the central-binomial kernel; Stirling turns it into $1/\sqrt{\pi m}$, two copies give $1/(\pi\sqrt{x(1-x)})$, and its integral is an arcsine — the same mechanism producing $\pi/4$ in the Catalan first-crossing problem.
8. **Trading lessons.** (a) P&L paths streak rather than oscillate. (b) “Time above high-water mark” is a poor edge signal at finite horizons. (c) A spread long away from fair value is not due to revert. (d) Maximum drawdowns cluster near the start or end of the sample, rarely the middle.
9. **Brownian counterpart.** For Brownian motion the occupation fraction is *exactly* arcsine at every t , not merely in the limit.

14 The Longest Run of Heads: A Tour of Approximations

14.1 Problem Statement

Thirty individuals each toss their own fair coin 162 times. For each person, we record the length of their longest unbroken run of heads. Then, across all 30 participants, we record the maximum of these longest runs. Find the expected value of this overall maximum.

Formally: let $X_1, X_2, \dots, X_n$ be i.i.d. fair coin tosses, each equal to H or T with probability $1/2$, and let

$$L_n = \max\{\ell \geq 0 : X_{i+1} = X_{i+2} = \dots = X_{i+\ell} = \text{H for some } 0 \leq i \leq n - \ell\}$$

be the length of the longest unbroken block of heads. Writing $L_{162}^{(1)}, \dots, L_{162}^{(30)}$ for the thirty participants' independent copies of L_{162} , the quantity of interest is

$$M = \max(L_{162}^{(1)}, L_{162}^{(2)}, \dots, L_{162}^{(30)}), \quad \text{and the goal is } E[M].$$

14.2 Why This Problem Is Interesting

The longest-run statistic is not a sum of independent random variables. This rules out the central limit theorem and every tool based on linearity of variance. Worse, **no simple closed form exists for $E[L_n]$** : the exact value can only be obtained by recursion or generating functions.

What *does* exist is a rich theory of approximations, each capturing the answer to a different level of accuracy. This section is a tour through four such methods, ordered from crudest to most precise. The pedagogical point is that *the same problem looks completely different depending on the tool you reach for*, and understanding the trade-offs between derivability, accuracy, and rigour is itself a skill. Roughly:

- **Method 1 (Poisson on the maximum).** Derivable from scratch in a minute; accurate to within about one unit, with a bias whose sign and size we will predict in advance.
- **Method 2 (Poisson on the pooled sequence).** A reformulation of Method 1 that reveals the key structural insight: 30 sequences of 162 tosses act like one sequence of 4860 tosses.
- **Method 3 (Schilling's asymptotic formula).** A one-line refined asymptotic with the correct constant, accurate to a tenth of a unit.
- **Method 4 (Exact Markov chain).** The exact answer, computable in microseconds. The rigorous benchmark against which the others are judged.

14.3 Method 1: The Poisson Approximation

14.3.1 Counting Potential Runs

Lemma 14.1 (Expected Number of Runs). *The expected number of starting positions at which a block of at least k consecutive heads begins, in n tosses of a fair coin, is*

$$e(k) = \frac{n+1-k}{2^k}.$$

Proof. For $1 \leq i \leq n - k + 1$, let A_i be the event that a block of k consecutive heads *begins* at position i :

$$A_i = \{X_i = X_{i+1} = \dots = X_{i+k-1} = \text{H}\}.$$

Note that A_i constrains only these k tosses — it does *not* require $X_{i-1} = T$ — so a single run of length $\ell > k$ triggers A_i at $\ell - k + 1$ distinct positions. (This double-triggering is harmless here, but it will return as the source of the Poisson bias.) Each event has $P(A_i) = 2^{-k}$, there are $n - k + 1$ of them, and linearity of expectation — which requires no independence — gives

$$E\left[\sum_{i=1}^{n-k+1} \mathbb{1}_{A_i}\right] = (n - k + 1) 2^{-k} = \frac{n + 1 - k}{2^k}. \quad \square$$

14.3.2 The Poisson Step

Let $W_k = \sum_{i=1}^{n-k+1} \mathbb{1}_{A_i}$ be the number of starting positions of k -runs, so that $E[W_k] = e(k)$ by Lemma 14.1, and observe the identity that connects the count to the maximum:

$$\{L_n \geq k\} = \{W_k \geq 1\}.$$

The *Poisson paradigm* (see the Appendix; the quantitative version is the Chen–Stein method) states that a sum of rare indicator events is approximately Poisson with mean $\sum_i P(A_i)$ provided two conditions hold:

- (i) each individual probability $P(A_i) = 2^{-k}$ is small; and
- (ii) the dependence is *local*: A_i and A_j are genuinely independent whenever $|i - j| \geq k$, because they then involve disjoint sets of tosses.

Both hold for k in the right tail (i.e. k noticeably above $\log_2 n$). Granting $W_k \approx \text{Poisson}(e(k))$,

$$p(k) := P(L_n \geq k) = P(W_k \geq 1) \approx 1 - e^{-e(k)} = 1 - \exp\left(-\frac{n + 1 - k}{2^k}\right).$$

For small k (at or below $\log_2 n$) the approximation is poor, but there $P(L_n \geq k) \approx 1$ anyway, which the formula also delivers; the approximation is tight precisely in the range that determines the answer.

Remark 14.2 (The Direction of the Error, Predicted in Advance). Condition (ii) fails at short range: for $|i - j| < k$ the events A_i, A_j are strongly *positively* associated, since one long run triggers several neighbouring starting positions at once (the double-triggering noted in the proof of Lemma 14.1). Occurrences therefore arrive in *clumps* rather than singly. A clumped counting variable has a larger zero-atom than a Poisson variable of the same mean, so $1 - e^{-e(k)}$ *overstates* $P(L_n \geq k)$, and the resulting estimate of $E[L_n]$ is *biased upward*. Method 3 will quantify the bias: almost exactly $+1$.

14.3.3 From One Person to Thirty

The longest runs of the 30 people are i.i.d. copies of L_{162} , so

$$P(M \geq k) = 1 - (1 - p(k))^{30},$$

and the tail-sum formula $E[M] = \sum_{k \geq 1} P(M \geq k)$ (truncated at $k = 25$, beyond which the terms are negligible) gives:

k	$e(k) = (163 - k)/2^k$	$p(k) \approx 1 - e^{-e(k)}$	$P(M \geq k)$
8	0.6055	0.4543	1.0000
9	0.3008	0.2598	0.99988
10	0.1494	0.1388	0.98869
11	0.0742	0.0715	0.89210
12	0.0369	0.0362	0.66911
13	0.0183	0.0182	0.42266
14	0.0091	0.0091	0.23878
15	0.0045	0.0045	0.12672
16	0.0022	0.0022	0.06508
17	0.0011	0.0011	0.03287
18	0.0006	0.0006	0.01646
19	0.0003	0.0003	0.00821
20	0.0001	0.0001	0.00408

Summing over $k = 1, \dots, 25$:

$$E[M]^{\text{Method 1}} \approx 12.47.$$

14.4 Method 2: The Pooling Principle

Method 1 treated the 30 people separately and combined them via the 30-th power. A more revealing reformulation gives the same answer up to higher-order corrections.

The Pooling Observation

The event “some person has a run of length $\geq k$ ” depends only on the rare occurrence of a long run *somewhere* among the $30 \times 162 = 4860$ total tosses, and Poisson rates of rare events are *additive*: thirty independent sequences contribute thirty separate rates that simply sum.

Pooling therefore reduces the problem to a single virtual sequence of $N := 4860$ tosses:

$$P(M \geq k) \approx 1 - \exp\left(-\frac{N + 1 - k}{2^k}\right),$$

one Poisson approximation in place of two nested ones.

Example 14.3 (Pooled Poisson for the Original Problem). With $N = 4860$:

$$E[M]^{\text{Method 2}} = \sum_{k \geq 1} \left[1 - \exp\left(-\frac{4861 - k}{2^k}\right) \right] \approx 12.58.$$

Remark 14.4 (Why the Two Methods Differ Slightly). The two methods are in fact the *same* approximation with slightly different rates. Method 1’s combination step is exact: since $p(k) = 1 - e^{-e(k)}$,

$$P(M \geq k) = 1 - (1 - p(k))^{30} = 1 - e^{-30 e(k)},$$

a pooled Poisson with rate $30(163 - k)/2^k$. Method 2 instead uses the rate $(4861 - k)/2^k$ of a genuinely continuous sequence, and the difference between the two rates is exactly

$$(4861 - k) - 30(163 - k) = 29(k - 1):$$

$k - 1$ phantom starting positions at each of the 29 seams between consecutive people's sequences — positions where the pooled formula lets a run straddle two different coins, which no real run can. Restoring this correction reproduces Method 1 to four decimals. The durable content survives intact: *the longest run across m sequences of length n behaves like one sequence of length mn* , because the seam correction, of relative size $\sim mk/mn$, vanishes for long sequences. This reduction is what lets the single-sequence formula of Method 3 answer the thirty-sequence question.

14.5 Method 3: Schilling's Asymptotic Formula

The Poisson approximation gets the scaling $\log_2 N + O(1)$ right but, as predicted in Remark 14.2, its constant is biased upward. The correct asymptotic constant was established by Schilling (1990), refining the law of Erdős and Rényi.

Schilling's Asymptotic Mean

Theorem 14.5. *For the longest run of heads in n fair tosses,*

$$E[L_n] = \log_2 n + \frac{\gamma}{\ln 2} - \frac{3}{2} + r(\log_2 n) + o(1),$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant and $r(\cdot)$ is a tiny periodic oscillation, $|r| < 10^{-5}$ in practice. Numerically,

$$\frac{\gamma}{\ln 2} - \frac{3}{2} = 0.8327 - 1.5 \approx -0.6673.$$

Remark 14.6 (References). Schilling, M. F. (1990), “The longest run of heads,” *The College Mathematics Journal* **21**(3), 196–207. The underlying growth law is due to Erdős and Rényi (1970), “On a new law of large numbers,” *J. d'Analyse Math.* **23**, 103–111; the extreme-value treatment with error bounds is Gordon, Schilling and Waterman (1986), “An extreme value theory for long head runs,” *Probab. Theory Related Fields* **72**, 279–287.

Remark 14.7 (Comparison with Poisson). The Poisson computation effectively produces the constant $\gamma/\ln 2 - 1/2 \approx +0.33$, which exceeds Schilling's by exactly 1. This +1 is the clumping correction of Remark 14.2 in closed form: a run of length ℓ contains $\ell - k + 1$ starting positions of k -runs, and treating those as independent occurrences overstates the tail by one unit of run length.

Example 14.8 (Schilling at $n = 162$ and on the Pooled Problem).

$$E[L_{162}] \approx \log_2(162) - 0.6673 = 7.3399 - 0.6673 = 6.673,$$

against the exact 6.682 of Method 4: an error of 0.009, about 0.15%. For the original problem, pool first (Method 2) and then apply the formula to $N = 4860$:

$$E[M]^{\text{Method 3}} \approx \log_2(4860) - 0.6673 = 12.2467 - 0.6673 \approx 11.58.$$

14.5.1 The Biased Coin

The same asymptotic extends to an unfair coin, a fact needed for every trading application below.

Schilling for a Biased Coin

Theorem 14.9. *If each toss is a head with probability $p \in (0, 1)$, the longest head run satisfies*

$$E[L_n] = \log_{1/p}(n(1-p)) + \frac{\gamma}{\ln(1/p)} - \frac{1}{2} + o(1).$$

For $p = 1/2$ this reduces to $\log_2 n + \gamma/\ln 2 - 3/2$, recovering Theorem 14.5.

Example 14.10 (Accuracy Check). Against exact values from the Markov chain of Method 4 (run with p in place of $1/2$), at $n = 252$:

p	formula	exact
0.50	7.310	7.316
0.45	6.399	6.404
0.40	5.607	5.612

The formula is accurate to three decimal places already at one year of daily data.

14.6 Method 4: Exact Computation via a Markov Chain

The exact distribution of L_n admits an efficient recursive computation. We give it both as the rigorous benchmark and because the technique generalises to many constrained-sequence problems — including, with $p \neq 1/2$, the biased-coin verification above.

14.6.1 The Run-Length State Machine

Fix a target k and track the length of the head-run currently in progress. This is a Markov chain on the states $\{0, 1, \dots, k\}$, killed the moment a run would reach length $k + 1$:

- from any state s , a tail resets to state 0 (probability $1/2$);
- from state $s < k$, a head advances to $s + 1$ (probability $1/2$);
- from state k , a head *exits the state space*: the forbidden run has occurred.

Theorem 14.11 (Exact Distribution). *Let T_k be the $(k + 1) \times (k + 1)$ substochastic matrix of the surviving transitions,*

$$(T_k)_{0,s} = \frac{1}{2} \quad (0 \leq s \leq k), \quad (T_k)_{s+1,s} = \frac{1}{2} \quad (0 \leq s \leq k-1),$$

all other entries zero; let $\mathbf{e}_0 = (1, 0, \dots, 0)^\top$ be the initial distribution (before any toss, the current run has length 0), and $\mathbf{1} = (1, \dots, 1)^\top$. Then

$$P(L_n \leq k) = \mathbf{1}^\top T_k^n \mathbf{e}_0.$$

Remark 14.12 (Reading the Formula). The two integers play entirely different roles: $k + 1$ is the size of the matrix — one state per possible current run length — while n is the exponent, one application of T_k per toss. Reading the product right to left:

- $\mathbf{e}_0$ starts all probability mass in state 0;
- each application of T_k advances one toss. Because T_k is *substochastic* — every column sums to 1 except column k , which sums to $1/2$ — probability mass *leaks out* exactly when a $(k + 1)$ -run is completed, and leaked mass never returns;

- $\mathbf{1}^\top$ adds up whatever mass survived all n tosses, which is precisely the probability that no run ever exceeded k : $P(L_n \leq k)$.

In practice one never forms matrix powers: iterating the vector costs $O(k)$ per step (one shift, one accumulation into state 0), hence $O(nk)$ overall — microseconds for $n = 162$. Replacing $1/2$ by p (heads) and $1 - p$ (tails) handles the biased coin with no other change.

Example 14.13 (The Smallest Case, Worked by Hand: $k = 1$). Take $k = 1$: we forbid any run of two heads, so $P(L_n \leq 1)$ is the probability that no HH ever appears. The state space is $\{0, 1\}$ (current head-run length), and the matrix is 2×2 :

$$T_1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix},$$

column 0 saying “from run-length 0, a tail stays at 0 and a head moves to 1,” column 1 saying “from run-length 1, a tail resets to 0 and a head *exits*” — hence the 0 entry, which is what makes the column sum $\frac{1}{2}$ rather than 1.

Iterating $v \mapsto T_1 v$ from $\mathbf{e}_0 = (1, 0)^\top$:

n	state 0	state 1	surviving mass = $P(L_n \leq 1)$
1	$\frac{1}{2}$	$\frac{1}{2}$	$1 = \frac{2}{2}$
2	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4} = \frac{3}{4}$
3	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{5}{8} = \frac{5}{8}$
4	$\frac{5}{16}$	$\frac{3}{16}$	$\frac{1}{2} = \frac{8}{16}$
5	$\frac{1}{4}$	$\frac{5}{32}$	$\frac{13}{32}$

Check against direct enumeration at $n = 3$: the strings without HH are TTT, TTH, THT, HTT, HTH — five of eight, $P(L_3 \leq 1) = \frac{5}{8}$ ✓. And the leaked mass $1 - \frac{5}{8} = \frac{3}{8}$ is exactly the probability that HH *does* occur within three tosses: HHH, HHT, THH, three of eight ✓.

The numerators 2, 3, 5, 8, 13 are unmistakable: writing $a_n = 2^n P(L_n \leq 1)$ for the number of HH-free strings, the chain’s two states give the two-term recursion $a_n = a_{n-1} + a_{n-2}$, so

$$P(L_n \leq 1) = \frac{F_{n+2}}{2^n},$$

with F_m the Fibonacci numbers. For general k the same mechanism produces a $(k + 1)$ -term (“ k -bonacci”) recursion, whose roots are algebraic numbers with no closed form — which is precisely why, as announced at the start of the section, no simple formula for $E[L_n]$ exists, and why approximation methods are the honest toolkit.

14.6.2 The Distribution of L_{162}

k	$P(L_{162} = k)$	$P(L_{162} \leq k)$
3	0.0107	0.0107
4	0.0796	0.0902
5	0.2173	0.3075
6	0.2682 (mode)	0.5757
7	0.2017	0.7774
8	0.1146	0.8920
9	0.0577	0.9497
10	0.0270	0.9767
11	0.0121	0.9888
12	0.0053	0.9941
≥ 13	0.0059	1.0000

The exact mean is $E[L_{162}] = 6.682$ (Schilling: 6.673; Poisson, per person: ≈ 7.67).

14.6.3 The Distribution of the Maximum

Independence across people gives the maximum's CDF by raising to the 30-th power, $P(M \leq k) = [P(L_{162} \leq k)]^{30}$:

k	$P(M = k)$	$P(M \leq k)$
≤ 8	0.0125	0.0125
9	0.0934	0.1059
10	0.2215	0.3274
11	0.2476 (mode)	0.5750
12	0.1856	0.7606
13	0.1134	0.8740
14	0.0624	0.9364
15	0.0326	0.9690
16	0.0166	0.9856
17	0.0084	0.9940
≥ 18	0.0060	1.0000

$$E[M]^{\text{Method 4}} = 11.49 \text{ (exact).}$$

14.7 Comparison of the Four Methods

Method	$E[M]$	Error vs. exact
Method 1: Poisson per person, combined	12.47	+0.98
Method 2: pooled Poisson on $N = 4860$	12.58	+1.09
Method 3: Schilling, $\log_2 N - 0.667$	11.58	+0.09
Method 4: exact Markov chain	11.49	0.00

What the Comparison Teaches

Schilling (Method 3) dominates on both speed and accuracy: one logarithm and one subtraction land within a tenth of a unit of the truth. When a number is wanted, pool to $N = mn$ and compute $\log_2 N - 0.667$.

Poisson (Methods 1 and 2) gets the scaling right and the constant wrong by +1 — and it is not in fact the fastest route to a number, since its tail sum requires a table. Its virtues are different, and real. It is *self-deriving*: reconstructible in a minute from $e(k) = (n+1-k)2^{-k}$, with no constant to memorise — and if the answer is then corrected by the known -1 bias, most of the gap closes. It delivers the *whole tail* $P(L_n \geq k)$, not only the mean. And it supplies the pooling argument without which the single-sequence formula of Method 3 could not address the thirty-sequence question. Methods 2 and 3 are complements, not rivals: *pooling comes from Poisson, precision from Schilling*.

The Markov chain (Method 4) is exact, cheap, and indifferent to asymptotics. Use it when n is small, when the full distribution is needed, or to verify anything.

Finally, the +1 bias is not academic: it overstates the typical extreme streak by a full unit, which in the applications below is the difference between “trigger the stop-loss” and “don’t.”

14.8 Connections to Trading

The longest-run problem is not a coin-toss curiosity. Its machinery transfers directly to four contexts, all through Theorems 14.5 and 14.9.

Remark 14.14 (Maximum Losing Streak in a Backtest). A strategy whose daily P&L is a fair coin produces losing streaks governed by Theorem 14.5:

$$E[L_n^{\text{loss}}] \approx \log_2 n - 0.667.$$

For $n = 252$ (one trading year): ≈ 7.3 consecutive losses expected (exact: 7.32). For $n = 1260$ (five years): ≈ 9.6 . A stop-loss that halts trading after k consecutive losing days will therefore be triggered *by chance alone*, with substantial probability, for any $k \leq \log_2 n$; calibrating k safely above the expected streak is essential, and the Gumbel-type right tail (each extra unit of k halves the probability) quantifies “safely.”

Remark 14.15 (Multiple-Testing Inflation). Backtesting m strategies over n days and inspecting the worst streak among them is exactly the original problem, and the pooling principle prices it:

$$E[\text{max streak}] \approx \log_2(mn) - 0.667.$$

For 30 strategies over one year ($mn = 7560$): ≈ 12.2 consecutive losses. For 100 strategies over five years ($mn = 126,000$): ≈ 16.3 . *At least one of thirty zero-edge strategies should be expected to lose twelve times in a row.* Reading such a streak as “strategy failure” is a multiple-testing error; the formula $\log_2(mn) - 0.667$ is the longest-run analogue of a Bonferroni correction, and the record-counting section makes the same point on a different statistic.

Remark 14.16 (Pattern Detection in Order Flow). Long runs of consecutive buys, or of same-direction ticks, are routinely read as momentum. The theory above supplies the null baseline: among $N = 4000$ trades, the longest buy-run expected under pure randomness is $\log_2 4000 - 0.667 \approx 11.3$. A run of 12 consecutive buys is thus *unremarkable*; genuinely anomalous runs are those exceeding the baseline by several units, each additional unit being exponentially less likely under the null.

Remark 14.17 (Edge Affects Frequency More Than Magnitude). Let q be the probability of losing a given trade. By Theorem 14.9 (heads $\rightarrow$ losses, $p \rightarrow q$), the longest losing streak over n trades has mean $\log_{1/q}(n(1-q)) + \gamma/\ln(1/q) - 1/2$. Over one year, $n = 252$:

$$q = 0.50 \Rightarrow 7.3, \quad q = 0.45 \Rightarrow 6.4, \quad q = 0.40 \Rightarrow 5.6,$$

(exact values 7.32, 6.40, 5.61). A genuine ten-point edge shortens the worst expected streak by under two trades: streak length is logarithmic in n and only mildly sensitive to q , because the base of the logarithm moves slowly. What the edge changes sharply is the *frequency* with which such streaks recur — not their size. Stop-losses and position sizing must therefore be calibrated to the realised q , and even a good strategy should budget for six consecutive losses in a typical year. (The Kelly section prices exactly this: a streak of length $\log_2 n$ at fraction f multiplies capital by $(1-f)^{\log_2 n}$.)

Key Takeaways

The Longest-Run Problem

1. **Logarithmic scaling.** The longest run in n fair tosses grows as $\log_2 n$ — not linearly, not as $\sqrt{n}$.
2. **Four methods, four answers.** Poisson gives ≈ 12.5 ; Schilling 11.58; exact 11.49.
3. **Pooling principle.** The maximum across m sequences of length n behaves like one sequence of length mn . Poisson additivity of rare-event rates is what justifies it.
4. **Poisson bias, predicted and priced.** Overlapping run events cluster, inflating the Poisson answer by $+1$ — the gap between the effective Poisson constant $\gamma/\ln 2 - 1/2$ and Schilling's $\gamma/\ln 2 - 3/2$.
5. **Schilling's formula.** $E[L_n] \approx \log_2 n + \gamma/\ln 2 - 3/2 = \log_2 n - 0.667$, accurate to ~ 0.01 already at $n = 162$; for a biased coin, $\log_{1/p}(n(1-p)) + \gamma/\ln(1/p) - 1/2$.
6. **Exact via Markov chain.** A $(k+1)$ -state run-length machine gives $P(L_n \leq k)$ in $O(nk)$; the matrix size is $k+1$, the toss count n is the exponent. Works verbatim for $p \neq 1/2$.
7. **Trading applications.** (a) Stop-losses: ≈ 7.3 consecutive losses per year are the baseline, not an anomaly. (b) Multiple testing: 30 strategies over a year expect a worst streak of ≈ 12.2 by pure chance. (c) Order flow: $\log_2 N - 0.667$ is the null for “unusual” runs. (d) Edge shortens streaks marginally and their frequency dramatically.
8. **When to use which method.** Fast number: pool to $N = mn$, then $\log_2 N - 0.667$ — one line, within 0.1. Constant not at hand, or the full tail wanted: Poisson, subtracting the known $+1$ bias. Exact, small n , or verification: the Markov chain.

15 The Kelly Criterion and Volatility Drag

15.1 The Motivating Problem

You are offered the following game. A biased coin lands heads with probability $p = 0.6$. Before each toss you may stake any fraction of your current capital on heads, at even odds: a win doubles the stake, a loss forfeits it. The game repeats indefinitely.

What fraction of your capital should you bet on each toss?

This is arguably the purest question in all of quantitative finance. You are handed a genuine edge — something every trader spends a career searching for — and the only remaining decision is *size*. Bet too little and the edge is wasted; bet too much and, as we shall see, the edge destroys you. The tension between these two failures, and its exact resolution, is the subject of this final section.

15.2 The Seduction of Expected Value

A natural first instinct: choose the fraction f that maximises expected wealth. Let W_n denote wealth after n bets with initial capital W_0 . Betting a constant fraction f , each toss multiplies wealth by $1 + f$ (probability p) or $1 - f$ (probability $q = 1 - p$), so

$$E[W_n] = W_0 (1 + (p - q)f)^n,$$

which is *increasing in f* . Expected wealth is maximised by $f = 1$: stake everything, every time.

Now observe what this strategy actually does. Betting everything, wealth after n tosses is $2^n W_0$ if *all* n tosses win, and 0 otherwise:

$$E[W_n] = (2p)^n W_0 \xrightarrow{n \rightarrow \infty} \infty, \quad P(W_n > 0) = p^n \xrightarrow{n \rightarrow \infty} 0.$$

With $p = 0.6$ and $n = 25$: expected wealth is 95 times the starting capital, while the probability of not being broke is 3×10^{-6} . **The strategy that maximises expected wealth leads to ruin with probability one.**

Why Expectation Fails Here

The expectation is real, but it is carried entirely by a single, exponentially improbable trajectory — the one that never loses. Averages are democratic across *scenarios*; a gambler lives along a single *trajectory*. When wealth compounds multiplicatively, the gap between “the average path” and “almost every path” becomes total: the mean grows while the typical outcome collapses.

This is the first deep lesson of the section, and it extends far beyond coin games: *for multiplicative processes, $E[W_n]$ is dominated by vanishingly rare blow-up paths and is a treacherous guide to decisions.* The correct question is not “which strategy has the highest mean” but “which strategy wins along almost every trajectory.”

15.3 The Right Objective: The Growth Rate

The resolution comes from taking logarithms — not as a choice of utility, but as the algebraic move that makes a limit theorem applicable. Betting a fixed fraction f , wealth compounds *multiplicatively*,

$$W_n = W_0 \prod_{i=1}^n (1 + f \epsilon_i), \quad \epsilon_i = \begin{cases} +1 & \text{w.p. } p, \\ -1 & \text{w.p. } q, \end{cases}$$

and the law of large numbers is a theorem about *sums*. The logarithm is the bridge between the two:

$$\log \frac{W_n}{W_0} = \sum_{i=1}^n Y_i, \quad Y_i := \log(1 + f \epsilon_i).$$

The hypotheses are immediate. Each Y_i is a fixed deterministic function of ϵ_i , so the Y_i inherit independence and identical distribution from the tosses; and each takes only two values, so the common mean — the *expected log-return of a single bet* — requires no integration:

$$E[Y_i] = p \log(1 + f) + q \log(1 - f) =: g(f), \quad \text{the same for every } i,$$

finite precisely when $f < 1$. The excluded case is not a technicality: at $f = 1$ a single loss gives $\log(0) = -\infty$, so $g(1) = -\infty$. The bet-everything catastrophe of the previous subsection reappears here as the divergence of the very quantity we are about to maximise.

The strong law of large numbers now applies directly:

$$\frac{1}{n} \log \frac{W_n}{W_0} = \frac{1}{n} \sum_{i=1}^n Y_i \xrightarrow{\text{a.s.}} g(f).$$

The mode of convergence is the whole point. “Almost surely” means along essentially *every* trajectory — not merely on average across scenarios, which is exactly the guarantee that $E[W_n]$ failed to provide. And the average log-return per bet, a quantity defined one line above as a mere expectation, turns out to govern the trajectory itself: exponentiating, $W_n \approx W_0 e^{ng(f)}$, so wealth grows or decays exponentially at the rate $g(f)$ — which earns it the name *growth rate*. A positive g sends $W_n \rightarrow \infty$ almost surely; a negative g sends $W_n \rightarrow 0$. (The convergence is of the *exponent*; the multiplicative level continues to fluctuate, which is why the comparison in Theorem 15.1 is stated as a ratio of two strategies’ wealths.)

Growth-Rate Optimality

Theorem 15.1. *Let f^* maximise $g(f)$, and let f be any other constant fraction with $g(f) < g(f^*)$. Then*

$$\frac{W_n^{(f)}}{W_n^{(f^*)}} = \exp\left(n\left[\frac{1}{n} \sum \log(1 + f \epsilon_i) - \frac{1}{n} \sum \log(1 + f^* \epsilon_i)\right]\right) \xrightarrow{\text{a.s.}} 0.$$

The growth-optimal bettor eventually — and permanently — overtakes every other constant-fraction strategy, along almost every trajectory.

Remark 15.2. [$E[\log W]$ versus $\log E[W]$] Maximising $g(f) = E[\log(\text{factor})]$ is *not* a utility assumption smuggled in through the back door; it is forced by the almost-sure asymptotics of multiplicative compounding. By Jensen’s inequality, $E[\log W] \leq \log E[W]$, always — and the gap between the two sides is exactly the gap between the typical trajectory and the mean. The bet-everything strategy maximises $\log E[W_n]$; the Kelly strategy maximises $E[\log W_n]$. Jensen’s gap is where the ruin hides.

15.4 The Kelly Fraction

Kelly Criterion, Binary Bet

Theorem 15.3. For the even-odds bet with win probability $p > 1/2$, the growth rate $g(f) = p \log(1 + f) + q \log(1 - f)$ is maximised at

$$f^* = p - q$$

— bet your edge. More generally, for a bet paying b -to-1 (win bf , lose f),

$$f^* = \frac{bp - q}{b} = p - \frac{q}{b},$$

the celebrated “edge over odds” formula.

Proof. Differentiate: $g'(f) = \frac{p}{1+f} - \frac{q}{1-f}$. Setting $g'(f^*) = 0$ gives $p(1 - f^*) = q(1 + f^*)$, i.e. $f^* = p - q$. Concavity of g ($g'' < 0$ throughout) guarantees a maximum. The general-odds case is identical with payoff bf on a win. $\square$

Example 15.4 (The Numbers for $p = 0.6$). Kelly bets $f^* = 0.2$: twenty percent of capital per toss. The optimal growth rate is

$$g^* = 0.6 \log(1.2) + 0.4 \log(0.8) \approx 0.0201 \text{ nats per bet} \approx 2\%,$$

so capital doubles roughly every $\ln 2/g^* \approx 34$ bets. Two further values on the curve deserve attention:

$$g(0.1) \approx 0.0150, \quad g(0.3) \approx 0.0148, \quad g(0.389) \approx 0, \quad g(0.4) < 0.$$

Betting *half* Kelly or *one-and-a-half* Kelly costs about the same modest slice of growth — but past $f \approx 0.39$ (roughly *twice* Kelly), the growth rate turns **negative**: a bettor with a genuine 60% edge who stakes 40% of capital per toss goes broke, exponentially fast, with probability one. *Overbetting converts an edge into ruin.*

15.5 Volatility Drag

Before treating continuous returns, we isolate the mechanism that makes overbetting lethal. It fits in one line:

$$(1 + x)(1 - x) = 1 - x^2 < 1.$$

An asset that rises 10% and then falls 10% does not return to par: it stands at 0.99. Up 50% then down 50% leaves 0.75. Gains and losses of equal *arithmetic* size do not cancel — the loss bites a larger base or, equivalently, the recovery works from a smaller one. Fluctuation itself has a cost, and the cost is quadratic in the fluctuation’s size.

Volatility Drag

Theorem 15.5. For a return R with mean μ and variance σ^2 (both small), the growth rate satisfies

$$g = E[\log(1 + R)] \approx \mu - \frac{\sigma^2}{2}.$$

The geometric (compounded) growth of an investment is its arithmetic mean return minus half its variance.

Proof. Taylor-expand $\log(1 + R) = R - R^2/2 + O(R^3)$ and take expectations: $E[\log(1 + R)] \approx \mu - E[R^2]/2 \approx \mu - \sigma^2/2$ for small μ, σ . (For log-normal returns the relation $g = \mu - \sigma^2/2$ is exact in the continuous-time limit.) $\square$

Possibly the Most Important Inequality in Trading

The term $\sigma^2/2$ is a tax that volatility levies on compounded wealth, paid regardless of the sign of the edge. A strategy earning an arithmetic 5% with 30% volatility compounds at $0.05 - 0.045 = 0.5\%$ — the edge is almost entirely consumed by drag. The same strategy levered $2\times$ has arithmetic mean 10% but drag 0.18: it *loses* 8% per year while being “twice as good” in expectation. Leverage scales the mean linearly and the drag *quadratically*; this asymmetry is the entire mathematics of position sizing, and Kelly is nothing more than its exact optimisation.

15.6 Continuous Kelly and the Growth Parabola

Consider now an asset (or strategy) with i.i.d. per-period returns of mean μ and variance σ^2 , and invest a constant fraction f of wealth (fractions $f > 1$ meaning leverage). The portfolio return per period is fR , so by Theorem 15.5,

$$g(f) \approx f\mu - \frac{f^2\sigma^2}{2}.$$

Continuous Kelly

Theorem 15.6. *The growth rate $g(f) = f\mu - f^2\sigma^2/2$ is a downward parabola, maximised at*

$$f^* = \frac{\mu}{\sigma^2}, \quad g^* = \frac{\mu^2}{2\sigma^2} = \frac{\text{Sharpe}^2}{2}.$$

Moreover $g(2f^*) = 0$: **betting twice Kelly earns exactly nothing**, and beyond twice Kelly the growth rate is negative.

Proof. Complete the square: $g(f) = \frac{\sigma^2}{2} [(f^*)^2 - (f - f^*)^2]$ with $f^* = \mu/\sigma^2$. The maximum, the value at the maximum, and the root at $f = 2f^*$ are all immediate. $\square$

Remark 15.7 (Reading the Parabola). Three features of $g(f) = \frac{\sigma^2}{2} [(f^*)^2 - (f - f^*)^2]$ repay contemplation.

First, the optimal growth rate is one half the squared Sharpe ratio — a beautiful identity linking the two most important numbers in strategy evaluation. A Sharpe-1 strategy, fully Kelly-sized, compounds at 50% of capital per unit time; a Sharpe-0.5 strategy at 12.5%.

Second, the parabola is flat at the top: the penalty for missing f^* is quadratic, so moderate sizing errors are cheap. Betting $0.75 f^*$ or $1.25 f^*$ each captures $g(f)/g^* = 1 - (0.25)^2 \approx 94\%$ of the optimal growth.

Third, the flatness is a trap when combined with uncertainty about μ . The interval of positive growth is $(0, 2f^*)$: its width is set by the *true* edge, which is never known. Overestimating μ by a factor of two — entirely routine, given how noisy edge estimates are — places a sincere Kelly bettor exactly at $2f_{\text{true}}^*$: zero growth. Any further optimism produces confident, systematic self-destruction. *The asymmetry of consequences (underbetting wastes some edge; overbetting can negate or reverse it) is the practical heart of the Kelly story.*

15.7 Fractional Kelly and the Price of Safety

Practitioners who use Kelly almost universally bet a fixed fraction $c \in (0, 1)$ of the Kelly stake — typically one-half or one-quarter. The parabola prices this choice exactly.

Fractional Kelly: Growth and Drawdowns

Theorem 15.8. *Betting cf^* (with $0 < c \leq 1$) in the continuous setting:*

- The growth rate is

$$g(cf^*) = c(2 - c)g^*.$$

- The probability that wealth ever falls to a fraction $a < 1$ of its starting value is

$$P(\text{drawdown to } a) = a^{(2-c)/c}.$$

In particular, **full Kelly** ($c = 1$): $P(\text{drawdown to } a) = a$ — a 50% chance of losing half the capital at some point, a 10% chance of losing 90%. **Half Kelly** ($c = 1/2$): 75% of the optimal growth, with drawdown probability a^3 — the chance of ever losing half the capital drops from 50% to 12.5%.

Proof. With $f = cf^*$, log-wealth is (in the diffusion limit) a Brownian motion with drift $m = c\mu^2/\sigma^2 - c^2\mu^2/(2\sigma^2) = c(2 - c)g^*$, proving the growth claim, and volatility $s = c\mu/\sigma$. For a Brownian motion with positive drift m and volatility s , the probability of ever falling below $-d$ (here $d = -\log a > 0$) is the classical hitting bound e^{-2md/s^2} — itself an optional-stopping computation with the exponential martingale e^{-2mX_t/s^2} , in the spirit of our random-walk sections. Substituting,

$$\frac{2m}{s^2} = \frac{2c(2 - c)\mu^2/(2\sigma^2)}{c^2\mu^2/\sigma^2} = \frac{2 - c}{c}, \quad P = e^{-\frac{2m}{s^2}d} = a^{(2-c)/c}. \quad \square$$

Why Everyone Bets Half Kelly

Half Kelly sits at an extraordinary point on the trade-off curve: it retains 75% of the growth while shrinking every drawdown probability to its *cube*. But the deeper reason for fractional sizing is not risk appetite — it is *epistemic humility*. The Kelly fraction is computed from an estimated edge, and estimated edges in markets are noisy, unstable, and biased upward by selection (the strategies you trade are the ones whose backtests looked best — recall the multiple-testing inflation of the longest-run section). Since betting $2\times$ the true Kelly yields zero growth, halving your computed stake is precisely the margin of error that protects against having overestimated your edge by a factor of two. Half Kelly is not timidity; it is a confidence interval expressed as a position size.

Remark 15.9 (Losing Streaks: a Bridge to the Longest Run). Even correctly sized, a Kelly bettor must survive streaks. In n bets with loss probability q , the longest losing streak is, by the theory of Section on the longest run, approximately $\log n / \log(1/q)$. For $q = 0.4$ and $n = 1000$ bets: streaks of ≈ 7 – 8 consecutive losses are *expected*, each multiplying capital by $(1 - f^*)^L = 0.8^8 \approx 0.17$ at full Kelly — an 83% peak-to-trough plunge within the normal operation of a winning strategy. (Half Kelly: $0.9^8 \approx 0.43$, a 57% plunge.) Fraction sizing never eliminates streak risk; it converts an unsurvivable drawdown into a merely harrowing one. A bettor — or a fund — that cannot tolerate the drawdowns its own sizing implies will be forced to abandon the strategy at the trough, converting a temporary excursion into a permanent loss. *Sizing and psychological (or institutional) drawdown tolerance are the same decision.*

15.8 Kelly and Information Theory

The criterion was born not on a trading floor but at Bell Labs. Kelly's 1956 paper interpreted a gambler with inside information as a communication channel, and found that the central quantity of Shannon's theory and the growth rate of compounding wealth are the same object.

Remark 15.10 (Growth Rate Equals Information). Substitute the Kelly fraction $f^* = p - q$ back into the binary growth rate. The two wealth multipliers simplify remarkably, because $p + q = 1$:

$$1 + f^* = 1 + p - q = 2p, \quad 1 - f^* = 1 - p + q = 2q.$$

The factor 2 is not inserted — it is what $1 \pm (p - q)$ becomes. Hence

$$g^* = p \log(2p) + q \log(2q),$$

and splitting each logarithm,

$$g^* = \underbrace{(p + q) \log 2}_{=1} + \underbrace{p \log p + q \log q}_{=-H(p)} = \boxed{\log 2 - H(p)}$$

where $H(p) = -p \log p - q \log q$ is the Shannon entropy of the coin.

The identity holds in any base; taking base 2 measures both sides in bits and sets $\log_2 2 = 1$:

$$g_{\text{bits}}^* = 1 - H_2(p),$$

the optimal growth rate is exactly the information the bettor holds about the outcome. A fair coin ($H_2 = 1$ bit) permits no growth — consistent with $f^* = 0$, no bet. A fully predictable one ($H_2 = 0$) yields 1 bit per toss, doubling wealth every time — consistent with $f^* = 1$ and multiplier $2p = 2$. For $p = 0.6$: $g^* = 1 - 0.971 = 0.029$ bits per toss, which at $\log 2 \approx 0.693$ nats per bit is $0.029 \times 0.693 = 0.0201$ nats — the 2% growth rate computed earlier. *The growth rate and the information content are the same number in different units.*

Kelly Betting Is Proportional Betting

The multipliers $2p$ and $2q$ have a reading that explains why entropy appears at all. Staking a fraction f at even odds is the same as allocating

$$a = \frac{1 + f}{2} \text{ of wealth to heads,} \quad 1 - a = \frac{1 - f}{2} \text{ to tails,}$$

with the winning side paying $2\times$ and the losing side nothing. At the Kelly fraction, $a = \frac{1+(p-q)}{2} = p$: *Kelly allocates to each outcome exactly its own probability.* Your wealth is then multiplied by $2 \times$ (the probability of whatever occurred), and g^* is the average logarithm of that multiplier. Betting your beliefs, and being paid the odds — that is all the formula says.

15.9 What Kelly Does Not Say

Intellectual honesty requires listing the assumptions doing the work. The criterion as derived assumes (i) the edge and odds are *known exactly*; (ii) bets are *repeated indefinitely* under identical conditions; (iii) capital is *infinitely divisible* and there are no transaction costs; (iv) the objective is *asymptotic* growth, with indifference to the path travelled. Every one of these fails in practice to some degree — edges are estimated, regimes shift, markets have frictions, and investors have finite horizons and finite pain thresholds. None of this demotes Kelly from its central role; it defines Kelly's role correctly: **an upper bound on rational size**. Positions above Kelly are indefensible under any assumptions; positions below it are a reasoned response to model risk, horizon, and drawdown tolerance. In practice the question “what is your size as a fraction of Kelly?” is better posed than “what is your size?”

Key Takeaways

Kelly and Volatility Drag

1. **Expectation misleads under compounding.** Maximising $E[W_n]$ prescribes betting everything and leads to ruin a.s.; the mean is carried by exponentially rare paths. For multiplicative processes, optimise the almost-sure growth rate $g(f) = E[\log(\text{factor})]$.
2. **Kelly fraction.** Even odds: $f^* = p - q$ (*bet your edge*). Odds b -to-1: $f^* = p - q/b$ (*edge over odds*).
3. **Volatility drag.** $(1+x)(1-x) = 1-x^2$: fluctuation taxes compounded wealth at rate $\sigma^2/2$, so $g \approx \mu - \sigma^2/2$. Leverage scales the mean linearly and the drag quadratically.
4. **Continuous Kelly.** $f^* = \mu/\sigma^2$, and $g^* = \text{Sharpe}^2/2$. The growth parabola is flat near the top and *crosses zero at exactly twice Kelly*: overbetting a true edge produces confident ruin.
5. **Fractional Kelly.** Betting cf^* earns $c(2-c)$ of the optimal growth with drawdown probability $P(\text{fall to } a) = a^{(2-c)/c}$. Half Kelly: 75% of the growth, drawdown probabilities cubed. Fractional sizing is a confidence interval on the edge, expressed as a position.
6. **Streak arithmetic.** Longest-run theory implies losing streaks of length $\sim \log n / \log(1/q)$ are expected; sizing determines whether they are survivable. Sizing and drawdown tolerance are one decision.
7. **Kelly is information.** $g^* = \log 2 - H(p)$: the optimal growth rate equals the bettor's information about the outcome, in bits. Edges are information; compounding is how markets pay for it.
8. **Kelly as an upper bound.** Real edges are estimated, not known. Kelly marks the ceiling of rational size; professional sizing lives deliberately below it.

A Mathematical Background

This appendix collects the tools that recur across several chapters. It is reference material, not a course: each entry states the result, gives the intuition that makes it usable, points to where the notes rely on it, and cites a source for the proof. Facts used only once are explained where they occur and are deliberately absent here.

A.1 Asymptotic Estimates

A.1.1 Stirling's Approximation

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \frac{1}{12n} + O(n^{-2})\right).$$

Intuition. The factorial grows faster than any exponential but slower than n^n ; Stirling pins it exactly between them, at $n^n e^{-n}$ times a $\sqrt{n}$ correction. The relative error is $\approx 1/(12n)$, so the approximation is already accurate to 0.8% at $n = 10$ and 0.17% at $n = 50$. The most frequent consequence in these notes is the central binomial coefficient

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}, \quad \text{equivalently} \quad u_{2n} := 2^{-2n} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}},$$

the return probability of the symmetric random walk. Every appearance of π in a combinatorial answer in these notes can be traced to this $\sqrt{\pi n}$.

Where it appears. Catalan numbers and the $\pi/4$ limit; the arcsine law (the $1/\sqrt{\pi n}$ decay of u_{2n} drives the whole continuous limit); first-passage asymptotics in the reflection-principle chapter; record counts for driftless walks.

Reference. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. I, Ch. II.9.

A.1.2 Harmonic Numbers and the Euler–Mascheroni Constant

$$H_n = \sum_{k=1}^n \frac{1}{k} = \ln n + \gamma + \frac{1}{2n} + O(n^{-2}), \quad \gamma \approx 0.5772156649.$$

Intuition. H_n is the discrete analogue of $\int_1^n dx/x = \ln n$, and γ is precisely the accumulated discrepancy between the sum and the integral. Two facts worth internalising: H_n diverges (so harmonic-scale phenomena never stop, they only slow down), and it diverges *logarithmically* (so doubling n adds only $\ln 2 \approx 0.69$). A useful mental check: $H_{100} \approx 5.19$, $H_{10^6} \approx 14.4$.

Where it appears. The coupon collector's mean nH_n ; the expected number of records H_n ; the mean number of cycles of a uniform random permutation (the same H_n , and not by coincidence); the constant $\gamma/\ln 2$ inside Schilling's longest-run formula.

Reference. Graham, Knuth and Patashnik, *Concrete Mathematics*, Ch. 6.3.

A.2 Recurring Distributions

A.2.1 The Geometric Distribution: Two Conventions

Two incompatible definitions circulate, and confusing them produces answers off by exactly one.

(A) Failures before the first success: $N \in \{0, 1, 2, \dots\}$, $E[N] = \frac{1-p}{p}$, (B) Trials including the success: $M \in \{1, 2, 3, \dots\}$, $E[M] = \frac{1}{p}$.
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Both share $\text{Var} = (1 - p)/p^2$, and $M = N + 1$. For $p = 1/2$: convention (A) gives mean 1, convention (B) gives mean 2.

How to choose. Ask whether the terminating trial itself contributes to the quantity being counted. If you are counting *how many times something happened before stopping* — rolls that scored, coupons collected, gains accumulated — use (A): the stopping trial produced nothing. If you are counting *elapsed time until an event* — draws until a specific card, tosses until a pattern — use (B): the final trial is part of the wait.

Memorylessness. Both versions satisfy $P(N > s + t \mid N > s) = P(N > t)$: the geometric is the unique discrete law with this property (the exponential is its continuous counterpart). This is why so many stopping problems collapse to a single line — the process forgets its history, so the answer cannot depend on it.

Where it appears. The dice-game valuation in the Kelly chapter (convention A, where the gain-accumulating rolls are counted and the exit roll is not); the card-to-top problem (convention B); coupon collector as a sum of geometrics; sample-buying draw counts.

A.2.2 The Poisson Paradigm

If $A_1, \dots, A_m$ are events each of small probability, with only weak local dependence, then the number of events that occur is approximately Poisson with mean $\lambda = \sum_i P(A_i)$. In particular

$$P(\text{no } A_i \text{ occurs}) \approx e^{-\lambda}, \quad \lambda = \sum_{i=1}^m P(A_i).$$

Intuition. The mean λ is exact by linearity of expectation — no independence required. What the approximation adds is the *shape*: rare, weakly dependent events distribute themselves like a Poisson count. The classic sanity check is the birthday problem: with n people, $\lambda = \binom{n}{2}/365$, giving $P(\text{no match}) \approx e^{-\lambda}$. At $n = 23$ this predicts 0.500 against the exact 0.493.

When it fails, and how. The approximation treats overlapping opportunities as independent, and therefore *overcounts* when occurrences cluster. The longest-run chapter measures this precisely: a run of length ℓ contains $\ell - k + 1$ runs of length k , so the naive Poisson count inflates the expected longest run by approximately +1 relative to the truth. Poisson is the right tool for a quick estimate and the wrong tool when the constant matters.

Where it appears. Longest run of heads (Methods 1 and 2); the coupon collector's right tail; the Poisson(1) limit for the number of correctly addressed envelopes; record rarity heuristics.

Reference. Alon and Spencer, *The Probabilistic Method*, Ch. 8 (the Poisson paradigm and the Chen–Stein method, which quantifies the error).

A.2.3 The Beta Distribution

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad x \in (0, 1), \quad E[X] = \frac{\alpha}{\alpha + \beta},$$

where $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha + \beta)$ is the normalising constant.

Intuition. The Beta is the natural law of a random *proportion*. Its two parameters act as “pseudo-counts”: $\text{Beta}(\alpha, \beta)$ behaves like the posterior belief about a coin's bias after seeing $\alpha - 1$ heads and $\beta - 1$ tails. Large parameters concentrate the mass; parameters below 1 push it toward the endpoints.

Three appearances in these notes, and their unity. The Beta is the hidden thread linking three chapters that otherwise look unrelated.

(i) **Order statistics.** For n i.i.d. $\text{Unif}(0, 1)$ variables, $U_{(k)} \sim \text{Beta}(k, n - k + 1)$, whence

$$E[U_{(k)}] = \frac{k}{n+1}$$

— the “equispaced sample” formula of the order-statistics chapter is a Beta mean.

(ii) **The Pólya urn.** Starting from a white and b black balls with reinforcement c , the limiting white fraction is $\text{Beta}(a/c, b/c)$. This is the *de Finetti mixing measure* of the urn’s exchangeable sequence (see below).

(iii) **The arcsine law.** The arcsine distribution is $\text{Beta}(1/2, 1/2)$: since $B(1/2, 1/2) = \pi$, its density is

$$\frac{x^{-1/2}(1-x)^{-1/2}}{\pi} = \frac{1}{\pi\sqrt{x(1-x)}}.$$

The famous π of the arcsine law is nothing but the Beta normalising constant, and the U-shape is the signature of parameters below 1.

Reading these together: uniform sampling, reinforcement dynamics, and random-walk occupation times all produce random proportions, and all such proportions live in the Beta family.

Reference. Johnson, Kotz and Balakrishnan, *Continuous Univariate Distributions*, Vol. 2, Ch. 25.

A.2.4 The Gumbel Distribution and Extreme Values

$$F(x) = \exp\left(-e^{-(x-\mu)/\beta}\right), \quad E[X] = \mu + \beta\gamma, \quad \text{Var}(X) = \frac{\pi^2\beta^2}{6}.$$

Intuition. The Gumbel is to maxima what the normal is to sums. By the Fisher–Tippett–Gnedenko theorem, the rescaled maximum of n i.i.d. variables can converge to only three laws (Gumbel, Fréchet, Weibull), and *light* tailed variables — exponential, normal, geometric — land in the Gumbel class. Its shape is asymmetric: a sharp left side and an exponentially decaying right tail, which is what “heavy right tail” means for maxima. Note where γ appears again: the mean of a standard Gumbel is exactly the Euler–Mascheroni constant.

The double-exponential signature. Whenever a survival probability has the form $P(M_n \leq a_n + b_n x) \rightarrow \exp(-e^{-x})$, an extreme-value limit is at work. Recognising this shape is often faster than deriving it.

Where it appears. The longest run of heads (both the distribution of L_n and of the maximum over 30 players); the coupon collector’s completion time (the double-exponential tail of the previous chapter); the asymptotics behind Schilling’s constant $\gamma/\ln 2 - 3/2$.

Reference. Embrechts, Klüppelberg and Mikosch, *Modelling Extremal Events*, Ch. 3.

A.3 Structural Tools

A.3.1 Exchangeability and de Finetti’s Theorem

A sequence $X_1, X_2, \dots$ is *exchangeable* if its joint law is invariant under every finite permutation of indices: what matters is which values occur, not in what order.

de Finetti’s theorem. Every infinite exchangeable sequence of $\{0, 1\}$ -valued random variables is a *mixture of i.i.d. sequences*: there exists a random variable Θ on $[0, 1]$ such that, conditional on $\Theta = \theta$, the X_i are i.i.d. $\text{Bernoulli}(\theta)$.

Intuition. Exchangeable is strictly weaker than i.i.d. — exchangeable variables may be strongly dependent — yet de Finetti says the only way to build such dependence is to draw a random “bias” once and then flip independently. All the dependence is funnelled through a single hidden parameter. This is the mathematical justification for Bayesian modelling: a prior is not an arbitrary choice but the mixing measure that exchangeability forces to exist.

The workhorse consequence. Most uses in these notes need only exchangeability itself, not the full theorem, in the following form: *in an exchangeable sequence, any statement about relative ranks or positions is uniform over the symmetric group*. Hence the maximum of the first n observations is equally likely to occupy each of the n positions, giving $P(\text{record at } n) = 1/n$ in one line and *without any assumption on the underlying distribution*. This distribution-free character is what makes rank-based arguments so robust.

Where it appears. Records (the $1/n$ law and the independence of record indicators); the secretary problem and all its variants (the rank lemma $E[\text{true rank}] = r(n+1)/(i+1)$); the Pólya urn (whose limiting Beta($a/c, b/c$) is precisely its de Finetti mixing measure); the expected number of correct envelopes.

Reference. Aldous, *Exchangeability and Related Topics*, École d’Été de Saint-Flour XIII.

A.3.2 Martingales and the Optional Stopping Theorem

A process (M_t) adapted to a filtration $(\mathcal{F}_t)$ is a *martingale* if $E[M_{t+1} | \mathcal{F}_t] = M_t$: given everything known at time t , the expected next value is the current one — a fair game.

Optional stopping theorem. If (M_t) is a martingale and T a stopping time, then $E[M_T] = E[M_0]$, provided one of the following holds: (i) T is bounded; (ii) $(M_{t \wedge T})$ is uniformly integrable; or (iii) $E[T] < \infty$ and the increments of M are bounded.

Intuition. *No stopping rule turns a fair game into a favourable one.* You may quit whenever you like, using any information available at the time, and your expected wealth remains what you started with. The power of the theorem lies in reading this backwards: if you can construct a fair game whose value at a cleverly chosen stopping time encodes an unknown quantity, then $E[M_T] = E[M_0]$ solves for it.

The regularity conditions matter. Without them the conclusion is false: the doubling strategy on a symmetric random walk stops at $+1$ with probability one, so $E[M_T] = 1 \neq 0 = E[M_0]$. What fails is (iii)—the required capital is unbounded. When applying the theorem, verify a condition; the classic error is to skip it.

Standard martingales used in these notes. For a random walk S_n with i.i.d. steps of mean μ and variance σ^2 :

$$S_n - n\mu, \quad (S_n - n\mu)^2 - n\sigma^2, \quad e^{\theta S_n} / (E[e^{\theta X}])^n.$$

The first gives hitting probabilities, the second expected hitting times, the third exponential (drawdown) bounds.

Where it appears. Gambler’s ruin (all three martingales); the ABRACADABRA casino argument for pattern hitting times; the Pólya urn (its fraction is a bounded martingale, whence convergence); the drawdown probability $a^{(2-c)/c}$ under fractional Kelly.

Reference. Williams, *Probability with Martingales*, Ch. 10–12.

A.3.3 Generating Functions

For a non-negative integer-valued random variable X , or for a combinatorial sequence (a_n) :

$$G_X(z) = E[z^X] = \sum_{n \geq 0} P(X = n)z^n, \quad A(z) = \sum_{n \geq 0} a_n z^n.$$

Intuition. A generating function is a bookkeeping device that stores an entire sequence in one object, chosen so that the operations one wants to perform become algebra: *convolution of sequences becomes multiplication of series*. This is why decompositions of combinatorial objects (“a path is a first excursion followed by a shorter path”) translate into equations that can simply be solved.

The three operations that do all the work.

- (i) **Sums of independent variables:** $G_{X+Y}(z) = G_X(z)G_Y(z)$.
- (ii) **Moments:** $E[X] = G'(1)$ and $E[X(X-1)] = G''(1)$; more generally the r -th derivative at 1 gives the r -th factorial moment.
- (iii) **Asymptotics:** the growth of a_n is governed by the singularity of $A(z)$ nearest the origin. For a rational $A(z)$ with dominant pole ρ , one has $a_n \sim C\rho^{-n}$.

Where it appears. Catalan numbers and the $\pi/4$ result; the arcsine law (the key cancellation in the bivariate generating function); Guibas–Odlyzko pattern hitting times via the autocorrelation polynomial; the exact distribution of the longest run (dominant-pole analysis); the renewal generating function $U(z) = (1 - G(z))^{-1}$ in the size-biasing chapter.

Reference. Wilf, *generatingfunctionology* (elementary and free); Flajolet and Sedgewick, *Analytic Combinatorics* (the definitive treatment of the singularity analysis in (iii)).